

발 간 등 록 번 호

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Policy Research 2023-22-03



2023 K-Innovation Partnership Program with Indonesia

Policy Consultation on the Design and Implementation of Research & Innovation Infrastructure and the Funding Policy for Improvement of the Science, Technology and Innovation System in Indonesia (1st year)

2023 K-Innovation Partnership Program with Indonesia

Title	Policy Consultation on the Design and Implementation of Research & Innovation Infrastructure and the Funding Policy for Improvement of the Science, Technology and Innovation System in Indonesia
Institute	Science and Technology Policy Institute (STEPI)
Partner country	Republic of Indonesia
In cooperation with	National Research and Innovation Agency (BRIN)
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Project Coordinator	Shinae Kang, Researcher, Division of Global Innovation Strategy Research, STEPI

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Executive Summary

The 2023 K-Innovation partnership program with Indonesia represents a significant initiative aimed at supporting the National Research and Innovation Agency (BRIN) of Indonesia in conducting comprehensive assessment of its existing research and development (R&D) support policies and programs. This policy advisory project, implemented by the Science and Technology Policy Institute (STEPI), serves as a platform for exchanging knowledge and experience in the realm of R&D support between South Korea and Indonesia.

The project adopts a multifaceted approach, comprising policy consultation, knowledge transfer, capacity-building workshops, and tailored training programs. These components are designed to provide a deeper understanding of R&D support mechanisms and their potential applicability in the Indonesian context. A critical aspect of this initiative involves sharing actionable insights and strategic recommendations based on South Korea's extensive experience in R&D support policies.

To facilitate practical knowledge exchange, STEPI, in collaboration with leading Korean R&D support institutions, has organized a series of interactive workshops and capacity-building sessions. These programs are aimed at enhancing the competencies of Indonesian stakeholders in the design, implementation, and evaluation of R&D support policies. By facilitating such an exchange of knowledge, the collaboration aims to empower Indonesian policymakers and stakeholders with the equipment and expertise required to enhance the country's R&D ecosystem.

The outcomes of the policy consultation and knowledge exchange activities culminated in a significant “dissemination” forum in Cibinong, Indonesia. This forum was strategically integrated as a special session within Indonesia's national research and innovation event,

Expo INARI, held in September 2023. This forum provided a platform for the discussion of policy recommendations and facilitated a dialogue on international cooperation, knowledge transfer and R&D support system.

In conclusion, the 2023 K-Innovation partnership program with Indonesia is an exemplary instance of international cooperation in the pursuit of fostering innovation and research excellence. Through this initiative, South Korea and Indonesia have embarked on a transformative path, with the shared goal of advancing R&D support policies and programs. This exchange of knowledge, policy insights and practical experience positioned Indonesia on a trajectory towards enhancing its R&D ecosystem and, consequently, accelerating its progress in the realms of science, technology, and innovation.

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CHAPTER 1

Project Overview

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Chapter 1. Project Overview

1. Project Background

Innovation has become a catalyst for advanced countries to harness their national economies, thereby facilitating the rapid transformation of developing nations into developed economies. South Korea, officially known as the Republic of Korea, is a shining example of a country that has harnessed innovation to drive economic development. Its remarkable growth trajectory has placed it on par with the Organization for Economic Cooperation and Development (OECD) nations, underscoring the critical role of research and development (R&D) and technology in accelerating national economic progress. The South Korean government's proactive engagement through its science-technology-innovation (STI) policy is a prominent measure that has fueled breakthroughs in the country's journey to economic development and prosperity.

At the heart of this innovation-driven approach lies a national innovation system (NIS), which serves as a functioning framework wherein both R&D and non-R&D endeavors stimulate economic activities. It is essential to recognize that innovation extends beyond traditional R&D activities: Rather, it encompasses a broad spectrum of activities, including reverse engineering, imitation, adoption, diffusion, modification, and learning by doing. Effective strategies for transferring innovation from developed to developing countries are pivotal in ensuring sustainable progress. South Korea has excelled in learning from developed nations such as Japan, the United States of America, and various European countries. This practice can serve as a valuable model for developing nations like Indonesia to emulate and adapt to their specific contexts.

In light of these considerations, the collaborative effort between the governments of South Korea and Indonesia assumes great significance. The primary objective of this collaboration is to facilitate the transfer of successful practices from South Korea's STI policy to Indonesia's own STI policy framework. This transfer of knowledge and best practices holds the potential to drive innovation, foster economic growth, and promote sustainable development in Indonesia, ultimately contributing to the prosperity and well-being of its citizens.

The 2023 K-Innovation initiative with Indonesia represents a continuation of the collaborative efforts between South Korea and Indonesia that have been ongoing since 2014. In light of the ever-evolving landscape of science, technology, and innovation (STI) at the global level, a new "project concept paper" (PCP) developed in 2023 aims to adapt to the current state of STI issues in Indonesia. This initiative seeks to leverage the expertise of South Korea's Science and Technology Policy Institute (STEPI) to offer valuable insights and recommendations about the STI challenges facing Indonesia.

The rapidly changing dynamics of STI issues on a global scale and the significant transformations occurring within Indonesia's STI policy institutions underscore the urgency of policies to enhance Indonesia's national innovation system (NIS). Notably, Indonesia has faced challenges in keeping pace with other ASEAN countries, so establishing the National Research and Innovation Agency (BRIN) represents a significant development in this regard. This initiative aims to address the challenges that have persisted for the last two decades.

The primary focus of the 2023 K-Innovation collaboration with Indonesia revolves around consulting and addressing two pivotal aspects of Indonesia's National Innovation System (NIS), namely, the Research and Innovation (R&I) infrastructure policy and the R&I funding policy. In particular, the K-Innovation 2023 project will place special emphasis on two facets of the R&I infrastructure policy:

- I. Analysis of the Research & Innovation (R&I) infrastructure policy in Indonesia, with a specific focus on both laboratory and non-laboratory facilities.

- II. The elaboration of a feasible Research & Innovation (R&I) infrastructure policy aimed at providing robust support to the national system in Indonesia.

To effectively address the challenges, the Government of Indonesia must design robust research and innovation (R&I) infrastructures and establish funding policies that fulfil specific functions, including development of well-governed R&I infrastructure frameworks and well-structured R&I funding mechanisms. To achieve these objectives, collaborative efforts between the Korean and Indonesian teams have focused on five key activities: theoretical conceptualization, focused group discussions (FGD); in-depth interviews; capacity-building workshops in Korea; and a dissemination forum. These activities collectively are designed to collectively contribute to enhancing Indonesia's STI landscape and promoting sustainable development across the country.

2. Project Objectives

The 2023 K-Innovation ODA program with Indonesia has three primary objectives. The first objective is to assist Indonesian policymakers, researchers, and policy analysts in the design and enhancement of research and innovation (R&I) infrastructures managed by BRIN. The second objective is to support these stakeholders in identifying and addressing the challenges related to R&I funding within BRIN. The last objective is to strengthen STI policy research capacities through a range of activities including STI policy consultation, collaborative STI research, training or policy workshops, and STI dissemination forum.

From 2023 onwards, the long-term goal is to lay a firm foundation for STI cooperation and collaboration between South Korea and Indonesia. Such will not be limited solely to cooperation and collaboration between STEPI and BRIN but will open up to other public or private agencies in Korea and Indonesia concerned about addressing STI issues. The Republic of Korea is an essential partner for the Government of Indonesia in providing STI policy consultation practices. In the 1960s, South Korea and Indonesia had similar histories regarding the early years of their independence. Both countries were categorized as having low-growth economies and difficulties in achieving prosperity. South Korea has undergone one of the most remarkable economic transformations and industrial development in the past 60 years. Beginning as an agriculture-based economy in the 1960s, it has become a country renowned for its advanced technologies, achieving a high-ranking global economic position by 2023, while Indonesia is growing slowly as a lower-middle-class economy. Therefore, transferring the lessons learned from South Korea to Indonesia is necessary.

The term of “science-technology-innovation (STI) policy”, interchangeable with “innovation policy”, refers to intentional and unintentional governmental efforts to create, establish, promote, and trigger innovation in a region or country. Innovation policy definitions are various, and do not aim merely to define or explain how to commercialize R&D products for the market, but also how social, educational, economic, and political systems are pivotal to formulating and implementing innovation policy. According to Patanakul and Pinto (2017),

STI, or innovation policy, focuses more on the role of companies as a leading actor. It is a government action to support the national business climate and promote entrepreneurship and innovation activities. Further, it aims to improve capacity building for national firms.

Innovation policy goals include the following aims: to combat private underinvestment in R&D; to provide more incentives to the organizations conducting R&D; to increase the diffusion of technology; to develop human resources in the S&T field; to protect intellectual property rights; to synergize industrial and trade policies; and to increase the supply of technologies to local companies (Mani, 2002). In this policy consultation, STI, i.e., innovation policy, focuses on developing and improving the quality of research and innovation (R&I) infrastructures and facilitation and promoting the policy for developing and improving the quality of research and innovation (R&I) funding mechanisms. R&I is the key that BRIN commits to bringing R&D products to the market, leading to supporting a knowledge-based economy (KBE) in Indonesia.

[Table 1-1] Characteristic of Policy for R&I Infrastructure

No.	Policy for R&I infrastructure support	Sources
1	Construction of scientific infrastructure, support for high-technology R&D and promotion of private research labs, industrial R&D, technology standards, and technology licensing.	Yoon (2018)
2	Establishment of public R&D laboratories that are fully equipped for technical research, technical assistance, certification, and quality control.	World Bank (2010)
3	Largely derived from supply side rather than demand side policies.*	European Union (2015)
4	Largely focused on directive intervention rather than facilitative intervention.**	Wang (2018)
5	Framing of policies on R&D for growth and the innovation system.	Schot & Steinmueller (2018)

Notes:

*Demand-side innovation policies comprise measures for stimulating private demand for innovation, intelligent and pre-commercial procurement policies, innovation inducement prizes, and standardization and regulation. Supply-side policies include fiscal incentives and direct support for R&D and innovation, access to finance for innovative ventures, skills upgrades and human resources policies, entrepreneurship policy, technical services and advice, cluster, collaboration, and networking policies (European Union, 2015).

**Directive intervention, which aims to achieve pre-determined results by making changes in investment and production patterns in selected industries; and facilitative intervention, which aims to create positive environments for private enterprises by providing public goods such as infrastructure and education (Luedde-Neurath 1998 in Wang, 2018).

[Table 1-2] Characteristics of Policy for R&I Funding

No.	Policy for R&I infrastructure support	Sources
1	Subsidies for adoption, government procurement, R&D subsidies, R&D tax incentives.	Yoon (2018)
2	Tax incentives for R&D, grants for R&D projects, accelerated depreciation of R&D equipment, exemption from duties on imported inputs into R&D, venture capital to facilitate commercialization of research results.	World Bank (2010)
3	Mainly derived from supply side rather than demand side policies.*	European Union (2015)
4	Directive intervention rather than facilitative intervention.**	Wang (2018)
5	Framing of policies on R&D for growth and the innovation system	Schot & Steinmueller (2018)

Notes:

* Demand side innovation policies comprise measures for stimulating private demand for innovation, intelligent and pre-commercial procurement policies, innovation inducement prizes, and standardization and regulation. Supply side policies include fiscal incentives and direct support for R&D and innovation, access to finance for innovative ventures, skills upgrades and human resources policies, entrepreneurship policy, technical services and advice, cluster, collaboration and networking policies (European Union, 2015).

**Directive intervention, which aims to achieve pre-determined results by making changes in investment and production patterns in selected industries; and facilitative intervention, which aims to create positive environments for private enterprises by providing public goods such as infrastructure and education (Luedde-Neurath 1998 in Wang, 2018).

The characteristics of R&I infrastructure and funding policies, as shown in points 3 and 4 on both tables above, are similar to the role of government in dealing with STI policy. Referring to Sun et al. (2019), under the top-down model, the government establishes programs and instruments to promote interaction and collaboration among academicians, business, and government actors, whereas with the bottom-up model, it facilitates the creation of a collaborative network among academicians, businesses, and government actors (market competition model). R&I infrastructure and funding policies are governmental investments and interventions designed to attract private sectors to invest in and conduct R&I activities. However, in point 5 on both tables above, the role of government is based on both the top-down and bottom-up models.

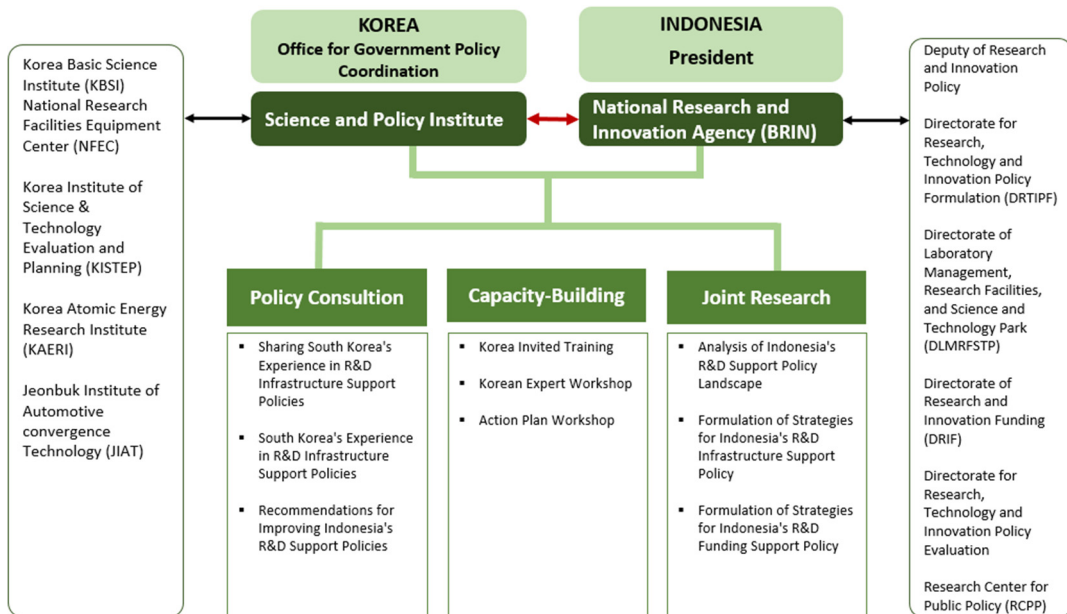
3. Project Framework

The 2023 K-Innovation Partnership program consisted of five key activities: a kick-off seminar, a policy consultation project, a joint research program aimed at producing a policy consulting report, a capacity-building training program for policymakers, policy analysts and researchers, and a dissemination workshop. The meetings and workshops were held offline in two countries, Indonesia (Jakarta City) and South Korea (Seoul and Sejong City). The first phase consisted in holding a kick-off seminar with the STEPI team and BRIN to discuss the details of the activities and to evaluate the previous year's activities. The kick-off meeting, which was held at BRIN's building in Jakarta (March 2023), consisted of presentations by Korean experts on Korean practices and cases related to the themes of R&D infrastructure and funding, with the participation of Indonesian experts, and officials from BRIN, non-governmental organisations, and universities.

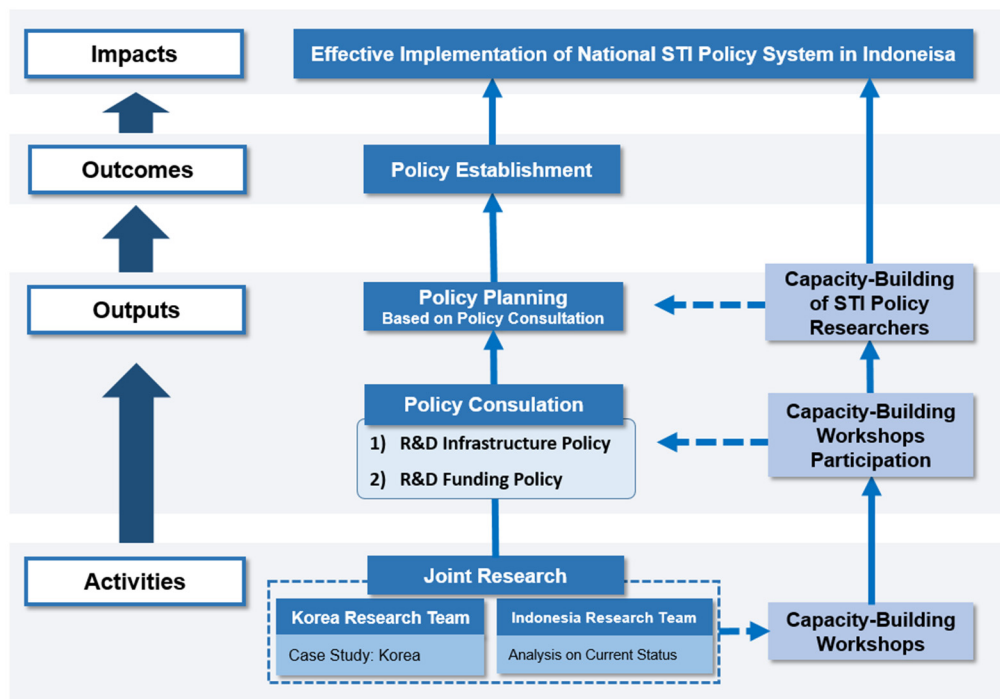
The Korean experts collaborated with Indonesian policymakers, policy analysts and researchers for joint policy consultation and research on two themes: 1) designing and improving public service in the field of research and innovation (R&I) infrastructures delivered by BRIN; 2) designing and improving public service in the field of research and innovation (R&I) funding as delivered by BRIN. The BRIN teams, under the supervision of the Korean experts, assigned focal points to various groups composed of Indonesian policymakers, policy analysts, and researchers. In post-pandemic period, kick-off meetings, training, and workshops were held under a hybrid system comprising offline and online means.

The offline or physical activities were the main channel for sharing the Indonesian situation and Korean practices and exchanging policy ideas and suggestions. The last activity of K-Innovation was a dissemination workshop wherein the research teams could present and receive feedback on the results of their joint researches from STI research community of Indonesia and Korea. Having completed the dissemination workshop, Korean and Indonesian researchers will work together to publish academic papers based on the results of their joint research.

[Figure 1-1] Schematization of the 2023 Project Structure



[Figure 1-2] Detailed Framework of the 2023 Project



4. Project Team

Led by Dr. Eun Joo Kim of STEPI, eighteen policymakers, researchers, and policy analysts from Korea and Indonesia participated in implementing the 2023 K-Innovation Project. The Korean team invited two Korean experts for a policy consultation on designing and improving research and innovation (R&I) infrastructures and funding in Indonesia. Armed with an in-depth understanding of Indonesia's STI environment and situation, and more particularly its STI governance and policy, the Korean experts were able to facilitate collaboration with BRIN.

Dr. Youngjoo Kim, who possesses a wealth of expertise in R&D infrastructure policy, also participated in supervising the policy consultation and research on R&I infrastructures in Indonesia. Dr. Jiho Hwang, who possesses abundant expertise in R&D funding policy, also participated in supervising the policy consultation and research on R&I funding in Indonesia. Ms. Shinae Kang acted as a project coordinator of the 2023 K-Innovation Project with the assistance of STEPI researchers, and the Indonesian joint research group supervised by Dr. Young Jo Kim. Korean advisers headed the Indonesian policy consultation and research teams, each of which was assigned a person in charge to ensure better communication and coordination among the Indonesian researchers. Dr. Dudi Hidayat served as the Indonesian counterpart and leader of the Indonesian teams.

Dr. Boediastoeti Ontowirjo took the advisor role on 2023 K-Innovation for the Indonesian teams. Dr. Dudi Hidayat was the coordinator for 2023 K-Innovation and also headed the Indonesian team on policy consultation for the research and innovation (R&I) infrastructure and funding facilitation. In terms of the R&I infrastructure policy consultation, the members of the task force included Dr. Tjahjo Pranoto, Ms. Nurbaiti Marsas Prilitasari, Ms. Yustina Nita Sulistami, Ms. Umi Salamah, and Mr. Anugerah Yuka Asmara. Likewise, the members of the research team for R&I funding included Ms. Supadmi, Dr Siti Annisa Silvia Rosa, Mr. Aditya Wisnu Pradana, Dr. Juhartono, and Ms. Nungki Indrianti.

[Table 1-3] Project Team

Name	Position/Organisation	Role
Dr. Eun Joo Kim	Principal Researcher, STEPI	Project Manager
Dr. Deok Soon Yim	Senior Research Fellow, STEPI	Participant
Ms. Shinae Kang	Principal Researcher, STEPI	Project Coordinator
Dr. Boediastoeti Ontowirjo	Deputy Chairman, Deputy of Research and Innovation Policy, BRIN	Counterpart and Advisor
Dr. Dudi Hidayat	Director, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Counterpart and Team Leader
Team 1. Research and Innovation (R&I) Infrastructure Policy		
Dr. YongJoo Kim	Lead Manager/Senior Researcher, National Research Facilities and Equipment Centre (NFEC), Korea Basic Science Institute (KBSI)	Team leader
Dr. Tjahjo Pranoto	Senior Engineer, Directorate of Laboratory Management, Research Facilities, and Science and Technology Park (DLMRFSTP), BRIN	Indonesian Team
Ms. Nurbaiti Marsas Prilitasari	Analyst of Science and Technology Utilization, Directorate of Laboratory Management, Research Facilities, and Science and Technology Park (DLMRFSTP), BRIN	Indonesian Team
Ms. Yustina Nita Sulistami	Coordinator, Directorate for Research, Technology and Innovation Policy Evaluation (DRTIPE), BRIN	Indonesian Team
Ms. Umi Salama	Policy Analyst, Directorate for Research, Technology and Innovation Policy Evaluation (DRTIPE), BRIN	Indonesian Team
Mr. Anugerah Yuka Asmara	Coordinator, Research Centre for Public Policy (RCPP), BRIN	Indonesian Team
Team 2. Research and Innovation (R&I) Funding Policy		
Dr. Jiho Hwang	General Director/Senior Research Fellow, Korea Institute of S&T Evaluation and Planning (KISTEP)	Team Leader
Dr. Juhartono	Coordinator, Directorate of Research and Innovation Funding (DRIF)	Indonesian team
Mrs. Nungki Indrianti	Policy Analyst, Directorate of Research and Innovation Funding (DRIF)	Indonesian Team
Dr. Siti Annisa Silvia Rosa	Policy Analyst, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Indonesian Team
Mrs. Supadmi	Policy Analyst, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Indonesian Team
Ms. Rahmayanti Pertiwi	Policy Analyst, Directorate for Research, Technology and Innovation Policy Formulation (DRTIPF), BRIN	Indonesian Team
Mr. Aditya Wisnu Pradana	Researcher, Research Centre for Public Policy (RCPP), BRIN	Indonesian Team

5. Project Activities

5.1 Kick-off Seminar

- Date/Venue: Thursday, 16 March 2023 / Jakarta, Indonesia BRIN
- Purpose: To hold an inaugural seminar to promote mutual understanding of the project's theme and annual planning agreement between the two organisations.
- Participants: Over 80 people, including STEPI' project team, commissioned researchers, BRIN researchers, and other Indonesian stakeholders.
- Main Contents:
 - Presentation of the purpose and detailed plan of the 2023 policy advisory.
 - Status of and support system for R&D funding and infrastructure support policies in Indonesia and Korea.
- Key Points:
 - Classification of the research infrastructure in Korea - large and small items of equipment, operational and maintenance aspects.
 - Characteristics of governmental department restructuring in Korea in the 1990s and success factors in decentralization.
 - Korea's R&D investment strategy and ways of transferring underutilised and unused equipment.
 - Correlation between government promises, political conditions, and the allocation and distribution of R&D budgets.
 - Methodology of Korea's future forecasting strategy and system for setting science and technology policies and R&D missions based on that strategy.

[Table 1-4] Kick-off Seminar program

Time	Programs	Presenters
09.00-09.10	Introduction of 2023 K-Innovation with Indonesia	Dr. Eun Joo KIM, Principal Researcher, Division of Global Innovation Strategy Research, STEPI
09.10-09.20	Summary of the Current Status of Indonesia's R&D Support System	Dr. Dudi Hidayat, Director, Policy Formulation for Research, Technology and Innovation / Deputy for Research and Innovation Policy, BRIN
09.20-09.30		Photo Session
09.30-10.00	Current status of Indonesia's R&D Infrastructure management system	Dr. Tjahjo Pranoto, Senior Engineer Laboratory Management, Research Facilities, and S&T Park / Deputy for Research and Innovation Infrastructure, BRIN
10.00-10.30	Current status of Indonesia's R&D Funding management system	Dr. Ajeng Arum Sari, Director, Research and Innovation Funding / Deputy for Facilitation of Research and Innovation, BRIN
10.30-11.00	Korea R&D Infrastructure Management system	Dr. Yongjoo KIM, Lead Manager, Research Infrastructure Policy Team, National Research Facilities & Equipment Centre, Korea Basic Science Institute (KBSI)
11.00-11.30	Korea R&D Budget Allocation, Laws and Regulations	Dr. Jiho Hwang, Chief Director, Office of Future Technology Strategy, Korea Institute of S&T Evaluation and Planning (KISTEP)
11.30-12.00	Q&A and Discussion	<Session Chair> Dr. Deok Soon Yim, Senior Research Fellow, Office of Sustainable Innovation of Policy Research, STEPI
12.00-13.30		Lunch
13.30-15.30	In-depth discussion on topics related to R&D infrastructure and funding	

[Figure 1-3] Photos of Kick-off Seminar



5.2 Capacity-Building Workshop: Invitational program

- Date/Venue: 15 May to 20 May 2023 / Korea
- Purpose: The Capacity Building Workshop on Improvement of Research Infrastructure and Funding in Indonesia has two objectives: The first is to support policymakers, researchers, and policy analysts in identifying the research and innovation infrastructure issues faced by BRIN, while the second is to support Indonesian policymakers, researchers, and policy analysts in designing and improving the research and innovation infrastructures delivered by BRIN by learning from South Korea's best practices. These objectives are to be carried out in order to promote mutual economic cooperation between Indonesia and South Korea by contributing to the development of Indonesia's science, technology and innovation (STI) capacity and improvement of the related systems and conditions.
- Participants:

[Table 1-5] List of participants in the invitational program in Korea

	Name	Position
1	Dr. Boediasoeti Ontowirjo	Deputy Chairman for Research and Innovation Policy
2	Dr. Yan Rianto	Acting Deputy Chairman for Research and Innovation Infrastructure
3	Dr. Dudi Hidayat	Director for Policy Formulation of Research, Technology, and Innovation
4	Dr. Ratih Damayanti S.Hut	Acting Director of Scientific Collection Management
5	Dr. Mohammad Subekti	Acting Director of Nuclear Facility Management
6	Dr. Tjahjo Pranoto	Senior Engineer, Directorate of Laboratory Management, Research Facilities, and Science and Technology Park (DLMRFSTP)
7	Nurbaiti Marsas Prilitasari	Coordinator/Analyst of Science and Technology Utilization, DLMRFSTP
8	Anugerah Yuka Asmara	Coordinator/Researcher, Research Centre for Public Policy
9	Yustina Nita Sulistami	Coordinator/Policy Analyst, Directorate for Research, Technology and Innovation Policy Evaluation
10	Rahmayanti Pertiwi	Policy Analyst, Directorate for Policy Formulation of Research, Technology, and Innovation

- Main Contents:
 - Sharing of South Korea’s policy experiences through visits to major R&D infrastructure support institutions in Korea.
 - Formulation of action strategies for improving Indonesia’s R&D infrastructure support policies.
 - Establishment of a network of South Korean and Indonesian experts in science and technology innovation.

[Table 1-6] program module

Module	Type	Details
STI Policy in Korea	Seminar 1	South Korea’s STI development - with the focus on the utilization and impact of international cooperation
	Seminar 2	South Korea’s STI development models and policy instruments
R&D Support System and Policy in Korea	Study Visit	Science and Technology Policy Institute (STEPI)
	Study Visit	Korea Basic Science Institute (KBSI) National Research Facilities Equipment Center (NFEC)
	Study Visit	Korea Atomic Energy Research Institute (KAERI)
	Study Visit	Jeonbuk Institute of Automotive Convergence Technology (JIAT)
Action Plan Workshop	Workshop	Group work action plan for the R&D support policy

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[Figure 1-4] Photos of the invitational program in Korea

	
Science and Technology Policy Institute	National Research Facilities Equipment Center
	
Korea Atomic Energy Research Institute	Jeonbuk Institute of Automotive Convergence Technology
	
KAERI Advanced Radiation Technology Institute	Seminar on STI Policy in Korea
	
SK T.um ICT Future Technology	Closing Ceremony

- Performance outcomes: Survey of Satisfaction and Self-efficacy
 - ① Survey Content: Relevance and satisfaction with the training (site visits, workshops, seminars) in relation to the training objectives, and personal assessment of one's capacity enhancement.
 - ② Survey Results: All the respondents, based on the total response criteria, provided positive responses regarding their satisfaction with the training and the personal efficacy surveys.
- Overall Results of Open-ended Surveys:
 - ① Survey Content: Overall assessment of the invitational training and suggestions for improvement.
 - ② Survey Results: The participants increased their understanding of Indonesia's situation through South Korea's experience. They would like more opportunities for discussions and problem-solving, as well as more insights into Korea's STI policy, including its talent management systems.

[Table 1-7] Results of the personal efficacy survey

Category	No.	Question	Strongly Agree	Agree	Neutral	Disagree	Strongly Disagree
Satisfaction Survey	1	Overall satisfaction	70%	30%	-	-	-
	2	[Site Visit] Were the contents of the workshop program appropriate to the project objective?	80%	20%	-	-	-
	3-1	[Seminar 1] The contents of the seminars were appropriate to the project objective.	60%	40%	-	-	-
	3-2	[Seminar 2] Were the contents of the seminars appropriate to the project objective?	60%	40%	-	-	-
	4	Were the action plan materials used in the workshop helpful for developing a strategic plan?	60%	40%	-	-	-
Personal efficacy survey	5-1	I am more fully aware of the characteristics (*Policies, Governance, Stakeholders) of the National R&D support (Infrastructure & Funding) system than I was before the workshop.	60%	40%	-	-	-
	5-2	I am more fully aware of the current status of the National R&D support (Infrastructure & Funding) system in Indonesia than I was before the workshop.	90%	10%	-	-	-
	5-3	I have a better understanding of the policy and procedures necessary for developing the National R&D support (Infrastructure & Funding) system compared to before the workshop.	60%	40%	-	-	-

5.3 Joint Research Workshop

- Date: Tuesday, 19 September 2023 / Jakarta, Indonesia (BRIN)
- Purpose: To share the results of the analysis of the 2023 policy research topics and discuss next year's policy advisory direction.
- Participants: All members of the Korea-Indonesia International Collaborative Research Team.
- Key Highlights:
 - ① Discussion of the current status of R&D infrastructure support in Indonesia and other related issues.
 - The action plan developed during the invitational training held in Korea in May 2023 provided a useful framework for assessing the current situation and identifying issues. However, in-depth discussions and advisory sessions with the Korean research team are still needed to formulate detailed solutions.
 - The absence of effective procurement policies has led to issues of redundancy in infrastructure support, while the lack of specialized personnel for research facility management has posed challenges for equipment maintenance. Furthermore, insufficient research facility equipment has hindered access for many researchers.
 - Sharing among BRIN researchers of the results of the survey on R&D infrastructure services.
 - Limitations of the existing research management system (ELSA): Lack of information and diversity in research fields.
 - Poor access to research facility equipment due its long distance from actual research sites.
 - Challenges stemming from frequent turnover and lack of expertise among research facility equipment management personnel.
 - Insufficient manuals and guidelines for using research facility equipment.
 - South Korea periodically formulates mid- to long-term plans for research facilities and equipment every five years. The third plan began this year with the focus on

establishing an ecosystem in which research institutions can manage and take responsibility for facilities autonomously.

- Under the first plan, the goal was to establish investment and management systems for research facility equipment, resulting in the formulation of four major strategies and fourteen detailed tasks. Guidelines for managing research facility equipment based on legislation were developed, along with the ZEUS platform, equipment registration, on-site visits, and surveys via online platforms.
- The second plan aimed to create a foundation of shared use of facilities and equipment, resulting in the formulation of three major strategies and twelve detailed tasks. It supported the consolidation of research equipment for shared use within universities and established a system for collecting and using research facility equipment fees.
- For national-level supervision and management of research facility equipment, it is important for Indonesia to conduct a step-by-step assessment of the current status of research facility equipment establishment, analyse the problems, and progressively develop practical solutions.
- To that end, it is recommended that Indonesia:
 - a) Conduct a fact-finding study on the status of establishment and operation
 - b) Solicit the concerns of researchers on-site through various methods (surveys, public hearings, etc.);
 - c) List possible solutions by matching them with the issues raised by administrators;
 - d) Propose a three- to five-year medium-term plan and an annual execution plan;
 - e) Explain the relationship between Korea's science and technology management system and R&D funding, the history of R&D investment and management systems in Korea, research and development governance systems, and Korea's R&D funding system and processes; and
 - f) Present recommendations for improving Indonesia's R&D funding system in the future.

- Future prediction and data analysis are required in R&D project planning. This is because predicting future technologies and analysing data on research and development activities are necessary prior to formulating science and technology policies or R&D project plans. Future technology prediction involves predicting which new technologies will emerge in the future, while data analysis on research and development activities provides information about our current status and position.
- Flexible incorporation of technology trends is essential when allocating R&D budgets. When allocating R&D budgets, close alignment with national science and technology policies is required, and consideration of changing technology trends in various fields is essential. In Korea, R&D budget allocation incorporates the National Science and Technology Basic Plan (prepared by the Ministry of Science and ICT every five years, providing the basic direction for national science and technology policy for the next five years), the National Fiscal Management Plan (prepared annually by the Ministry of Economy and Finance, including projections for the government's R&D budget size for the next five years), and the next-year research and development investment direction and priorities (prepared annually by the Ministry of Science and ICT).
- In Indonesia, private sector R&D investment is relatively low. Although the government cannot fully control private sector R&D activities, it can stimulate private sector R&D investment through tax incentives, among other measures. In Korea, during the 1960s and 1970s, R&D was primarily led by the government. However, in order to promote research and development activities by private enterprises, laws such as the Technology Development Promotion Act were enacted to strengthen the tax benefits for research and development. Similarly, in the 1980s there was a significant increase in research activities by private companies.

[Figure 1-5] Photos of the joint research program



5.4 Dissemination Forum

- Date/Venue: Wednesday, 20 September 2023 / Indonesia Cibinong Science Centre (INARI, Research and Innovation Expo)
- Purpose: To organise an Impact Dissemination Forum for presentation of the results of the 2023 K-Innovation Partnership Program and policy consultation.
- Participants: Researchers from Korea and Indonesia and over 100 BRIN researchers.

[Table 1-8] program of the Dissemination Forum

Time	Contents	Presenters
14:00-14:10	<Opening Remarks>	Dr. Eun Joo Kim (*Project Manager), Principal Researcher, Division of Global Innovation Strategy Research, Science and Technology Policy Institute (STEPI)
14:10-14:25	<Session 1> Large-scale research facilities: How to ensure their effectiveness	Dr. Sungil Park, Principal Researcher, Department of Nuclear Policy and Strategy, Korea Atomic Energy Research Institute (KAERI)
14:25-14:40	<Session 2> Policy tools for South Korea's research infrastructure	Dr. Yongjoo KIM, Senior Researcher, National Research Facilities & Equipment Centre (NFEC), Korea Basic Science Institute (KBSI)
14:40-14:55	<Session 4> Status of the policy for managing and utilizing the research infrastructure in South Korea	Ms. Sujeong Seo, Researcher, National Research Facilities & Equipment Centre (NFEC), Korea Basic Science Institute (KBSI)
14:55-15:10	Break	
15:25-15:40	Action plans for effective management of the R&D infrastructure	Dr. Tjahjo Pranoto, Senior Engineer, Directorate of Laboratory Management, Research Facilities, and Science and Technology Park (DLMRFSTP)
15:40-16:30	<Q&A Session> Q&A and Discussion Session	Chaired by Dr. Deok Soon Yim, Senior Research Fellow, Division of Global Innovation Strategy Research, Science and Technology Policy Institute (STEPI)

- Key Highlights:
 - Scientific and technological development is the cornerstone of a nation's competitiveness, which originates from the outcomes enabled by the research infrastructure. Since 2000, South Korea has concentrated on constructing diverse research infrastructures in a drive to expand its national research capacity. Despite these efforts, however, the outcomes did not match the level of investment, leading to the realization that judicious allocation of resources is crucial.

- Consequently, policies were devised to enhance investment efficiency through problem analysis, and various research infrastructure policy tools were developed accordingly. To facilitate this, diverse information on research infrastructures was collected and disseminated, and distribution channels and utilization systems were established to enhance the capabilities of researchers. As a result, investment efficiency increased, while the investment ratio decreased.
- Indonesia can also benefit from South Korea's experience and develop various policy tools based on this, with the ultimate goal of improving efficiency.
- Technologically advanced countries are striving to stay ahead in the fierce competition over science and technology through large-scale research facilities. However, such facilities require substantial investments, making them inaccessible to everyone and thus emphasizing the importance of international cooperation.
- Developing countries like Indonesia should prioritize investments in universities and talent development through international collaboration before investing in facilities. From the perspective of national policy, the focus should be on prioritizing facilities that serve multiple users.
- South Korea's research infrastructure management and utilization system aims to enhance research productivity by managing introduced research infrastructures efficiently and promoting their joint utilization. This is accomplished through four main policies:
 - a. Key policies in the management stage include management ethics education, policy explanation meetings, and fact-finding surveys. These policies are currently being implemented to establish autonomous management systems at research institutions and to promote an ethical culture.
 - b. Key policies in the utilization stage include community/consortium support, policy forums, and professional training. These policies are being implemented to enhance the specialized operational capacity of research institutions and to improve research and development outcomes.
- Key policies in the utilization and disposal stage include the integrated management of research facility equipment fees, the promotion of shared

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utilization, and support for equipment transfers. These policies are designed to establish a stable foundation for the use of infrastructures and equipment at research institutions and to promote their shared utilization.

- The policies underpinning research infrastructure data/systems - which are essential for their management, utilization, and disposal - include the ZEUS platform and survey analysis. These can provide the computational infrastructure required to support the stable management and use of research infrastructures at research institutions.

[Figure 1-6] Photos of the Dissemination Forum in Indonesia





6. Project Schedule

[Table 1-9] Project Schedule

Activity	2022												2023
	1	2	3	4	5	6	7	8	9	10	11	12	1
Project Planning													
Organizing the Project Team													
Kick-off Seminar													
International Joint Research													
Training Workshop													
Submission of Interim Report													
Dissemination Forum													
Submission of Final Report													
Presentation of Final Report													
Evaluation of Final Report													
Publication													

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CHAPTER 2

Current Status of the R&I Infrastructure Policy in Indonesia

Dr. Tjahjo Pranoto

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Coordinator, Research Centre for Public Policy, BRIN



Chapter 2. Current Status of the R&I Infrastructure Policy in Indonesia

1. Overview of General Issues Regarding R&I Infrastructure Facilities

1.1 Research Infrastructure and Economic Development

Fabiano (2020: 1-15) asserted that research and innovation infrastructure is critical to support the implementation of research activities aimed at improving a country's socio-economic development and the quality of people's lives. This can be assessed from sectoral and essential aspects, such as “ease of access to and availability of research facilities and infrastructures”, “increasing the quality and capacity of human resources”, “collaboration and transfer of knowledge or correlation of knowledge with work demands and global or international insights”. Research infrastructure includes the facilities, equipment and technical support necessary for high-quality research. The results of using research infrastructure as a resource for research institutions are the primary input in the research and innovation ecosystem. Research infrastructure plays an important role in supporting and improving the implementation of research and the creation of innovations and technologies.

Countries or regions that have a robust research infrastructure tend to be more competitive on the global stage (Zendulková, 2022: 47-56). Countries with robust research infrastructure can attract investment in research and development and encourage knowledge-based economic growth. (Širá, 2020, p. 4). The resolution of certain problems in science and technology requires special facilities, thereby encouraging efforts to reach a better understanding of problems in the fields of science and technology and to create solutions that will have a positive impact on society. Institutions that can make significant research

contributions to gain a better reputation at both the national and the international level. Eder (2011: 647-650) stated that when human resources are capable of carrying out the latest research, but where the availability of research infrastructure is limited or inadequate, this will have such consequences as research activities being hampered or research activities having to be postponed, even though research results are needed immediately to answer pressing needs and current challenges.

Infrastructure is the backbone of any nation's development and prosperity. Without well-developed infrastructure, it is impossible to deliver basic services such as education, healthcare, transportation, and energy to citizens. In Indonesia, an archipelago with a population of 270,000,000 people spread across 17,000 islands, infrastructure development is of paramount importance. As such, governments must develop and implement policies that ensure the proper planning and implementation of infrastructure projects while also addressing such issues as quality assurance, safety standards, cost control, and environmental protection. By focusing on these areas of policymaking for infrastructure projects, governments can create a better future for their citizens. Therefore, it is important to strengthen infrastructure in order to support economic development and basic services.

Good infrastructure not only improves the quality of life for citizens but also attracts foreign investors and promotes economic growth. To achieve this goal, Indonesia needs to formulate a clear infrastructure development policy that considers such factors as cost-effectiveness, sustainability, and environmental impact. The development policy must be strategically planned and implemented in order for the country to reap its benefits in the long run. Due to rapid technological advances, infrastructure can now be developed to enable ever more complex and expansive research projects. From dealing with massive datasets to exploring artificial intelligence algorithms, the right infrastructure is essential for today's researchers.

Developing and maintaining a good infrastructure is essential for any research. Without an efficient and reliable infrastructure, research is likely to suffer. From the internet to laboratories, universities need an effective infrastructure to conduct their studies. As

technology advances, new equipment is being developed that can help researchers in their work, especially in playing a role to support and improve the implementation of research and the pursuit of technological innovations.

The existence of a strategic research infrastructure with adequate and sophisticated equipment will make it easier for researchers in the country to conduct research and create innovations and technologies. However, if Indonesia's human resources can carry out the latest research, but the availability of equipment is limited or inadequate, it will have unfortunate consequences, such as hampering or delaying research activities, and research results will be needed immediately to answer the latest needs and challenges. A country's infrastructure can significantly influence its ability to conduct research. A decent infrastructure will enable researchers to access data and resources more easily, thereby boosting their productivity and efficiency. On the other hand, inadequate infrastructure can impede their progress and lead to an inefficient implementation of research activities. Therefore, it is important for nations to prioritize investments in infrastructure development if they hope to progress in the field of research activities.

2. Indonesia's Vision 2020-2024

The Indonesian National Medium Term Development Plan (RPJMN) 2020-2024 sets forth the four main pillars of national development as follows:

- ① Human development and mastery of science and technology.
- ② Sustainable economic development.
- ③ Equitable development.
- ④ National security and good governance.

These four pillars will serve as the foundation for realizing Indonesia's vision of becoming one of the world's largest economic powers by 2045 ("Indonesia Maju 2045"). This shift in the focus of national development is not accidental. In the period 2030-2040, Indonesia is predicted to receive a 'demographic bonus', namely when the number of productive age population (15-64 years) is greater than the number of people under 15 years or over 64 years. This productive age population group is predicted to reach 64% of the total national population, which is projected to reach 297 million people (Bappenas, 2017). However, in order for Indonesia to gain the maximum benefit from this, the increase in human resources (HR) must be balanced with an increase in quality and innovation capacity.

As a key step in increasing Indonesia's national research and development capacity, the government issued Act No. 11/2019 on the National System of Science and Technology (National Science and Technology Law). This 2019 Act was enforced to replace Act No. 18/2002 in order to address a number of deficiencies that prevented the 2002 law from serving as an optimal framework for the contribution of science and technology to national development. By issuing Act No.11/2019, the Government of Indonesia has demonstrated greater attention and commitment to supporting developments in the field of research and innovation at the next stage of Indonesia's development.

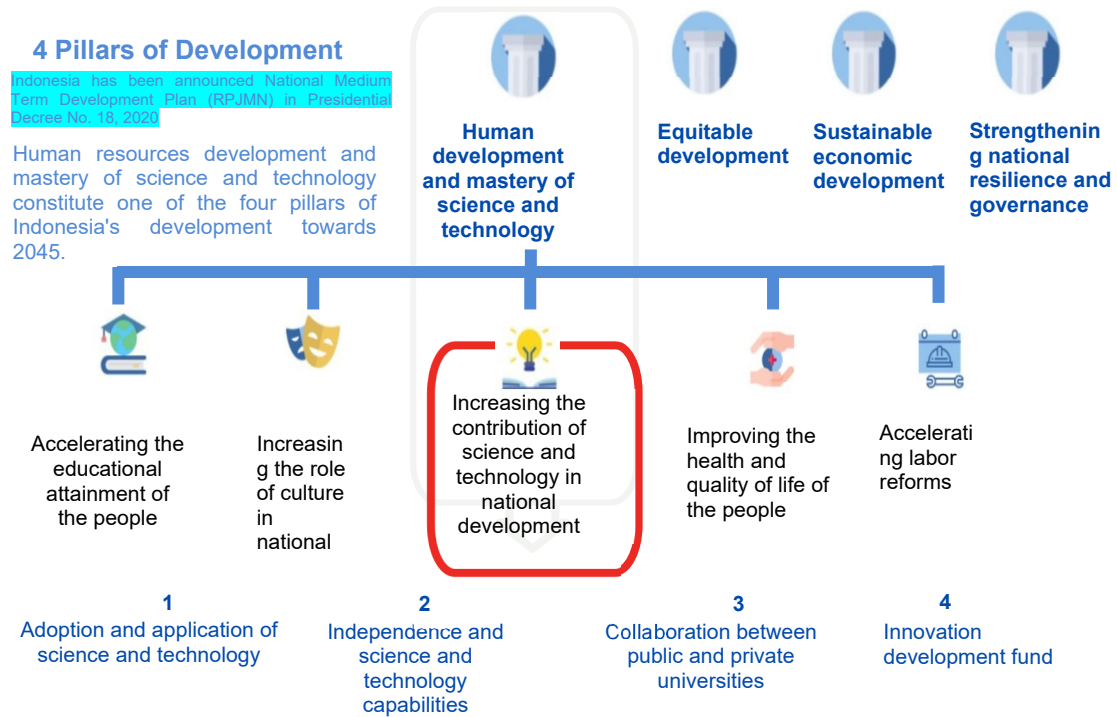
Indonesia's development vision for the five-year period 2020-2024 is presented in the National "Long-term Development Plan (RPJPN) 2005-2025" as the final stage, i.e. that of "realizing an independent, advanced, fair and prosperous society through accelerated development in various fields, the economic structure has been developed based on a competitive advantage in various regions supported by high-quality, competitive human resources."

In the National Medium-Term Development Plan Document, Indonesia's 2020-2024 vision is the realization of an advanced Indonesia that is "Sovereign, Independent, and Nation character Based on the *"Gotong Royong"* (solid and help each other), which will be achieved through the 2nd *Nawacita* Mission, which includes improving the quality of the Indonesian people, productive, independent and competitive.

Indonesia is triggered to be more productive, competitive, and flexible country in facing these changes. Therefore, Indonesia is preparing major stages under the President's direction. The development of human resources and mastery of science and technology, one of the four pillars of development (Figure 2-1), is the key to Indonesia's future. Industrial cooperation and hardworking human resources who are dynamic, productive, and skilled, and are masters of science and technology. Indonesia will continue to improve the quality of education and also build a talent management institution. The government will identify, facilitate, and provide educational and self-development support for Indonesian talents, while highly talented members of the Indonesian diaspora must be supported in order to make a significant contribution to accelerating Indonesia's development. Indonesia will also set up a specific institution to manage these talents and manage great talents who can help the country to compete globally.

Act No. 11/2019 on the Science and Technology System has been stipulated in the development of science and technology, which states that the process of organizing R & D should be carried out through educational activities and the research, development, assessment and application of science and technology.

[Figure 2-1] Pillars of Indonesia's Development in 2045



Source: Law of BRIN No. 6 Year 2023, p.31.

3. BRIN's Research and Innovation Infrastructures

Following up on Act No. 11/2019 above, the Indonesian Government, through Presidential Regulation No. 78/2021, has established and made BRIN as a governmental research institution in Indonesia. BRIN is a government institution under, and responsible to, the President (Article 2). BRIN has integrated roles of organising government tasks in the fields of research, development, assessment, and application, as well as invention and innovation, nuclear administration, and national space administration (Article 3). This Presidential Regulation also regulates the integration of research institutions under the governmental institution through (Article 65):

- ① The duties, functions, and authorities of work units that carry out research, development, and application of science and technology within ministries and institutions henceforth shall be transferred to BRIN.
- ② The transfer of duties, functions, and authorities provided in the foregoing paragraph (1) shall be followed by the transfer of the employees, facilities, financing, and assets involved in the execution of these duties and functions, and henceforth shall become the employees, equipment, financing, and assets of BRIN.

The integration of research and development institutions across ministries and institutions has greatly expanded the authority of the Deputy of Research and Innovation Infrastructure (DIRI). Through an integration process lasting approximately one year, the current distribution of research infrastructure throughout the country (laboratories, botanical gardens, field stations, earth stations, research vessels, science-technology park (STP)) is presented in Table 2-1.

[Table 2-1] Number of R&D Facilities

	Facilities	Number
1	Laboratories	41
2	Botanical Gardens	5
3	Field Stations	25
4	Observation Stations	7
5	Research Vessels	5
6	Science and Technology Parks	4

Source: BRIN internal source.

As the main structure, this organisational structure includes three roles, i.e. policy function, funding function, and R&D function. Research infrastructure management is included in the funding function; whose role is to provide researchers with research facilities. The Deputy assigned with this task (DIRI- *Deputy for Research and Innovation Infrastructure*) has the main tasks and functions as stated in the Head of BRIN Decree 1/2021:

■ Article 84

The Deputy for Research and Innovation Infrastructure (DIRI) shall be responsible for organising the formulation and implementation of policies in the field of research and innovation infrastructure.

■ Article 85

As regards the duties referred to in Article 84, the Deputy for Research and Innovation Infrastructure (DIRI) shall perform the following functions:

- ① Formulation and implementation of policies in the field of research and innovation infrastructure.
- ② Integration, coordination, synchronisation of programs, budgets, and work plans for policy implementation in the field of research and innovation infrastructure management.

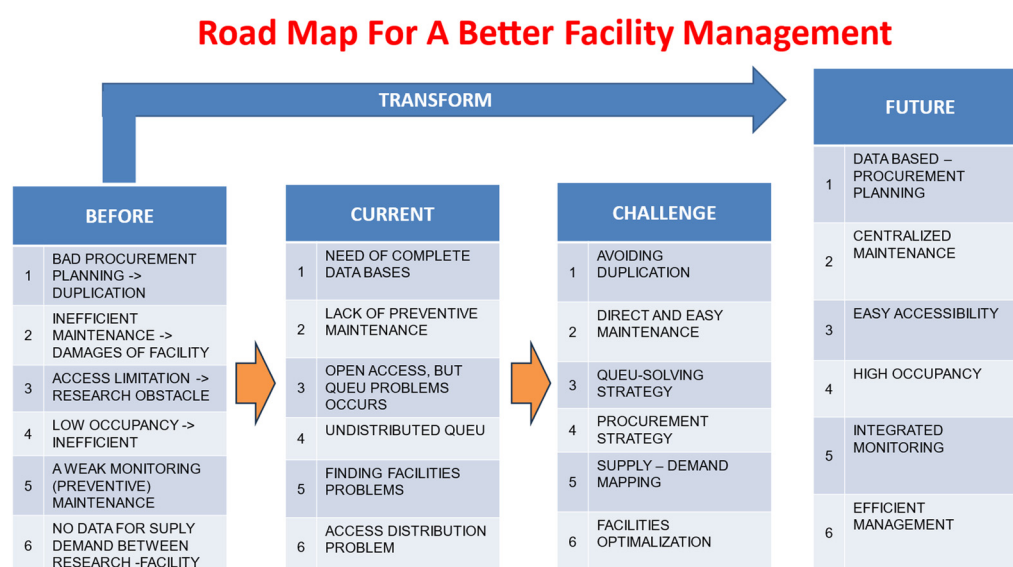
- ③ Implementation of policies in the field of research and innovation infrastructure management, which includes science and technology parks, botanical gardens and scientific collections, research vessels, research laboratories and facilities, nuclear facilities, and space facilities.
- ④ Optimisation of the research and innovation infrastructure.
- ⑤ Provision of technical guidance and supervision of the management and optimisation of the research and innovation infrastructure.
- ⑥ Monitoring, evaluation, and reporting in the field of management and optimisation of the research and innovation infrastructure.
- ⑦ Performance of other functions as assigned by the Head.

4. Policy Issues related to Research and Innovation (R&I) Infrastructure

The Deputy for Research and Innovation Infrastructure (DIRI)'s tasks and duties are delivered through the targets imposed on DIRI (the following seven targets). The mapping of current problems must be portrayed correctly and thoroughly, and it is crucial to formulate a strategy for achieving the targets. With more than 10,000 researchers currently employed at BRIN, an appropriate governance policy is needed to achieve the set target (*“transforming research management to Increase critical mass”*).

Figure 2-2 illustrates current conditions and future (expected) conditions and shows the root of the problem and its impacts.

[Figure 2-2] Road Map for Better Facility Management



Source: Author's elaboration.

During the first year of its establishment, BRIN did not have a complete database of all centrally managed equipment, which resulted in the procurement of the same equipment

even with different specifications (accuracy, speed, measurement range, quality, technology, other parameters), while the existing equipment could still meet the researchers' objectives. The proposal procedure is done openly, and there must be sufficient evidence that the proposed equipment is available in the database system, so that duplication can be detected more quickly.

The equipment procurement review process should be conducted openly and rigorously by researchers according to their field classification, with reference to the procurement priority list. As researchers have different levels of knowledge and understanding of the latest technological developments, this open review process simultaneously results in a natural knowledge exchange among researchers.

There should be database system in supporting the integration of research equipment management, which includes the specifications, functions, maintenance history data, calibration data, and distribution locations that should be connected to the state property management data. With the presence of this database system, efficient and effective research infrastructure management can be achieved

The current research infrastructure management process is still improving despite several obstacles, including a lack of personnel for managing the research equipment infrastructure, given that some personnel who have mastered the technical equipment have moved to work units that focus on research centres instead of infrastructure management services, which means that the formation of laboratory management structures is not yet perfect.

Given BRIN's goal of focusing on research activities, preventive maintenance and calibration are important points in ensuring that the data generated by research infrastructure facilities are traceable, accurate and precise. Sudden or fatal damage to equipment can occur if a defective component is detected too late, hampering R&I infrastructure service for users originated from internal and external BRIN. This is certainly a concern for leaders, especially policy makers when implementing maintenance service standards from all sides (different quality, price, delivery time, availability of spare parts, and service standards).

Improvement of the management mechanism of research infrastructure equipment must also be carried out for all equipment whether it is in prime condition, needs repairing, or is idle or damaged. The centralized integration of certain equipment is expected to increase budget efficiency, especially concerning equipment that require certain space specifications or environmental conditions that meet the prevailing standards, so as to minimize potential damages. Certain items of equipment are categorized as characterization equipment, process equipment, or preparation equipment. Advanced equipment such as characterization equipment for implementation can only be carried out by tool operators, possibly accompanied by researchers, but additional expenses must be paid if there are several operators.

The establishment of a maintenance team unit (responsible for equipment centrally) is also necessary to continue improving the competencies required to fully understand the application of maintenance service standards. Induction training is required before anyone can engage in activities using research infrastructure facilities, while rewards and punishments can also be applied to all classes of users of research infrastructure facilities, this to establish a culture of effective cooperation between research infrastructure managers and facility users.

Prior to the incorporation of BRIN, the use of infrastructure facilities that used to be compartmentalized within each agency and dominated by certain research groups led to an exaggerated sense of ownership among certain researchers. This was also due to a lack of openness regarding information about each existing research facility, even within the same institution, causing information to be hidden. Even researchers appointed as equipment supervisors tended to overuse equipment for their research projects while preventing other researchers from using the equipment promptly. In contrast, the current condition is open, making it easier to gain access to equipment, while all the internal researchers at BRIN, as well as external service users, can obtain information on the distribution of equipment.

E-Layanan Sains (ELSA), is a website-based system that has been created and developed by BRIN to support the utilization of service-based research infrastructure facilities that are accountable and use the “first in, first out” system. ELSA continues to develop and improve along with the laboratory services at BRIN, which have different characteristics. ELSA has been recognized for its excellence, receiving an award in the “innovation” category from the Indonesian Ministry of Civil Servant Empowerment and Bureaucratic Reform (KEMENPANRB) in 2022.

Strategies that could be carried out at this time to improve the performance of laboratory services include:

- ① Create integration-based laboratory service standards to support the improvement of services related to the use of research infrastructure, such as making laboratory service standards for SEM, DSC, XRD, FTIR, XRF, AAS, GC MS, PCR, Tensile Test, and ICP.
- ② Increase the competency of human resources responsible for implementing, supervising, and managing laboratories (both technical and non-technical).
- ③ Provide training for supporting the management of human resources, such as the ELSA user flow process for the public service team under the Bureau of Public and General Communication and Secretariat - BRIN, currently the first step service of ELSA BRIN which establishes direct communication with service users related to services in BRIN including laboratory services contained in DIRI.
- ④ Conduct assessments in the form of external survey activities for several laboratories accredited by the National Accreditation Committee in Indonesia to maintain laboratory quality assurance and implement internal audit activities in all BRIN’s laboratories managed by the Directorate of Laboratory Management, Research Facilities, and Science and Technology Park (DPLFRKST) - BRIN laboratories, both accredited and unaccredited.
- ⑤ Provide information related to the person in charge of the laboratory, contacts, and

equipment in order to facilitate researchers' access to the laboratory.

In addition to research infrastructure, the implementation of integration-based laboratory services also continues to be carried out in line with the direction of the laboratory integration policy implemented by the leader. However, in the process of achieving or realizing excellent services, there are still several obstacles in the field, including the reduced number of human resources (HR) engaged in service activities, such as technical implementation (for example operators, supervisors) and service management (for example customer service and personnel in charge of managing laboratories) due to work unit displacement.

The surge in services was quite pronounced at several laboratories in the Directorate of Laboratory Management, Research Facilities, and Science and Technology Park (DPLFRKST)-BRIN in 2021, i.e. above 800 services for the last quarter. This increased significantly, especially in 2022, to above 20,000 service applications per year, with incoming applications amounting to more than 18,000 services by the end of 2023. Service applications will continue to increase as laboratories are better managed.

[Table 2-2] Challenges to Providing R&D Facilities

No	Before	Explanation
1	Inefficient procurement strategy and redundant processes	Overlapping procurement activities among agencies result from insufficient coordination of procurement planning. Consequently, procurement redundancies and disparities in procurement technology between agencies arise.
2	Inadequate maintenance practices leading to facility deterioration	Inconsistent maintenance services characterized by varying quality, cost, turnaround time, and service criteria. This results in suboptimal budget allocation and substandard maintenance outcomes.
3	Restricted accessibility, impeding research progress	• Segmented and incomplete information between agencies - even between research centers within the same agency. • Facilities and equipment are dominated by a small number of research groups, creating an unwarranted sense of ownership among these researchers. Impacts: Other researchers (with similar competencies) are unaware of the existence of relevant equipment for their research. Some who are aware are unable to utilize it due to non-standardized rules and regulations; gaining access is problematic.
4	Underutilization and inefficient use	The equipment is underutilized directly due to the issues outlined in point #3. Researchers' lack of knowledge or inability to access equipment hinders efficient utilization. Consequently, research activities are impeded by insufficient equipment accessibility.
5	Inadequate monitoring and absence of preventive maintenance	The absence of equipment usage hour tracking elevates the risk of unexpected equipment failures. Consequently, the execution of preventive maintenance is challenging.
6	Insufficient supply and demand data communication across research centres	Inadequate monitoring prevents the collection of supply and demand information related to procurement and researcher requirements, making it difficult to achieve optimal equipment usage based on supply and demand dynamics.

Source: Author's analysis

What is the best strategy for the Deputy for Research Infrastructure and Innovation to address the current state of research infrastructure in Indonesia? Looking at the root causes of the problem (Figures 2-2 & Table 2-2), the challenges faced by the Deputy for Research and Innovation Infrastructure are as follows:

- ① Budget inefficiency in procurement, operations, and maintenance.
- ② Poor efficiency of facility utilisation.
- ③ Researchers' limited access to facilities.
- ④ Challenge: insufficient personnel and equipment reliability.

Challenge no. 4 seems to be the most important one. However, this point is a complementary instrument for executing the policy concepts that led to the previous points. To manage tens of thousands of equipment and services, the use of digital/internet technology as a management instrument is necessary. Nevertheless, the policy is the key to this matter, i.e. what kind of policy should be adopted to manage research facilities in terms of the following:

- ① Prioritization of facility procurement.
- ② Facility maintenance strategy.
- ③ Servicing of facilities.
- ④ Optimisation of the occupancy/usage rates of facilities.
- ⑤ Maintenance of all services following laboratory licenses/standards.
- ⑥ Equipment operator management (training etc.).
- ⑦ Others.

5. Executing Agencies in the Research and Innovation (R&I) Infrastructure

The directorates of the Deputy for Research and Innovation Infrastructure, as policy implementers of the Deputy for Research and Innovation Infrastructure, are directly responsible for managing the research infrastructure. The four Directorates include the Scientific Collection Management; Research Ship Fleet Management; Laboratory, Science and Technology Park Management; and Nuclear Power Facility Management Directorates. Institutionally, the duties and functions of these four directorates are outlined in the Decree of the Head of BRIN, Number 1, 2021 for Organisation and Work Procedures, as described below.

5.1 Directorate of Scientific Collection Management:

The Directorate of Scientific Collection Management has the task of overseeing the formulation and implementation of policies and providing technical guidance and supervision in the field of scientific collection management.

■ Article 91

In carrying out the duties provided under Article 90, the Directorate of Scientific Collection Management shall have the following functions:

- a. Preparation of policy formulation and implementation in the field of scientific collection management.
- b. Management of collections of plants.
- c. Management of collections of zoological specimens.
- d. Management of collections of botanical specimens.
- e. Management of collections of microorganisms.

- f. Management of biodiversity collections.
- g. Management of seed banks.
- h. Management of historical collections.
- i. Management of other scientific collections.
- j. Provision of technical guidance and supervision in the field of scientific collection management.
- k. Monitoring, evaluation, and reporting related to the management of scientific collections.
- l. Implementation of other functions assigned by the Deputy for Research and Innovation Infrastructure.

5.2 Directorate of Research Vessel Management

The Directorate of Research Vessel Management has the task of overseeing the formulation and implementation of policies and providing technical guidance and supervision in the field of research vessel fleet management.

■ Article 94

In carrying out the duties provided under Article 93, the Directorate of Research Vessel Management shall have the following functions:

- a. Preparation of policy formulation and implementation.
- b. Management of the fleet of research vessels.
- c. Management of shipping.
- d. Management of expedition program.
- e. Management of measuring instruments and research vessel equipment.
- f. Provision of technical guidance and supervision in the management of research vessels.

- g. Monitoring, evaluation, and reporting in the management of research vessels.
- h. Performance of other functions assigned by the Deputy for Research and Innovation Infrastructure.

5.3 Directorate of Laboratory Management, Research Facilities, and Science and Technology Park

The Directorate of Laboratory Management, Research Facilities, and Science and Technology Park has the task of overseeing the formulation and implementation of policies and providing technical guidance and supervision in the areas of laboratory management, research facilities, spatial organisation, and science and technology.

■ Article 97

In carrying out the duties provided under Article 96, the Directorate of Laboratory Management, Research Facilities, and Science and Technology Areas shall have the following functions:

- a. Formulation and implementation of policies in the areas of laboratory management, research facilities, spatial administration, and science and technology.
- b. Quality control, standards, and accreditation.
- c. Management of laboratory services, research facilities, and space administration facilities.
- d. Operational management and maintenance of laboratories, research facilities, and space administration facilities.
- e. Operational management, maintenance, and facility of science and technology areas.
- f. Provision of technical guidance and supervision in the areas of laboratory management, research facilities, space administration facilities, and science and technology.
- g. Monitoring, evaluation, and reporting in the areas of laboratory management,

research facilities, space administration facilities, and science and technology.

- h. Performance of other functions assigned by the Deputy for Research and Innovation Infrastructure.

5.4 Directorate of Nuclear Facility Management

The Directorate of Nuclear Facility Management has the task of overseeing the formulation and implementation of policies and providing technical guidance and supervision in the field of nuclear facility management.

■ Article 100

In carrying out the duties provided under Article 99, the Directorate of Nuclear Facility Management shall have the following functions:

- a. Policy formulation and implementation in the field of nuclear facility management.
- b. Quality control, standards implementation, and accreditation.
- c. Implementation of nuclear implementation facility services.
- d. Operational management and maintenance of nuclear implementation facilities.
- e. Implementation of safety, security, and protection of installation of nuclear facilities.

6. Evaluation of the R&I Infrastructure Policy and Its Issues

Kurniawan (2017: 2) stated that the development and management of research infrastructure in Indonesia faces several obstacles. Even now, obstacles related to a lack of funding are often encountered. The availability of sufficient and sustainable funding support is essential to building and maintaining a modern and effective research infrastructure. Sometimes, an inadequate budget limits the ability of research institutes etc. to expand their facilities and update their research equipment. In addition, research infrastructure often experiences limitations in terms of the latest technology and equipment, which can hinder researchers' ability to conduct high-quality, innovative research. (Juhlin, 2013, p. 9)

The BRIN policy evaluation study team has conducted a survey of 400 researchers regarding BRIN's research facilities. The survey, which comprised an online questionnaire and interviews with five research coordinators, aimed to map problems in the management of research infrastructure. Based on these problems, appropriate solutions could be derived and analysed as a means for policymakers to determine appropriate policies.

[Table 2-3] Mapping of issues facing BRIN's research infrastructure

No.	Component	Sub-component	Questionnaire Statement of Respondents' Perspective (400 respondents)	Resume of interview (5 respondents)
1	Organisation	Appropriate Structure	1. The scope of the funding service conforms to the nature of the research of the research centre / research group (RC/RG). 2. The scope of the infrastructure service via the ELSA platform conforms to the nature of the research of the RC/RG 3. The scope of the infrastructure service outside ELSA conforms to the nature of the research of the RC/RG. 4. Generally, researchers have found it easier to gain access to	Comments on the service unit in terms of appropriate structure: • Research infrastructure services do not fully conform to the RC/ RG characteristics. (The scope of ELSA is yet to cover all kinds of analysis/testing services.) • Infrastructure management does not yet meet researchers' requirements, as facilities are located in distant places. (Facilities or labs are in distant locations, putting risk on the sample; labs have been relocated to locations

No.	Component	Sub-component	Questionnaire Statement of Respondents' Perspective (400 respondents)	Resume of interview (5 respondents)
2			the research infrastructure since the introduction of centralised management at BRIN.	far from their previous ones [i.e. before BRIN].)
		Role Clarity	5. Researchers understand the roles of the (regional) coordinator, director and deputy of the funding and infrastructure service unit. 6. Researchers are informed about the long-term research funding and infrastructure management and development plan. 7. There is no overlap in the functions and implementation of the funding and infrastructure service units, either in high or middle-level management.	Comments on the service unit in terms of clarity of roles: • Changing of Lab coordinators, disrupting the flow of knowledge transfer. Researchers are not informed properly regarding the long-term plan for infrastructure management. (I.e. what are the facility/lab classifications; what are the strategies; what are the priorities for lab management?)
3		Key Individuals	8. Services and bureaucracy in funding and infrastructure services are quick and easy. 9. Service units can solve researchers' complaints and problems effectively. 10. The (regional) coordinators and unit representatives in each region actively communicate about updated business processes in funding and infrastructure services.	Comments on the service unit in terms of key individuals: • Persons in charge of research facilities and labs are yet to have sufficient knowledge/competence or numbers. (Technical knowledge about labs, professionalism, technicians or lab coordinators are overwhelmed by the number of service demands.) • Lack of clarity on authority of persons in charge of certain functions. (Lab supervisors tend to play a rather passive role, and lab coordinators tend to wait for instructions.) • Weak role of coordinators and middle-level management in communicating high-level policy (Bottleneck in services, passive role of coordinators in communicating any updates or fresh information with users/researchers.)

No.	Component	Sub-component	Questionnaire Statement of Respondents' Perspective (400 respondents)	Resume of interview (5 respondents)
4	Process Management	Knowledge Sharing	11. Researchers are able to obtain information about the status and progress of proposed funding and infrastructure services. 12. Researchers receive appropriate and clear information during the socialization of funding and infrastructure services. 13. Researchers receive enough information and assistance regarding funding proposals. 14. Researchers can access the results of evaluations of funding and infrastructure services.	Comments on the service unit in terms of knowledge sharing: • Lack of coordination among service units. (Coordination with the RC about lab management, coordination with another directorate to provide services.) • Lack of information on the status of services and procurement progress. (incomplete information, long chain of information) • Not all researchers receive socialization of infrastructure services. (Lack of range of information for researchers in several areas outside Java, require periodic socialisation.)
5		External Focus	15. Researchers understand clearly the standard procedure for funding and infrastructure services. 16. Researchers can communicate with and provide suggestions to service units in various ways. 17. Researchers receive feedback from the service unit regarding their questions/suggestions within a specified time.	Comments on the service unit in terms of external focus: • Communication with user/ researchers are not going well (researcher finding it hard to communicate with contact person, changing of lab supervisor, slow response from contact person) • Information is not conveyed properly on the ELSA system (user-friendliness, information about facilities not covered in ELSA)
6		Standard Procedure	18. Standard procedures for funding and infrastructure services are well-documented and easily accessed. 19. Information regarding business processes or standard procedures for funding and infrastructure services are easily understood. 20. Researchers understand that the service unit is making	Comments on the service unit in terms of standard procedure: • Lack of supply chain management (There is a long queue for service and material procurement.) • Standard procedures for some processes are not yet available or published.

No.	Component	Sub-component	Questionnaire Statement of Respondents' Perspective (400 respondents)	Resume of interview (5 respondents)
			improvements to the funding and infrastructure services.	
7	Process Management	IT Support	21. All information regarding funding and infrastructure services needed by researchers is available online and is up to date on the service homepage. 22. Every researcher can access necessary information about funding and infrastructure services on its homepage. 23. The information system for accessing funding and infrastructure services is safe, reliable, and easy to use. 24. The service unit has provided a contact line for researchers in various forms (contact person, online chat, e-mail, etc.).	Comments on the service unit in terms of external IT support: • Need to Access to the centralized infrastructure database (Incomplete database of research facilities and labs, facility/lab classification, improved logbook)

Source: Author's analysis

[Table 2-4] Results of Respondents' Perspective

Sub-Component	SD	D	A	SA	SD-D	A-SA
1. Appropriate Structure	9%	47%	41%	3%	56%	44%
2. Clarity of Role	5%	45%	47%	3%	50%	50%
3. Key Individuals	7%	46%	44%	3%	53%	47%
4. Knowledge Sharing	4%	36%	56%	3%	40%	60%
5. External Focus	5%	36%	57%	3%	40%	60%
6. Standard Procedure	4%	34%	58%	4%	38%	62%
7. IT Support	2%	21%	72%	6%	23%	77%

Notes: SD: Strongly Disagree; D: Disagree; A: Agree; SA: Strongly Agree; SD-D: Strongly Disagree-Disagree; A-SA: Agree-Strongly Agree.

Source: Author's analysis

Analysis of Problems with the Research Infrastructure Service:

I. Organisation

■ Appropriate structure

- Infrastructure arrangements that do not meet the needs of researchers, with items of equipment located far apart; equipment and research facilities spread across many BRIN campuses over long distances.
- Infrastructure services do not yet support the various characteristics of research activities.

■ Role clarity

- Researchers need to learn the outline and road map for infrastructure arrangements.
- Researchers need to know the plans regarding the research infrastructure development concepts to be implemented, resulting from an agreement between the infrastructure management and BRIN's leadership, so that the research group team can develop a research scheme that aligns with the development plan. Researchers also need to learn the outline and road map for infrastructural arrangements.

■ Key individuals

- Lab operators, supervisors and managers often change, thus hampering the flow of knowledge.
- The competencies and numbers of operators, supervisors and lab managers are deemed inadequate.
- Insufficient and restricted authority granted to the human resource (HR) personnel assigned specific duties. Insufficient human resources leading to the absence of customer service and liaison functions.

The issues outlined above emerged due to the need for human resources with new tasks and functions. The tasks of operating equipment, changing equipment specifications, and

maintenance that researchers previously carried out are now done by a new type of human resource, notably operators, supervisors, and lab managers. One identified issue concerns the frequent changes of operators or supervisors of research equipment. Service users feel that a knowledge gap appears between operators, supervisors, and lab managers whenever changes occur, potentially hampering the smooth use of research equipment and further impeding research activities. The results of confirmation with the Deputy for Innovation Research Infrastructure (DIRI) indicate that there are limitations in the DIRI's ability to recruit competent operators, supervisors, and technical managers. This is related to the second issue pointed out in this sub-component, which concerns the numbers and competences of operators, supervisors, and technical managers. The current urgent need is met by recruiting researchers from research organisations who have knowledge and experience of the relevant equipment. However, this step is a short-term policy designed to address urgent problems and ensure that research equipment services can still be provided.

II. Process Management

■ Knowledge Sharing

- Inadequate information sharing and poor coordination between units in service delivery.
- There is a lack of universal comprehension of the new research infrastructure utilization procedures among researchers. Lack of transparency in non-tax revenue management within internal service managers.

The policy issue regarding knowledge sharing arose from the desire to orchestrate research and innovation services related to aspects of organisational governance in various BRIN work units. At the level of director, significant policy information related to BRIN is obtained from the “Weekly meeting”. Lack of coordination among managers has caused incomplete information and policy. Assuming that a Board of Executives’ (BOE) forum and policies are produced, it is hoped that there will be information related to an organisation that is comprehensive and integrated, and which can be understood by the team in the technique level.

■ External focus

- Miscommunication within management
- Unclear flow of information
- Ineffective socialisation

Policy issues related to external focus arise because researchers - as users - feel that they do not receive precise information about the availability of equipment or how to access it, the location of the lab needed for testing, or who the researcher must coordinate with to ensure the use of the equipment according to its intended purpose. This is because the research team felt they had to discuss preparations for analysis with the lab manager first of all, not to mention any changes to the PIC yet to be notified to the researchers. Then, during service implementation, problems may arise with the verifier's response regarding clarification of sample preparation.

Another problem arises when a research activity consists of a series of procedures that must be carried out in different regional locations, and where moving and transporting the samples requires considerable effort and could damage them.

■ Standard Procedure

- The standard procedures for implementing the services are inadequate or have not been socialized.
- Standard procedures for repairing and preventing equipment damage and service supply chain management have not been prepared.

The policy issue related to standard procedures is as follows: Although researchers generally can access all research facilities, many items of research equipments cannot be used because they cause errors, need to be correctly calibrated, are damaged, have no solvent, or have no columns, etc.

■ IT Support

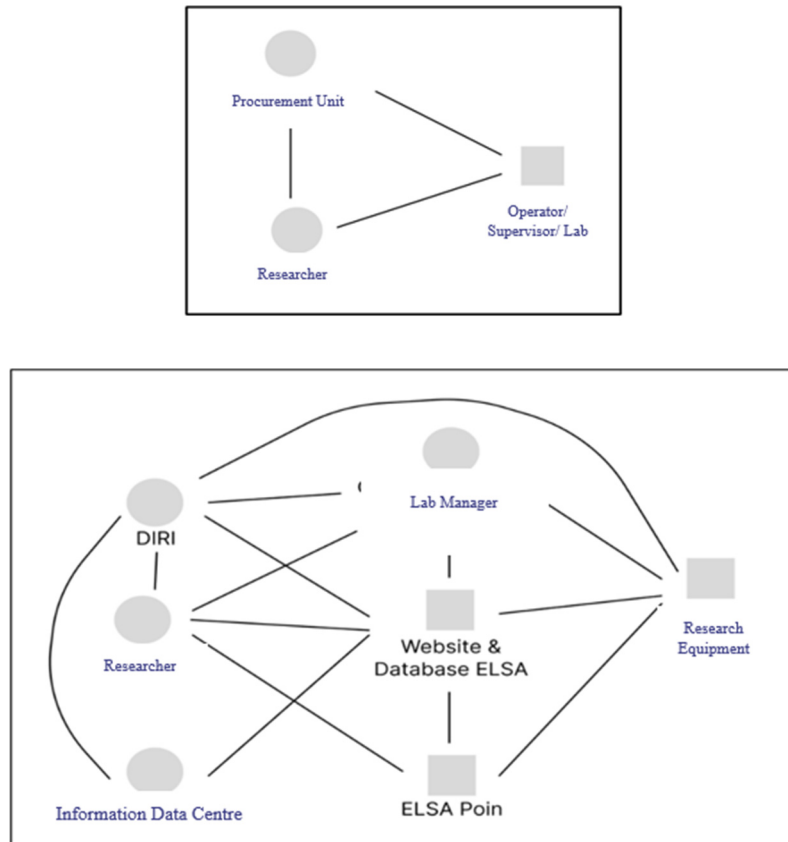
- Access to centralized information and unreliable databases

This policy issue arose because researchers felt that the tool database was incomplete as a source of information, as a data matrix/tool classification, or as a logbook. The critical information needed concerns the availability of equipment (i.e. available but in use, unavailable, etc.), capability, and condition (damaged, etc.). Apart from that, information regarding new applications for the purchase of new infrastructure facilities is also needed.

In contrast to a layman's understanding that centralized management of research and innovation services will make business processes more manageable, for the centralized management of research and innovation services at BRIN to run effectively, the service process requires more actors and more equipment with more complex interactions, regardless of which ecosystem is thought to have a better impact.

The policy issues that arise in this case concern the impacts of creating a new ecosystem and a new paradigm for centralized infrastructure services. Where there are the perspectives of both human and non-human actors, such as information systems (databases and websites), new human resources such as laboratory operators or laboratory managers who did not previously exist, or new regulations and procedures for the use of research infrastructure such as ELSA points, various types of adjustments will have to be made by both the service providers and the service recipients.

[Figure 2-3] Diagram showing the differences between the old and new ecosystems of research and innovation services



Source: Author's Analysis

With new actors emerging in the centralized research infrastructure services ecosystem, new associations must be well established so that the services can run effectively. Associations are formed through intense interactions and must be facilitated by supportive structures and regulations. Therefore, it is necessary to provide policy interventions to support the transition from the new ecosystem and to facilitate interactions and coordination between the actors in the ecosystem.

The relationship between the respondents' perceptions about centralized research and innovation infrastructure services and policy umbrellas

[Table 2-5] Recommendation for continuous improvement of BRIN's research infrastructure

Policy Umbrella	Sub-component	SD	D	A	SA	SD-D	A-SA
Preparation of regulations and policies that support new ecosystems and research and innovation services	Appropriate/suitable structure	9%	47%	41%	3%	56%	44%
	Clarity of roles	5%	45%	47%	3%	50%	50%
	Personnel liaison (balance between centralization and decentralization)	7%	46%	44%	3%	53%	47%
Strengthening the interaction between management units and the service management system's users and service managers	Knowledge sharing	4%	36%	56%	3%	40%	60%
	External focus	5%	36%	57%	3%	40%	60%
Facilitating easy interactions between users of research facilities, and research and innovation	Standard procedures	4%	34%	58%	4%	38%	62%
	Information technology support	2%	21%	72%	6%	23%	77%

Notes: SD: Strongly Disagree; D: Disagree; A: Agree; SA: Strongly Agree; SD-D: Strongly Disagree-Disagree; A-SA: Agree-Strongly Agree.

Source: Author's analysis

The table explains that appropriate/suitable structure of organisation (44%), role clarity of responsibility (50%), and personnel liaison (47%) (i.e. the balance between centralization and decentralization) are the main issues facing BRIN, whereas knowledge sharing, external focus, standard procedures, and information technology support are valued at more than 50%. For new institutions, appropriate structure of the organisation, role clarity of responsibility, and key individual are the critical issues to which BRIN must respond and resolve promptly in order to create an established and robust organisational structure and management system.

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CHAPTER 3

Current Status of the R&I Funding Policy in Indonesia

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Chapter 3. Current Status of the R&I Funding Policy in Indonesia

1. Overview of the Indonesian Government's R&D Budget

Research and development are the backbone of a nation's progress. All developed nations have effective research and development programs and have proven that their progress is supported by the development of science and technology. There are several instances of industrialised countries that have made substantial advances in research, technology, and innovation, for example: Switzerland, the United States, South Korea, Germany, and Singapore (Aryani, 2021). To support economic growth, governmental methods that stimulate innovation, such as the allocation of state funding for science and technology, research and innovation, must be prioritised, while policymakers should emphasise the establishment of an S&T-based policy ecosystem. Furthermore, the success of innovation is dependent upon government leadership of technological development as well as state support in the form of policies and budgets.

Indonesia's ranking, based on the 2023 Global Innovation Index study, compared to Indonesia's degree of innovation with Singapore and Malaysia, is shown in the table below.

[Table 3-1] Respective Rankings and Scores of Indonesia, Malaysia and Singapore in the Global Innovation Index, 2023

Pillars of the Global Innovation Index	Singapore 		Indonesia 		Malaysia 	
	Overall Ranking: 5 th		Overall Ranking: 61 st		Overall Ranking: 36 th	
	Rank	Score	Rank	Score	Rank	Score
Institutions	1	98.4	70	48.7	29	68.7
Human capital and research	2	63.2	85	25.8	32	44.3
Infrastructure	8	63.1	69	39.2	51	46.5
Market sophistication	6	67.4	37	45.0	18	53.2
Business sophistication	3	69.4	77	25.6	36	38.8
Knowledge and technology outputs	10	55.3	61	23.7	37	32.2
Creative outputs	18	46.0	68	23.8	47	30.7

Source: Author's elaboration from the Global Innovation Index Database, WIPO, 2023, p. 58.

According to the 2023 Global Innovation Index Study, Indonesia ranked reasonably highly at 61st in the category of “innovation” out of 132 nations. However, Indonesia's ranking is still low when compared to that of Malaysia (36th) and Singapore (5th). Several factors, including institutional support, human capital and research, and business sophistication, must be improved by the Indonesian government.

When it comes to Asian rankings, Indonesia ranked 12th in South East Asia, East Asia, and Oceania, behind Singapore (1st), Malaysia (8th), Vietnam (10th), Thailand (9th), and the Philippines (11th), and just ahead of Brunei Darussalam (14th) (WIPO, 2023, p. 19).

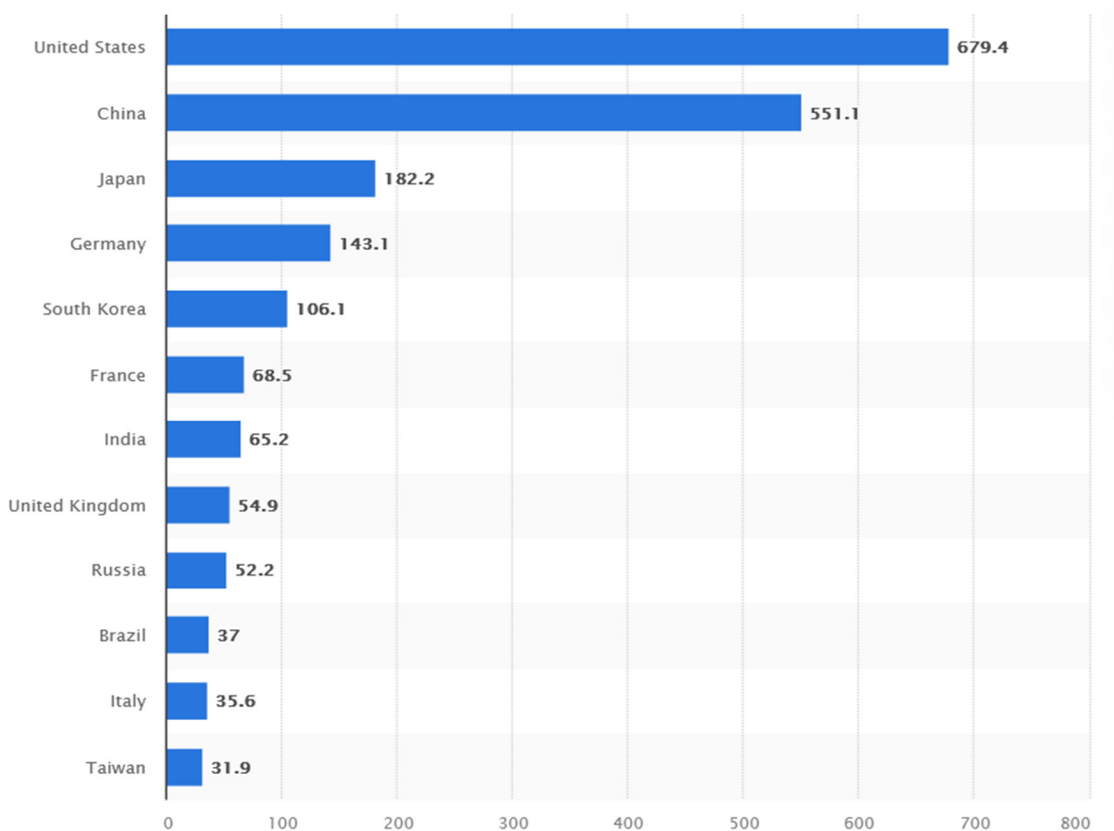
To enhance and raise competitiveness and the quality of economic growth based on science and technology (science and technology), the OECD and the World Bank recommend that countries invest around 2% of GDP in research and development. In fact, Indonesia's science and technology and R&D funds are currently still in the range of 0.3% to 0.1% of GDP, lagging far behind neighbouring countries such as Singapore, Malaysia and Thailand. Limited R&D funds will make it difficult for Indonesia to develop quality economic growth and become a

developed country with a GDP above USD 12,000. This means that Indonesia remains caught in the “middle income country trap.” (DIPI, 2022)

Many countries appear devoted to improving their country by establishing strong data and research bases, as evidenced by the substantial amount of funding they invest in research and development. The United States ranks top, with a gross annual research expenditure of US\$679.4 billion. This sum represented a rise from the previous year's amount of US\$643.8 billion in 2021. In 2022, US research expenditure was expected to account for 3.07% of GDP.

The table below shows the leading countries' research and development (R&D) expenditure in 2022 (unit: US\$ billion).

[Figure 3-1] Twelve Countries with the Highest Research and Development Budget/Gross Expenditure (2022)



Source: <https://www.statista.com/statistics/732247/worldwide-research-and-development-gross-expenditure-top-countries/>

China ranks second overall with a gross R&D expenditure of \$551 billion. This sum also represents an increase over the previous year's budget of US\$507 billion (2021). Japan comes in third place, with a gross expenditure of US\$182.2 billion in 2022. This sum is also higher than the previous year's gross budget of US\$176.6 billion. Germany and South Korea rank fourth and fifth, with a gross research expenditure of US\$143.1 billion and US\$106.1 billion, respectively, in 2022. The figures for the other seven countries are shown in the graph above.

Indonesia ranks 34th out of the forty nations covered by *R&D World*. In 2022, its budget stood at US\$8.2 billion. According to R&D World, Indonesia had the lowest research budget-to-GDP ratio at just 0.24% in 2022.

Through various measurements, the Indonesian government has proved its commitment to the advancement of research and innovation. The National Medium-Term Development Plan (RPJMN) is one example of this policy, as it includes research and innovation priorities as part of the national development plan. Furthermore, the government has also strengthened the National Research and Innovation System in order to realise the Vision of Advanced Indonesia 2045. This includes the creation of innovation funds, which will be a foundation of Indonesia's growth in 2045.

In addition, under Presidential Regulation No. 78 of 2021, the government established BRIN (National Research and Innovation Agency) to coordinate research and innovation activities in Indonesia. With the establishment of BRIN, research and development funds that were formerly distributed between research institutions and ministry development have been united under a single BRIN facilitating scheme. Research funding is of course one of the essential components for improving innovation and economic growth.

2. Entities Involved in the Government's R&D Budget Allocation

In Indonesia, the government's allocation of the research and development (R&D) budget involves a number of organisations and agencies, with Bappenas (National Development Planning Agency) playing a major role in its planning. As a national development planning body, Bappenas is in charge of developing and executing national development policies, especially those linked with R&D and innovation. To achieve its goal of becoming a developed country by 2045, the Indonesian government has enhanced the national research and innovation system. Through inter-government collaboration, academia, and industry, Bappenas and other stakeholders are working to improve the triple helix of research, technology, and innovation. As they are also involved in creating national development plans that include research and development as a core component, Bappenas exercises significant influence over how the government allocates its R&D funding. To increase innovation, raise competitiveness, and promote sustainable development in Indonesia, they also make sure that the funds allotted for research and development are used efficiently.

According to Law No. 17 on State Finance (2003), the role of planning institutions such as the National Development Planning Agency (Bappenas) cannot fully synchronise the planning of programs and activities with budget planning. The planning and budgeting carried out by Bappenas are limited to indicative budget allocations, while the definitive amounts are determined by the Ministry of Finance. Subsequently, the implementation of budget planning is conducted by the Ministry of Finance.

The Ministry of Finance also plays a role in financing research and development and is responsible for awarding money to various ministries and R&D institutes, which also participate in the national budget planning process.

The national expenditure of Indonesia (APBN) is drafted annually by the Ministry of Finance, after which the research funding component is carried out by Ministries and institutions

granted the authority and functions required to manage research funds. Each ministry and institution are required to propose a plan for its budget for the following year. According to the Directorate General of Budget (DJA), Ministry of Finance, the small amount of the budget available for research is also the result of budget proposals submitted by various ministries and institutions. The DJA of the Ministry of Finance does not itself have the right to propose the budget amount; rather, it only reviews and approves the proposals.

The main organisation in charge of managing and distributing Indonesia's funds for research and development (R&D) is the National Research and Innovation Agency (BRIN), which plays a crucial role in advancing the nation's R&D industry. BRIN manages national research grants, including those from the government budget, and can grant funds to research institutes, universities, and academics to help them with their research. BRIN can also fund research and innovation development initiatives that have the potential to boost Indonesia's competitiveness and innovation.

BRIN monitors and evaluates the implementation of government-funded R&D initiatives. It guarantees that the given budget is used as planned and that the desired outcomes are obtained. BRIN also fosters research and development collaboration with foreign partners, and can provide financial assistance for international collaboration initiatives involving Indonesian scholars.

Apart from the abovementioned roles, BRIN promotes innovation and the transfer of technologies derived from the results of research to the industrial sector. In this respect, it helps to enhance collaboration between research institutions, businesses, and government.

The function of BRIN with regard to R&D budget distribution is critical to fostering the growth of research and innovation in Indonesia. BRIN collaborates closely with numerous stakeholders in directing R&D funds to initiatives that benefit national development, increase competitiveness, and help to resolve the country's social and economic challenges.

In addition to BRIN, the Ministry of Education, Culture, Research, and Technology (Kemendikbudristek) plays a significant role in the development of research and innovation in Indonesia. The Ministry has the following tasks (Kemendikbudristek, 2023):

- ① Encourage innovation and scientific advancement through research and development initiatives executed by educational and research institutes under the auspices of the Ministry of Education and Culture.
- ② Establish the budget for research and development using funds from the State Revenue and Expenditure Budget (APBN).
- ③ Increase the national ecosystem of research and innovation by promoting partnerships between state-owned businesses (BUMN) and institutions.
- ④ Focus on developing research in such areas as the green economy, digitisation, and human resource development in the future.
- ⑤ Strengthen collaboration with the Indonesian diaspora in order to use global networks to boost innovation.
- ⑥ Create an innovative culture via collaborative efforts with all Indonesians.

3. Allocation and Adjustment System of the Indonesian Government's R&D Budget

3.1 Allocation and Adjustment System of the Indonesian Government's R&D Budget: Before BRIN

There are various sources of R&D funds in Indonesia, including the State Revenue and Expenditure Budget (APBN), Regional Revenue and Expenditure Budget (APBD), endowments, business entities, and other sources - both valid and non-binding. The majority of Indonesia's R&D expenditure budget comes from sources of government funds. According to the Ministry of Research, Technology, and Higher Education (2017), only 43.74% of the total GERD of Rp30.78 trillion (0.25% of PDB) in 2016 was dedicated to research activities, while the rest was allocated for non-research purposes. (STEPI & LIPI, 2020)

In Indonesia, the management of R&D funding operates through two mechanisms: institutional funding and project funding. The institutional funding mechanism adheres to the national planning regulations outlined in Law No. 25 of 2004, which pertains to the National Development Planning System. The budgeting process follows the guidelines stipulated in Law No. 17 of 2003, which relates to State Financing. All public R&D institutions receive a portion of the state budget (APBN) allocated specifically for their R&D activities, based on the programs they have planned in advance. The allocation of funding is documented in the Entry List of Budget Implementation (DIPA).

The second R&D funding mechanism is project funding, where funds are assigned to a group or an individual to conduct a specific R&D activity with clearly defined parameters of scope, budget, and timeframe. This type of funding is typically granted upon submission of a project proposal outlining the intended research activities (Lepori et al., 2007). The funds are often administered through intermediary institutions responsible for managing R&D funds and generally entail a certain level of operational autonomy (Steen, 2012).

Project funding is obtained from three primary sources. First, it originates from the Entry

List of Budget Implementation (DIPA) of the Ministry of Research and Technology (MoRT), which offers various program schemes such as the National System of Research Incentives (Insinas) and Industrial Technology Development (PPTI). Second, there is the research and development fund mechanism administered by LPDP under the Ministry of Finance, which operates within the RISPRO program scheme. Lastly, there is the Indonesian Science Fund (Dana Ilmu Pengetahuan Indonesia/DIPI), a research funding institution managed by an independent, non-structural organisation known as AIPI (Akademi Ilmu Pengetahuan Indonesia/Indonesian Academy of Sciences).

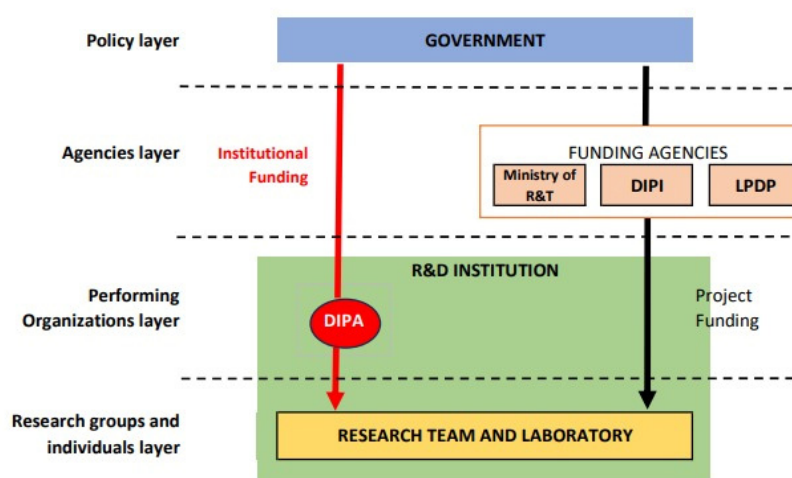
The MoRT allocates a portion of its budget from DIPA to fund research programs conducted by research institutions. This funding is distributed through various competitive schemes or by establishing research consortia. The distribution mechanism is typically based on competition, with proposals evaluated for their feasibility. Each scheme has specific terms, conditions, and policies outlined in the relevant ministerial regulations. For instance, the Insinas scheme focuses on R&D activities between technology readiness levels (TRL) 1-5, whereas the PPTI scheme supports R&D between TRL 4-6, and the program of Innovation assists R&D projects that have already reached the prototype stage and are moving towards commercialisation.

Another source of funding is DIPI, which operates as an autonomous research funding institution under the AIPI to enhance the national research ecosystem by providing competitive, flexible, and sustainable research funding. The funding schemes are designed to enhance the quality of cutting-edge fundamental research, thereby strengthening Indonesia's competitiveness globally. DIPI was officially launched on March 30, 2016 by the Minister of Finance in accordance with Presidential Regulation No. 9 of 2016, which amended the AIPI statutes. The establishment of DIPI is supported by the Ministry of Research, Technology and Higher Education, the National Development Planning Agency (Bappenas), and the Ministry of Finance through LPDP, as well as a number of countries including the United Kingdom, the United States, and Australia.

Another source of R&D project funding is administered by the Educational Fund Institute of

Management (LPDP). LPDP operates as a department within the Ministry of Finance and is responsible for managing the educational endowment fund, which was established in 2010 and disbursed through educational scholarship programs. The R&D funding support is delivered through a competitive mechanism known as the national research project (RISPRO). LPDP oversees the overall management, regulations, and operational structure of this program.

[Figure 3-2] R&D funding mechanism in Indonesia (before BRIN)



Source: STEPI (2020), p. 80.

3.2 Allocation and Adjustment System of the Indonesian Government's R&D Budget: After BRIN

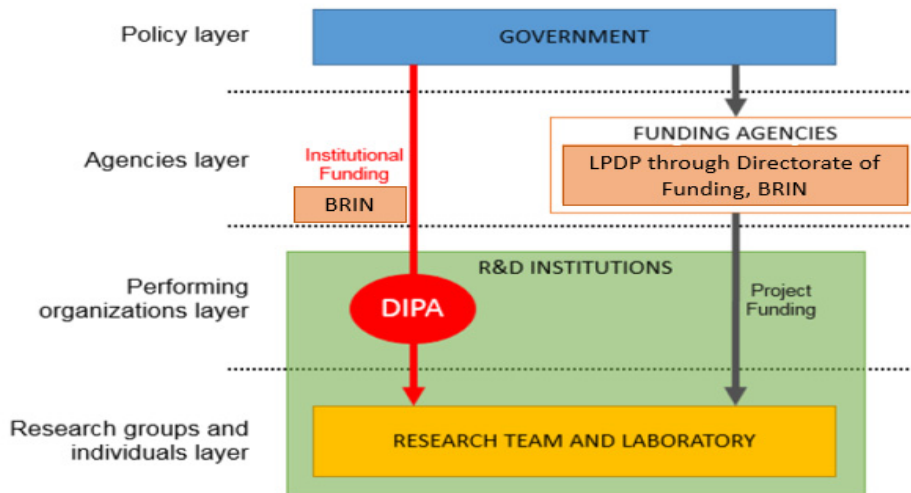
After BRIN was formed in 2021, five research institutions - including the Ministry of Research and Technology, LIPI (Indonesian Institute of Sciences), BBPT (Agency for the Assessment and Application of Technology), BATAN (National Nuclear Energy Agency), and LAPAN (National Institute of Aeronautics and Space) - were merged into one entity called the "National Research and Innovation Agency" (BRIN). Additionally, other research units within the ministries were also merged into BRIN. Research funding still operates through two mechanisms: institutional funding and project funding. However, under institutional funding, research funding is centralised and allocated to BRIN instead of other ministries.

This is done in accordance with Law No. 11 of 2019, which stipulates that research activities shall be centralised under BRIN. The establishment of the National Research and Innovation Agency aims to solve the problems associated with the inefficient use of research funds in Indonesia (Dsulfikar, 2019). The second mechanism, project funding, involves endowments for research funding provided by LPDP (Endowment Fund for Education).

BRIN has a strategic financing program to fund research and innovation activities, that the funding scheme is competitive and available to all stakeholders, including academics at research institutes, universities, and businesses. This competitive scheme stated in BRIN's funding website (<https://pendanaan-risnov.brin.go.id>) and Decree of The Head of National Research And Innovation Agency Number 196/1/HK/2022 Concerning Guidelines for Research and Innovation for Onward Indonesia. Endowment money and the APBN are two sources of support.

Institutional funding is allocated to each research organisation within BRIN, while project funding from LPDP is managed by the Directorate of Funding at BRIN. Currently, both institutional funding and project funding operate through a competitive mechanism. Researchers who seek research funding for their studies must submit a research proposal. Institutional funding requires the involvement of BRIN researchers, while project funding managed by the Directorate of Funding is open and can be awarded to researchers from universities, the private sector, or other R&D institutions.

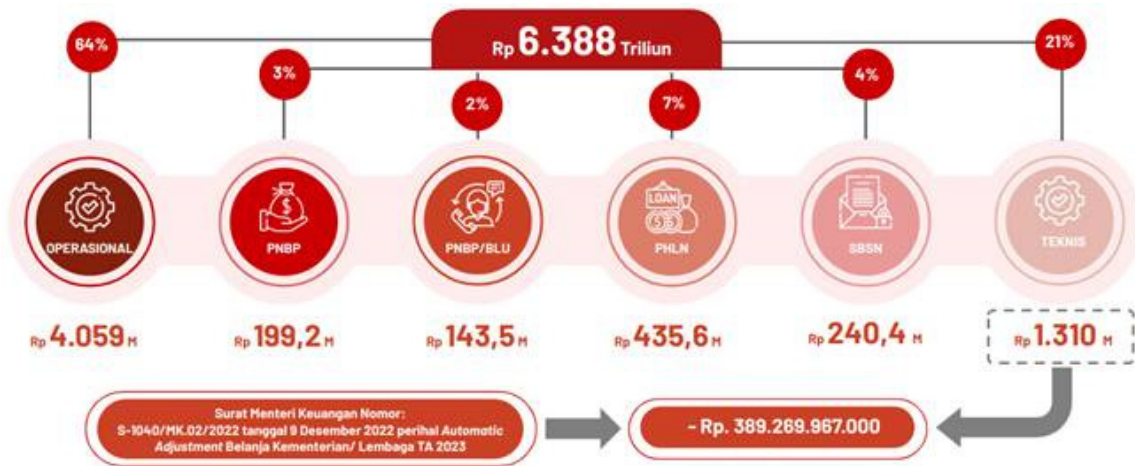
[Figure 3-3] R&D funding mechanism in Indonesia after BRIN



Source: Author's elaboration referring to [Figure 3-5].

The overall national research budget decreased in 2023 compared to 2022. The national research budget amounted to Rp 3.1 trillion or 0.016 percent of GDP in 2022, but it decreased to only Rp 2.2 trillion or 0.010 percent of Gross Domestic Product (GDP) in 2023. The budget allocation for BRIN in the fiscal year 2023 is illustrated with a total of Rp 6.388 trillion. This allocation does not differ significantly when compared to the total budget allocation in the fiscal year 2021 for the five entities that were fully integrated into BRIN, namely the Ministry of Research and Technology (Kemenristek), BATAN, BPPT, LAPAN, and LIPI.

[Figure 3-4] BRIN's Research Budget in 2023



Source: BRIN internal data, 2023.

Note: PNBP-Penerimaan Negara Bukan Pajak (non-tax state revenues), BLU-Badan Layanan Umum (Public Service Agency), PHLN-Pinjaman dan Hibah Luar Negeri (foreign loans and grants), SBSN-Surat Berharga Syariah Negara (State Sharia Securities), Teknis (technical), Surat Menteri Keuangan Nomor S-1040/MK.02/2022 tanggal 9 Desember 2022 perihal Automatic Adjustment Belanja Kementerian/ Lembaga (Letter of the Minister of Finance No. S-1040/MK.02/2022, dated December 9, 2022 regarding Automatic Adjustment of Ministry/Agency Expenditures).

From that budget allocation, BRIN needs to allocate 64% for operational expenses, which include employees' salaries and routine needs such as utilities (electricity, telephone, internet, water), outsourcing for cleaning, security, and drivers, fuel, operational vehicles, stationery, subscription expenses (journals, satellite imagery for national needs), as well as maintenance of office facilities.

Due to the integration of the five entities (Ministry of Research and Technology, LIPI, BPPT, BATAN, and LAPAN) and the consolidation of research and development units from seventy-two ministries and agencies (K/L), BRIN's management now has to oversee approximately 15,000 civil servant employees in 52 office locations and 100 other non-office locations. BRIN also needs to support the operation of three research reactors, a fleet of research vessels, a fleet of remote sensing aircraft, and various other research infrastructures.

The structure of BRIN differs markedly from that of other ministries and agencies (K/L) endowed with a similar budget but with a significantly smaller number of civil servant employees and tasks/functions that do not require a massive physical infrastructure. On the

other hand, as a research institution, BRIN has highly technical tasks and functions, and must bear the cost of maintaining and operating various research infrastructures, which can be substantial. Additionally, BRIN is also obliged to facilitate not only BRIN researchers but also researchers throughout Indonesia.

The Non-Tax State Revenue (PNBP) budget for BRIN is derived from revenues generated through external collaborations, and is entirely allocated to support the operation and maintenance of infrastructure exclusively for research purposes. Therefore, BRIN no longer utilises a purely rupiah (RM) budget for the maintenance and operation of its research infrastructure. Rather, the PNBP sourced from the Public Service Agency (BLU) of the Technology Services Centre (Pusyantek) at BRIN is entirely generated from service users and is used to finance the services provided to its partners. Meanwhile, the budget sourced from Foreign Loans/Grants (PHLN) and Shariah Securities of the State (SBSN) is allocated for new productive research infrastructure expenditures, which serve as investments for future assets.

Therefore, out of the total budget allocated to BRIN, the funding source from pure rupiah (RM) for programs is only 21%, or Rp 1.31 trillion. This adjustment is in accordance with the President's directive contained in the letter sent from the Minister of Finance (No. S-1040/MK.02/2022) to all ministries and institutions on December 9, 2022, regarding the implementation of automatic adjustment (de-facto budget reduction in advance) to mitigate the global crisis in 2023. BRIN experienced an automatic adjustment of Rp 389 billion, which could only be taken from the RM allocation of BRIN. Therefore, in real terms, the remaining RM budget for BRIN's programs is Rp 921 billion, which is allocated to support various programs, including research material expenses for BRIN's researchers in the twelve research organisations, researcher mobility, and the development of human resources, as well as infrastructure expenses related to buildings and research instruments aside from the financing acquired through SBSN and PHLN. BRIN allocates a significant portion of its budget to research infrastructure because it is considered that this will become a productive asset in the long term. Research infrastructure components represent the highest cost in almost all research activities.

4. Future Strategy for Funding Research and Innovation

4.1 Future Strategy for Funding Research and Innovation

According to the United Nations' Educational, Scientific and Cultural Organisation latest data on Indonesia's research and development budget, the country's gross domestic expenditure on research and development in 2018 was only around 0.23%, showing an actual decrease in its ratio of research and development spending to GDP from the previous two years, i.e. 0.245% in 2016 and 0.238% in 2017. These figures fall depressingly short of the UN Educational, Scientific and Cultural Organisation's recommended science and technology budget ratio of 2%. It is reported that a limited budget for science and technology results in restricted research facilities, insufficient funds for operation and maintenance, and low incentives for researchers. In addition to the low budget for science and technology, one of the research and innovation problems in Indonesia concerns the overlapping of research outcomes leading to wastage of state finances.

The merging of research institutions into BRIN is expected to be able to consolidate the Indonesian research ecosystem and render it more effective and efficient. So far, the issue of low research budgets has been caused by overlapping research conducted across multiple research institutions previously, resulting in the budget being squandered ineffectively. Therefore, the incorporation of all government research institutions into BRIN can prevent the duplication of research and budget wastage. The limited budget resources require BRIN to be smart and selective in determining which activities will be prioritised, and where these activities will eventually have an impact nationally. However, research funding needs to be professionally managed.

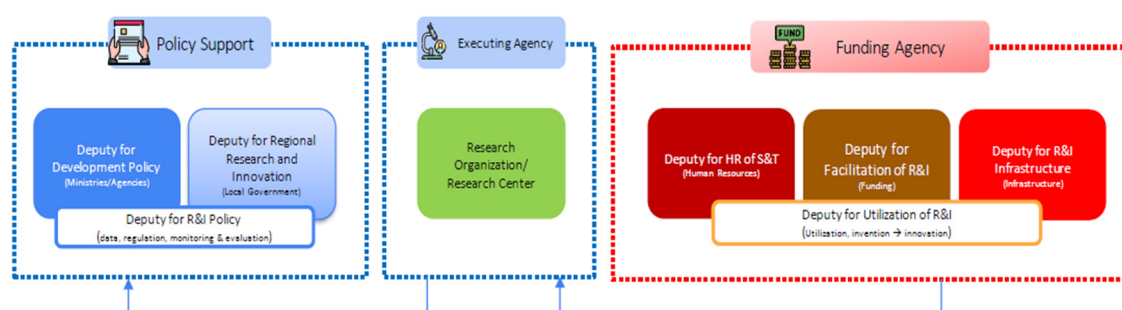
In addition to making budgets more effective, BRIN encourages private sector involvement in the world of research. Strengthening the role of the private sector in research is expected to overcome real-life problems. All the facilities and resources owned by BRIN could be utilised through collaboration mechanisms in an effort to change the composition of

government and private involvement in Indonesia in terms of research funding. BRIN's target is to ensure that 80% of research and innovation budget comes from the private and non-governmental sectors. Collaboration is one of the strategies for dealing with limited constraints, both in terms of limited human resources and limited funding. Through optimal collaboration with external parties, both with investors and strategic industries, it is expected that significant contributions will be made in technology down-streaming efforts. Embracing the private sector is or can be a key driver of research and innovation development, as seen in Switzerland, which ranked number one in the 2021 Global Innovation Index in terms of knowledge and technology outputs.

4.2 Scheme for Funding Research and Innovation

In general, BRIN's main roles and functions are to support evidence/science-based development policies at ministries, agencies, and local governments, and to strengthen the national research and innovation ecosystem as both a funding agency and the executing agency for research and innovation activities. Institutionally, these duties and functions are shown in the figure below.

[Figure 3-5] Classification of BRIN's roles and functions based on its organisational structure



Source: Sari (2023).

Furthermore, the realisation of innovative products is an indicator of the success of the research and development process - from the creation of an invention to its acceptance by stakeholders. Thus, the key to enhancing the roles of universities, and industrial and

business entities is collaboration. Through collaboration, BRIN's research facilities can be utilised together to create optimal research outputs. Research collaboration involving stakeholders, both domestically and internationally, will undoubtedly strengthen the research ecosystem in Indonesia through facilitation mechanisms. Currently, the dominance of governmental involvement in research in Indonesia, rather than the private sector, contrasts markedly with the wider global research landscape. Therefore, BRIN aims to change the contribution of private sector involvement in research in Indonesia become larger. This effort will begin with initiating schemes that offer funding and facilitation openly for private sector.

Efforts to enhance the participation of government, universities, and industrial and business entities in research and innovation entail increasing collaboration in the development and utilisation of scientific knowledge products. Some of the efforts undertaken include improving the innovation products of research-based startup tenants, increasing the number of innovations utilised by the public and industry/business entities, and increasing patents. Innovation is a crucial pillar for competing with other countries worldwide. Currently, many countries around the world rely on innovation within their borders to boost economic growth and other sectors. As a funding agency, BRIN's research funding and facilitation does not only finance research activities for researchers at BRIN, but is openly accessed by diverse parties on a competitive basis, including researchers from other research institutions, universities, and industries. The funding schemes for facilitation are open throughout the year, are carried out through self-management and contractual processes, and involve a selection process for each funding scheme. BRIN's strategy for ensuring that research and innovation facilitation activities are aligned with the broader government agenda is based on the competitive scheme with selection method on research proposal. This activity is one way to synergise the existing programs within BRIN.

Research funding procedures need to be improved with the aim of creating a better research ecosystem. The process of selecting research for funding is carried out online and is open to everyone all year round, while the evaluation of proposals is based on the track record of the researchers. In addition, funding is available only for the direct costs of

research activities, and not for supporting activities. All schemes of funding facilitation are open to the public regardless of the affiliation on a competitive basis.

The following diagram represents the research and innovation facilitation program.

[Figure 3-6] Research and Innovation Funding Facilitation



Source: Author's elaboration referring to <https://pendanaan-risnov.brin.go.id/>

4.2.1 Research Collaboration Centre

The Research Collaboration Centre is expected to become a centre or hub of collaboration in conducting international standard research and innovation in specific fields in a multi-and interdisciplinary manner with a very high standard of results that are relevant to the needs of science and technology users. The program focuses on specific featured themes with specific targets in either science or industry. The Research Collaboration Centre brings together researchers from universities with researchers from BRIN to work together. There are two types of Research Collaboration Centre. First, there is the Research Collaboration Centre of Science and Technology, which involves universities and research units within BRIN. This type is more based on the development of science and technology. The second type is the Industrial Research Collaboration Centre, which is a combination of the first type and involvement with industries and business entities. In addition to pursuing the development of science and technology, it also emphasises the development of product and services resulting from technology. With this type of Research Collaboration Centre and the research units of companies are able to submit proposals to it.

4.2.2 Sailing Days

The purpose of the so-called “Sailing Days” is to provide open and inclusive access to BRIN’s research vessel facilities to researchers, lecturers, and students wishing to acquire data and/or collect specimens. The purpose of activities designed to support these “Sailing Days” is to facilitate multidisciplinary research aimed at producing significant scientific findings in the fields of biodiversity and non-biodiversity at the macro, micro, and molecular levels and promoting their sustainable use. This may include coastal and oceanic research by researchers and partners in the fields of mapping and marine geosciences, oceanography and atmospheric science, marine biology and fisheries.

4.2.3 Health Innovation Product Testing

The facilitation program schemes for health innovation product testing such as (preclinical trials, clinical trials, and medical device tests) are open to BRIN’s researchers and researchers from external agencies, research institutions, universities, and companies that engage in research and development. Research that can be facilitated under the health innovation product testing facilitation program are research that have been scientifically proven from among drugs, phyto-pharmaceuticals, vaccines, and medical devices.

4.2.4 Agriculture Innovation Product Testing

This scheme is a facilitation program for testing agricultural and livestock innovation products and fisheries that is open to BRIN researchers and other agencies from private research institutions, universities, and industries/companies that engage in research and development. As this scheme finances the cost of testing, the budget is not given to the researcher but rather to the test institute; and this process can continue for years on the understanding that each test will not necessarily be successful. With the facilitation fund for agriculture innovation product testing, it is expected to encourage scientific proof, safety, and the benefits of agricultural and livestock products and equipment, and ensure that the results of research lead to product innovation. This is certainly expected to encourage synergism and government collaboration - in this case BRIN, universities, and industry - so

that in the future the results of testing agricultural and livestock innovation products can provide certainty regarding the usefulness of research and innovation results in the agricultural sector as well as in the fields of animal husbandry and fisheries.

4.2.5 Research and Innovation for Onward Indonesia (RIIM) Competition

The RIIM research funding focuses on research and innovation in the fields of food, health, and energy for national self-sufficiency, as well as funding in other fields such as aviation, space, biology, environment, electronics, informatics, manufacturing, nanotechnology, materials, earth, maritime, nuclear power, social and humanities, archaeology, languages, literature, governance, economics, and public welfare. Research and Innovation for Onward Indonesia uses the “National Research Priorities 2020-2024” as a policy reference. The National Research Priority program is directed as an effort to recover the national economy by down-streaming the appropriate results of technology research, reducing national dependence on imported products, and promoting the commercialisation of research results and pioneering technologies and increasing their added value. “National Research Priority Funding 2022-2024” focuses on ten areas of research including food, energy, health, transportation, engineering, defence and security, maritime, social humanities, education, and arts and culture, as well as multidisciplinary and cross-sectoral areas. Research and Innovation for Onward Indonesia can be accessed by researchers in Indonesia from diverse research institutions, universities, business entities, and community organisations that have the capacity and competences required to meet the demand for predetermined research and/or innovation themes. The Research and Innovation Program for Onward Indonesia is aimed at improving the ability of Indonesian human resources or researchers to create new technology through research and innovation activities aimed at producing technological novelties, as evidenced by research outputs in the form of scientific papers and intellectual property rights.

4.2.6 RIIM Expedition

RIIM Expedition funding is funding that can be accessed by all researchers from research

institutions, universities, business entities, and community organisations - both from Indonesia and overseas - that collaborate with research institutions in Indonesia in research activities through the collection of data and/or specimens in the field. The fund recipient is obliged to submit the data obtained from RIIM expedition funding and must store it in the National Scientific Repository in accordance with the laws and regulations.

4.2.7 RIIM Start-Up

The RIIM research-based startup funding program provides financing for prospective startups based on BRIN's research results, with the aim of transforming them into profitable and sustainable startup companies. Under this scheme, BRIN provides an open opportunity for all Indonesian citizens to utilise and commercialise BRIN's research results. Participants who are selected will participate in mentoring for a maximum of six months, and those who pass the mentoring will receive funding of up to IDR 300,000,000 per year with multi-year funding (maximum of 2 years).

This scheme is an effort to give startup companies a more active role in strengthening the country's economy. In this way, the community understands that research and innovation are not only for researchers, but can also be utilised by all Indonesian people. In addition, research and innovation can also add value to micro-economic commodities.

4.2.8 RIIM Joint Fund

The RIIM Joint Fund is a research funding scheme based on collaboration with overseas funding agencies.

4.2.9 RIIM Invitation

This program has been closed.

4.3 Information System for Call for Proposal (CFP)

BRIN provides an information system as a medium for managing research proposals at BRIN. This information system can be accessed at <https://pendanaan-risnov.brin.go.id/>. With this system, researchers can register their proposals according to the required program funding scheme.

To sum up, the funding schemes managed by BRIN, and especially the Directorate of Funding, will evolve continuously so as to better support research activities with specific characteristics or aimed at specific outcomes: For example, research priorities in the fields of food, health, environment, and energy, as well as research that can facilitate collaboration among institutions or researchers, both domestically and internationally. In terms of information systems, the current system used in research funding management will also undergo continuous improvement to meet emerging needs during research management. At present, the information system is not operating optimally for research management activities, and still functions as little more than a repository of research data. In the future, it is expected that the system can be integrated into every stage of research management, from the phase of selection to the disbursement of research funds, as well the monitoring and evaluation of research activities.



CHAPTER 4

South Korea's R&D Infrastructure Policy

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Chapter 4. South Korea's R&D Infrastructure Policy

1. The Background to and Rationale for Policy Improvements

Stephan and Levin (1992) argued that researchers cannot do research without equipment, and that research productivity is a causal relationship between research infrastructure, research personnel, and research funding. This means that the performance and availability of research infrastructure to support the research field has a very important impact on the productivity of S&T R&D. In particular, recent R&D has been centred on state-of-the-art research infrastructure, which is supported by the state rather than individuals and organisations. According to John Peter Dickinson (1989), as the practice of modern science has evolved into a collective mega-science type of research, unlike in the past, hundreds of researchers in the research infrastructure tend to work on one project. In addition, while research infrastructure was mainly utilised for developmental research (OECD's research stages: basic, applied, and developmental) in the past, large-scale research infrastructure is now being used for purely theoretical disciplines (e.g. mathematics → multiscale large-scale computers), while social infrastructure that indirectly supports the huge research infrastructure has become necessary in recent years, which has led to the formation of science and industry complexes (e.g. Technopoles).

Many research infrastructures in various countries are currently being operated with the support of the state, and many research infrastructures in Korea also receive state support to fund their construction, operation, and maintenance.

As such, research infrastructure is a key component of science and technology infrastructure,

and its role in creating innovative research results is gradually expanding. In recent years, the discovery of new scientific knowledge and the development of innovative technologies have been achieved through high-quality scientific results derived from the utilisation of state-of-the-art S&T research infrastructures. In particular, new and fundamental approaches are needed to solve challenging problems, while securing an advanced research infrastructure is crucial for the adoption of new methods and differentiated approaches. Therefore, advanced countries are enhancing their competitiveness by securing advanced research infrastructures capable of creating world-class research results.

The term “research infrastructure” traditionally refers to physical devices required for research and development (R&D) and the spaces that support them; however, its meaning is gradually expanding. In the past, scientific and technological research was conducted by small groups of researchers using self-developed equipment. Still, in recent years, it has changed to a collective type of research centred on a large-scale research infrastructure. Therefore, while the research infrastructure of the past consisted solely of research equipment or the facilities needed to operate it, in modern times it has come to be referred to as “research infrastructure”, which encompasses large-scale or integrated research equipment, personnel, budget, and the data generated by the research infrastructure.

▪ What is “research infrastructure”?¹

This term refers to the facilities, resources, and related services used to conduct top-level research in a scientific field, including major scientific equipment or facilities, or databases for disseminating research data.

In particular, “research infrastructure” refers to the facilities, equipment, data, manpower, and systems necessary to conduct scientific and technological research, and it has both a direct and indirect impact on innovation, economic growth, and the solving of social problems, making it a very important factor in establishing and determining the direction of

¹ OECD GSF(2019), “Reference framework for assessing the scientific and socio-economic impact of research infrastructures” p. 6.

R&D investment. Research infrastructure is an essential element in creating the results of S&T research. Scientific and technological research requires innovative research facilities, equipment, and data; and a country can generate scientific and technological breakthroughs by providing this infrastructure quickly. The technologies discovered to operate a large research infrastructure (internet, web, big data analytics, etc.) in turn have led to the creation of new technologies and industries. Research infrastructure also plays an important role in promoting a country's economic growth and social development. Innovative research infrastructures create indirect jobs in a given region, assist the development of a culture of science and technology, and stimulate the interest of future generations in science and technology, thereby promoting the country's economic growth and social development.

Therefore, the adoption of systems of managing and using research infrastructure is expected to lay solid foundations for S&T research and development and create an efficient research and development environment, which will serve as the basis for a more solid growth engine for Indonesia.

2.

Understanding Indonesia's Science and Technology Research Infrastructure Policy

Indonesia's science and technology policy aims to emphasise research, development, innovation, and human resource development in the field of science and technology in order to strengthen the country's development and competitiveness. Indonesia established BRIN (in accordance with Presidential Decree No. 78/2021), whose role is to integrate governmental tasks in the fields of research, development, evaluation, and application, as well as invention, innovation, nuclear energy management, and national space management. In addition, the integration of research functions distributed across five research institutes and seventy-eight government ministries has created management demands for various research infrastructures.

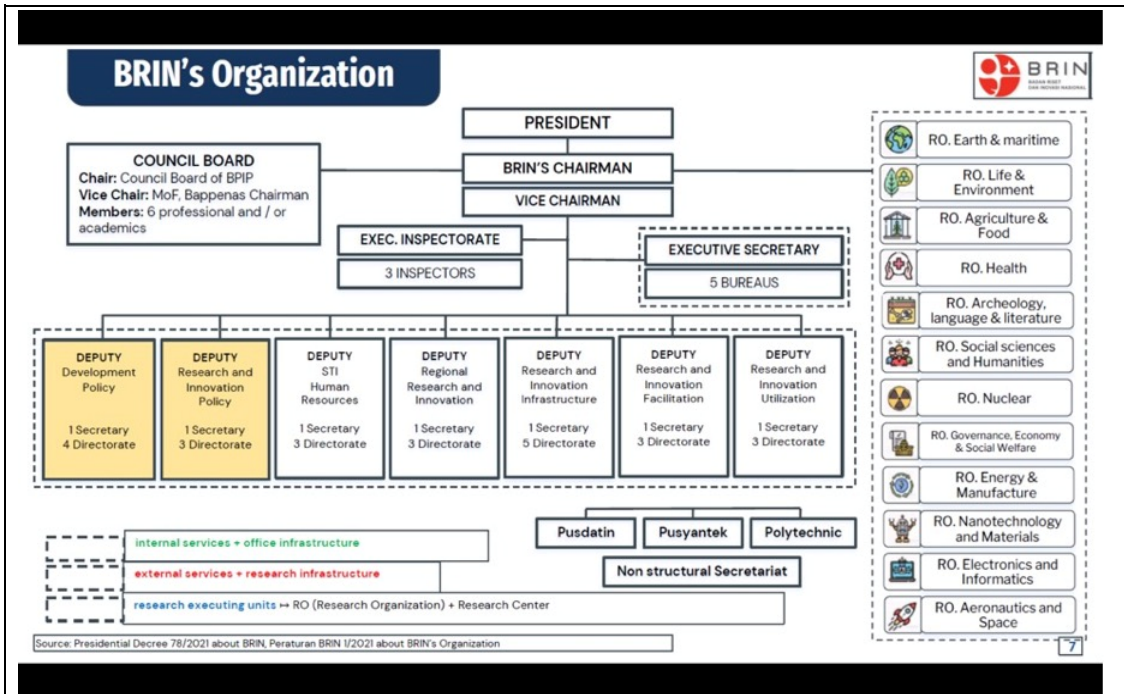
[Table 4-1] Status of BRIN's Research Infrastructure

	Type of facility	Number
1	Laboratories	41
2	Botanical Gardens	5
3	Field Stations	25
4	Observation Stations	7
5	Research Vessels	5
6	Science and Technology Parks	4

Source: BRIN's internal data.

To this end, BRIN systematically formulates and implements comprehensive policies, funding, and research infrastructure management.

[Figure 4-1] Duties of BRIN's Undersecretary for RI



[BRIN Decree 1/2021

Article 84. The Deputy Director General for Research and Innovation Infrastructure shall have the duties of organising the formulation and implementation of policies in the field of research and innovation infrastructure.

Article 85. In carrying out the duties referred to in the foregoing Article 84, the Deputy Director General in charge of research and innovation infrastructure shall perform the following functions:

- Formulate and implement policies in the field of research and innovation infrastructure.
- Integrate, coordinate, and cooperatively implement programs, budgets, and work plans for the implementation of policies in the field of research and innovation infrastructure management.
- Implement policies in the field of research and innovation infrastructure management, including science and technology parks, botanical gardens and scientific collection management facilities, research vessels, research laboratories and facilities, nuclear facilities and space facilities.
- Promote optimisation of the research and innovation infrastructure.
- Provide technical guidance and supervision in the management and optimisation of the research and innovation infrastructure.
- Monitor, evaluate and report on the management and optimisation of the research and innovation infrastructure.
- Perform other functions assigned by the President.

Source: BRIN's internal data.

After integrating the research infrastructure through BRIN and establishing a systematic policy, at this Year, the K-Innovation researchers identified the following problems in Indonesia's research infrastructure policy through interviews, field visits, and consultation activities such as creating a problem logic tree.

- (Failure and obsolescence) Much of Indonesia's research infrastructure is aging, which reduces the efficiency and effectiveness of science and technology research.
- (Insufficient supply) A typical response from interviews with several researchers is that there is insufficient supply relative to demand. This has been singled out as a typical case that hinders the conduct of S&T research.
- (Fragmentation) Indonesia's research infrastructure is concentrated in certain regions, which hinders the development of domestic S&T research.

BRIN used the problem logic tree technique to analyse the causes of this issue as follows:

- Overlapping procurement plans: There is a lack of comprehensive reviews of supply and demand among institutions, resulting in redundant constructions.
- Inefficient maintenance: Damages are caused to facilities and non-standardised maintenance services in terms of quality, price, delivery time, etc.
- Restricted access: Research obstacles and information silos prevent other researchers from fully utilising the facilities even though they are available.
- Underutilisation: Large amounts of equipment are not being utilised, and there is no clear understanding of demand.
- Weak monitoring: There is a lack of effective measures for preventive maintenance or regular inspections, resulting in underutilisation.
- Lack of supply and demand data among research facilities: Supply and demand data need to be understood.

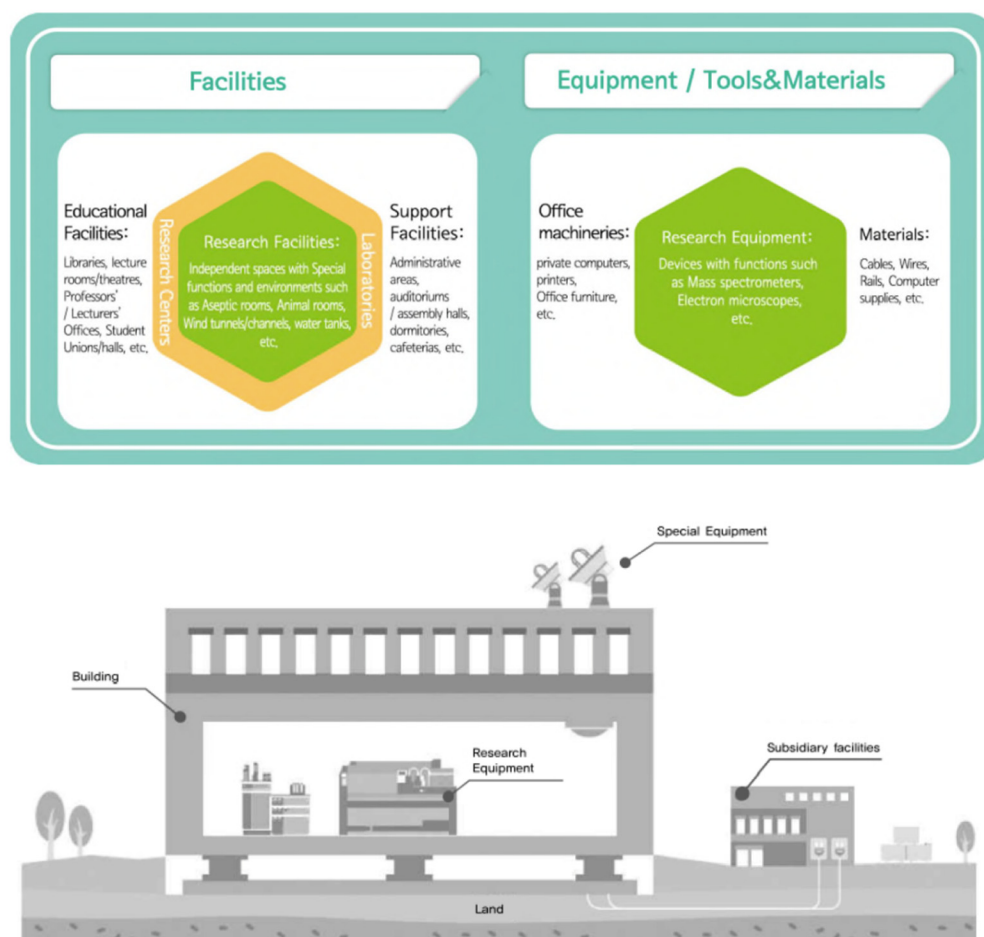
As a solution to these problems, BRIN is preparing the following goals and measures:

- (Goals) 1) Ensure efficient budgeting for facility/equipment procurement, operation, and maintenance; 2) improve the utilisation efficiency of facilities; 3) improve researchers' access to facilities; 4) support human resources and provide stable management.
- (Action Plans) 1) Prioritise facility procurement; 2) develop a facility maintenance strategy; 3) provide facility services; 4) optimise facility occupancy/use; 5) establish laboratory management standards; 6) provide management training for equipment operators, etc.

3. Status of South Korea's Research Infrastructure Policy

The current status of South Korea's research infrastructure policy to improve Indonesia's research infrastructure policy is as follows. In Korea, research infrastructure is defined as “research facilities and equipment that are directly used for research and development activities and that embody the functions and environments required for research and development” and is defined according to the relevant laws.

[Figure 4-2] Definition of research infrastructure



Source: Standard Guidelines for the Management and Utilisation of National R&D Facilities and Equipment, administrative rules, Ministry of Science Technology and ICT, No 2021-92(2021).

In the relevant laws and regulations of South Korea, national research infrastructure is defined as “research facilities and equipment costing more than thirty million won that are built or planned to be built through national research and development projects”, whereas research equipment is defined as “facilities and equipment that perform independent functions in research and development that cost more than one million won and have a durability of more than one year”, and research facility is defined as “an independent research space consisting of one of the following forms to support research and development activities (testing, analysis, measurement, education, training, etc.) for a specific purpose”.

- ① One large research facility;
- ② One system combining multiple research equipment;
- ③ One unit that integrates various items of research equipment in one place to facilitate their joint utilisation.

In Korea, the national research infrastructure is classified according to the scope of its use (sole utilisation, joint utilisation), utilisation status (normal operation, idle, underused, unused), and construction type (individual research facility equipment, system research facility equipment, development research facility equipment). In general, research infrastructure refers to research facilities and equipment that are built regardless of the financial sources. Still, in Korea, equipment built through national research and development projects (i.e. government funding) is designated as a policy target and is collectively referred to as “national research infrastructure”.

These “national research infrastructures” are usually acquired through individual research projects, have a longer lifespan than the period of the underlying R&D project (usually 5 to 10 years), and also have the characteristic of incurring certain amounts of operating costs and maintenance investment for continuous utilisation every year. In addition, they can be divided into state-of-the-art research equipment that is used for easy-to-use research, and general-purpose research equipment that is used for education and training. However, even state-of-the-art research infrastructure that is used for easy-to-use research can be used for

applied research or education depending on its lifespan, so we review and closely manage the possibility of continuous utilisation even after the end of easy-to-use research.

[Table 4-2] How the research infrastructure is utilised over its lifespan

Classification	(Early) Use in Leading Research	(Middle) Use in General Research	(Late) Use in Educational Research
Performance level	The world's first and most advanced specifications and performance.	Specifications and performance in high demand due to wide range of applications.	Specifications and performance suitable for confirming existing knowledge rather than creating new knowledge.
Purpose of utilisation	(Leading) Creation of unique and groundbreaking research data or research data in new areas.	(Catch-up) Creation of highly utilisable research data that is not the first in the world, but where similar research is actively being conducted.	(Dissemination) Creation of research data to confirm and verify existing knowledge or to disseminate it.
Power user	A group of power users leading the research field.	A large number of researchers and industrial workers in various research fields.	Educators and trainees involved in the dissemination of existing knowledge.
Construction method	(Top-down) Reviewed by a group of insightful experts in related fields.	(Bottom-up) Introduced through a demand survey of researchers in related fields.	(Hybrid) Introduced by matching suppliers and demanders for educational use.

Source: Author's elaboration.

Until the 1990s, South Korea invested relatively little in its research infrastructure. However, with the expansion of R&D investment since 2000, the amount of investment in research infrastructure has also increased. Research infrastructure policies have been promoted because the National Assembly, National Audit Office, and other organisations have requested measures for improving the efficiency of investment in the national research infrastructure. In other words, research infrastructure policies aim to secure investment efficiency.

Early research infrastructure policies promoted management-oriented policies represented by budgetary input control to improve investment efficiency.

The National Research Infrastructure Registration System, introduced in 2005, became the

basis for establishing a management system by identifying the construction status of the entire research infrastructure. Meanwhile, the National Research Infrastructure Integrated Review System, launched in earnest in 2013, presented an opportunity to establish a management foundation for overall budget investment.

This management-oriented policy implementation system, which includes review and registration of research infrastructures, can be very useful from the perspective of policymakers. Controlling the budgetary input of all research infrastructures built annually has made it possible to curb redundant construction and reduce the budget. Furthermore, it is possible to select appropriate research infrastructure performance that meets the target research and to calculate reasonable research infrastructure construction costs.

In addition, based on the budget control information and research infrastructure information registered after construction, SEUS was established in 2013 and operated accordingly. Subsequently, a system for transferring underused research infrastructure was established (2016-), followed by integration of the national research infrastructure reservation system (2017-), and the establishment of a reservation service and performance utilisation system centred on research facilities (2019-).

Despite these budget and registration policies, the results of collecting opinions from research sites, auditing external organisations, and conducting policy research in the early 2010s showed that the “consumers” (i.e. researchers) of research infrastructures felt that there was still a large shortage of research infrastructures available for research, and that access to and joint utilisation of such infrastructures was difficult. Meanwhile, the providers (i.e. infrastructure operators) of research infrastructures complained about the difficulties in securing professional operators and operating costs for their research infrastructures. Finally, research infrastructure policymakers (i.e. government agencies) pointed out that many research infrastructures were being neglected and that duplicate investment requirements persisted.

The fundamental goal of building and using research infrastructures is to conduct innovative research based on cutting-edge research infrastructures, while the fundamental investment

goal is not to build research infrastructures well but to utilise them well once built. Therefore, since 2013, Korea has established and promoted a five-year basic plan for the national research infrastructure encompassing its entire life cycle from construction to utilisation, operation, and disposal.

South Korea's Basic Plan for National Research Infrastructure aimed to improve the status and competitiveness of national science and technology by establishing and expanding efficient and effective R&D infrastructures based on interdepartmental cooperation, and by improving inefficiency issues such as duplicate purchases and the lack of joint utilisation of research infrastructure; and to make a qualitative leap in R&D projects through the establishment of mid- and long-term plans that took into account changes in the science and technology environment, future research needs, and ministry-specific characteristics.

[Table 4-3] Basis for establishing the Basic Plan for Equipment for National Research Facilities

Basic Act on Science and Technology

Article 28 (Expansion, upgrading, management, and utilisation of R&D facilities and equipment) ① The government shall establish and promote measures to expand, upgrade, manage, operate, jointly use, and dispose of R&D facilities and equipment that are necessary to promote efficient and balanced R&D.

Enforcement Decree of the Basic Act on Science and Technology

Article 42 (*Promotion of expansion, upgrading, etc. of R&D facilities and equipment*)

- ① The head of the relevant central administrative agency shall periodically survey the status of R&D facilities and equipment in order to efficiently promote the expansion, upgrading, management, operation, joint utilisation, and disposal of R&D facilities and equipment (hereinafter referred to as "expansion, upgrading, etc.") in accordance with Article 28 (1) of the Act.
- ② The head of the relevant central administrative agency shall formulate the following plans based on the results of the survey pursuant to the foregoing paragraph (1) and reflect them in the mid-and long-term plans for national R&D projects.
 - i. Plans for the expansion of R&D facilities and equipment necessary for relevant R&D projects.
 - ii. Plans for the management, operation, and joint utilisation of R&D facilities and equipment.
 - iii. Plans for upgrading R&D facilities and equipment.
 - iv. Plans for disposing of R&D facilities and equipment.

Source: ENFORCEMENT DECREE OF THE FRAMEWORK ACT ON SCIENCE AND TECHNOLOGY (2022).

Since the enactment of the 'Basic Act on Science and Technology' (16.01.2001), which is the highest law pertaining to science and technology in South Korea, the establishment of measures for upgrading the research infrastructure (Article 28 of the Act and Article 42 of the Enforcement Decree) has been specified, although not as a separate measure. Rather, it has been included in various statutory plans in the technology field, such as the Basic Plan for Science and Technology and the Comprehensive Plan for Basic Research Promotion.

In 2004, the law was amended to stipulate the establishment of a 'Plan for Securing, Advancing, and Jointly Utilising Research Facility Equipment'. However, it was not prepared as a practical measure because the current status of the entire research infrastructure had not been identified at that point in time. The infrastructure environment was not prepared, such as in Indonesia. In 2009, a dedicated agency was designated to manage information registration, deliberate, and conduct a status survey of the research infrastructure, and based on this, a comprehensive plan was established in 2013. The third plan is currently being promoted.

The 'National Research Facility Equipment Upgrading Plan' has the status of a mid-to-long-term development strategy that synthesises the mid-to-long-term plans related to research infrastructure established by each ministry, and it presents the vision, goals, and directions for the national research infrastructure for the next five years.

The 1st National Research Facility Equipment Upgrading Plan (2013-2017) was prepared to promote investment efficiency, operational advancement, capacity enhancement, and maximisation of the research infrastructure at the national level. The specific contents are as follows:

- A. (Investment Efficiency) To strengthen the efficiency of investment in the national research infrastructure, integrated deliberation of the budget for the research infrastructure was conducted (2016), and the system involving research institutes' own deliberations on the construction of research infrastructures was activated. In particular, a survey was conducted on the status of construction of national research infrastructures in order to determine the actual operation and utilisation

of research infrastructures and to induce the registration of unregistered and missing research infrastructures. This enabled identification of the overall investment status at the national level and improved the accuracy and updating of research infrastructure information.

- B. (Operational Advancement) We established a national research infrastructure management system by enacting (2016) the "Standard Guidelines for the Management of Equipment at National Research and Development Facilities" into law; supported the professional operation of research infrastructure by fostering young professional personnel (1,264 by 2022); and contributed to securing professional operational capabilities by providing operational support for high-value research infrastructures and the management of research infrastructure professionals.
- C. (Maximising Utilisation) By proposing and achieving the target rate of 'joint utilisation allowed' for research infrastructure, which indicates the level of joint utilisation of the research infrastructure foundation (Actual: 60.3%, Target: 60%), Ansys induced a change in the awareness of joint utilisation at research sites. Ansys increased the utilisation of research infrastructures by supporting the active transfer of research infrastructures, raised awareness of joint utilisation by establishing and operating a national research infrastructure online reservation service, and laid the foundations for the current platform-based utilisation system.
- D. (Enhancement of support capabilities) To support the utilisation of research infrastructures by companies, we strengthened the national R&D research base by integrating the research infrastructure of national research institutes, and we launched the Analytical Science Instrument Association (2017) to enhance the localisation capabilities of research infrastructures by companies.

The 2nd National Research Facility Equipment Advancement Plan (2018-2022) promoted policies designed to strengthen utilisation and support capabilities by distinguishing between research equipment, research facilities, and the research equipment industry in order to promote the use of national research infrastructures.

- A. (Improving the utilisation of research-supported research infrastructures) Improvements were made to enable researchers (institutions) to smoothly introduce (deliberate) the necessary research infrastructure in a timely manner and to use (maintain and maintain) it stably over a long period of time.

Ansys has established a full-cycle system so that research infrastructures can be quickly reassigned to other researchers (institutions). To this end, we have classified research infrastructures that should be reserved for sole use and research infrastructures whose joint use should be encouraged based on the size of each research infrastructure (small, medium, large) and whether it can be jointly used (general purpose/non-general purpose), and we have prepared the appropriate policies. In addition, when reviewing (deliberating) the feasibility of introducing research infrastructures, we have established a system for determining whether to build additional research infrastructures based on the demand for their 'utilisation', rather than the 'duplication' of existing research infrastructures. We have also established a policy for easing the burden on research sites by differentiating management methods based on the number of research infrastructures and the scope of their utilisation.

In particular, a system for securing operating expenses was established to ensure that research infrastructures which should be used exclusively by each researcher can be maintained and used stably, and to ensure that jointly used research infrastructures were centralised (Core-facility) under the responsibility of the institution, so that they can be operated jointly and used efficiently and professionally. All decisions on such matters are made by the Equipment Review Committee within each research institution so as to increase autonomy.

- B. (Improving Efficiency of Research Infrastructure Utilisation) By unifying utilisation/support information about the Research Infrastructure Utilisation Portal, like SEUS, we have strengthened support for the use of national research infrastructures by the private sector, such as companies and individual makers, and expanded customised services by characterising each R&D infrastructure by taking advantage of its strengths and providing linkage services for similar fields. In

addition, we have implemented a policy for managing large research infrastructures so that they can be built and operated stably. To this end, the research infrastructure utilisation system and the support systems scattered among ministries, regions, and institutions were unified in stages to strengthen their accessibility and efficiency, and differentiated online and offline channels for the joint utilisation of research infrastructures were continuously discovered and provided for each type of researcher (i.e. researchers at small, medium and large enterprises, research institutes, and universities). We have characterised the various R&D infrastructures in operation by each research support field and induced service linkages between similar fields, and supported improvement of the capabilities of R&D infrastructures by establishing a field-specific characterisation model. In addition, we have strengthened the management of large-scale research infrastructure construction to ensure efficient expansion, and established a project to support the operation of large-scale research infrastructures that require stable operation due to the high levels of research utilisation (joint utilisation).

- C. (Domestic Research Infrastructure Industry Development and Human Resource Development) Based on the results of the first plan, which laid the 'cornerstone' for the development of the domestic research infrastructure, a 'comprehensive support' system - ranging from research infrastructure (parts) development support to reliability evaluation and market development support - was established to support the promotion of a full-fledged research infrastructure industry, stable employment of research equipment experts, and the training of human resources tailored to the field (operation and development).

Through these secondary plans, we achieved the following results:

- A. Reduction of the rate of idle and under-used equipment by 30%: As of the performance target year of 2019, out of a total of 55,084 pieces of research equipment, 1,888 pieces of under-used equipment and 1,275 pieces of idle equipment accounted for 5.7% of the total, which is 7.9% lower than the original target of 13.6%.
- B. Core facilities (specialised research support facilities): From 2018 to 2021, a total of 60 centres were supported to create core research support facilities, including 13 and 20 through the Ministry of Education project, achieving twice the initial target of 30 core research support facilities.
- C. Establishment of a platform for using R&D-based equipment for each of the ten research support areas: We established the K-Facility Search and Reservation Service in SEUS, and used it to enable classification and search by each of the seven supported research areas, technology development stage, and current issues.

From this year, we are promoting the 3rd National Research Facility Equipment Upgrading Plan (2023-2027), the main contents of which are as follows:

[Table 4-4] Korea's Research Infrastructure Policy Changes

(1st•2nd) management-centred → (3rd) field-centred based on autonomy and responsibility				
1st Plan		2nd Plan		3th Plan
Establish a system for investing in and managing research infrastructures.	➔	Lay the foundations for using research infrastructure.	➔	Establish a field-centred management and utilisation system.
Unify the decentralised management system across ministries.		Link and integrate the utilisation systems of ministries and technologies into one.		Establish an institution-centred management and utilisation system institution.
• Systematically build and expand research facilities and equipment needed in the field. • Essential for securing national competitiveness in the era of technological hegemony and developing ultra-gap technologies establishing mid- to long-term plans based on local demand for large-scale research facilities and equipment.				

-
- Strengthen management and utilisation based on autonomy and responsibility.
 - Establish management and utilisation strategies tailored to the characteristics of research sites so that various kinds of research requiring research facilities and equipment can be carried out effectively.
 - Secure the capability to operate and develop research facilities and equipment.
 - Establish a systematic system for strengthening the capabilities of specialised operating personnel and lay foundations for the growth of the domestic equipment industry.
-

Source: Author's elaboration.

South Korea's research infrastructure policy promotion can be summarised as follows:

- A. (Identification of current status) To identify the current status of investment in research infrastructure, information on research infrastructure budget investment and construction infrastructure is registered, and an evaluation is conducted in the input process to monitor how much has been invested in the past and how much is currently being invested, and fine policy adjustments can be made accordingly.
- B. (Management System) We have established a system to clearly distinguish the roles of the stakeholders (central government, research institutes, research facilities, researchers, etc.) participating in the research infrastructure management system and to clearly establish management responsibilities and limits for each role and actively implement them.
- C. (Utilisation System) We are continuously identifying what is needed by the users (researchers) of research infrastructures and maintaining the factors that utilisation (supply and demand, utilisation methods, systems, etc.) to induce active utilisation.

It is concluded that the problems of Indonesia's research infrastructure can be solved based on this process and the resulting policy outputs.

4. Direction of Policy Advisory for Indonesia's Research Infrastructure

According to the causes and solutions of the problems identified by BRIN in Indonesia, it can be summarised as cause analysis and alternative suggestions to establish procurement priorities for redundant investments to solve problems, establish maintenance strategies for inefficient maintenance, optimise information services and usage for limited access and low utilisation, and strengthen the training of management operators for weak monitoring.

When judged from a policy perspective, this could be an efficient alternative or response to a visible phenomenon, but it is difficult to see it as a fundamental solution based on South Korea's experience over the past twenty years. This judgment can be inferred based on the following grounds.

- A. (Duplication of procurement plans, aging, insufficient supply, and fragmentation)
- Most of Indonesia's research infrastructure is aging, with insufficient supply compared to demand, and is concentrated in certain regions, thus hindering local science and technology research development. As a solution, the government is proposing to prioritise the procurement of research infrastructure.

The general method of procurement prioritisation is as follows.

- (Bottom-up) Demand competition for researchers (facilities) → peer evaluation → selection committee → prioritisation.
- (Top-down) Recommendation of BRIN by national strategic priorities → Peer review → Selection committee → Prioritisation.

In general, this process is required the current infrastructure operational status, infrastructure operation foundation (manpower, operation costs, and environment), strategic research infrastructure demand according to the national strategy, and the available financial resources for the construction and operation of infrastructure. These considerations are not found in the materials presented by

BRIN, and the action plan is set with too tight a schedule, which makes its effectiveness questionable. Of course, a method of piloting and then modifying and continuously supplementing the policy could also be used, but repeated implementation of the same policy may lower the credibility of the policy and reduce the response rate of the field.

- B. (Inefficient Maintenance) We are proposing a facility maintenance strategy to improve non-standardised maintenance procedures and methods. Based on the data presented by BRIN and the results of field interviews, we identified the following problems. The operational personnel of research infrastructures are mainly researchers, who tend to focus on operations rather than on maintenance; while the concepts of preventive maintenance, regular inspection, and operational management of maintenance are not clear. In addition, there are cases where maintenance costs are not explicitly secured and cannot be handled immediately when problems arise. This is due to a lack of clear management categories and a shortage of available personnel. In other words, since the managers of the infrastructure are mostly researchers whose main task is research and development, maintenance of the research infrastructure is handled as an incidental task. Even if a clear maintenance strategy could be established in the future, the same problem might occur unless adequate concepts of equipment operation personnel and policies are established beforehand.
- C. (Improving Access to Research Infrastructures) In order to optimise the use of research infrastructures, a plan to increase joint utilisation services and complement the information system was presented. As the majority of research infrastructure operators are researchers, they are faced with the problem of having to postpone their main duties in order to support shared utilisation. Therefore, most of them support shared utilisation in their spare time, and the higher the performance of the research infrastructure, the more serious this problem becomes. In addition, since the operators of research infrastructures are not regular service personnel, it is difficult for them to provide immediate results or a friendly service within the desired deadline. Therefore, to promote collaboration, it is

important to systematically foster research infrastructures that can be used collaboratively rather than providing collaboration information. According to BRIN's plans, most plans are implemented within three years, which is judged to be unrealistic considering the time required to develop human resources, relocate physical resources, and discover and launch the relevant services.

Therefore, in order to fundamentally understand the problem, we have listed various phenomena and grouped them together so as to identify the underlying causes and propose policy techniques (MECE analysis and logic tree) for obtaining solutions to them.

[Table 4-5] Analysis of Indonesian Policy Status and Solutions

Indonesia Action Plan for Innovation of S&T Research Infrastructure						
MECE		Problem Logic Tree				
Large Category	Medium Category	Source	What is (Phenomena)	What is (Detail phenomena)	Why is (Reason)	How is (Solution)
1. the efficient of government investments	1-1. the efficient of government new investments	Midterm Report	Duplication procurement	-	Bad procurement planning	?
1. the efficient of government investments	1-1. the efficient of government new investments	Midterm Report	Unequal procurement between technologies	-	Bad procurement planning	?
1. the efficient of government investments	1-2. the efficient of government existing Investments	Midterm Report	Damages of facility	-	Inefficient maintenance	?
1. the efficient of government investments	1-2. the efficient of government existing Investments	Midterm Report	Damages of facility	-	weak monitoring to (preventive) maintenance	?
2. the effectiveness of government investments	2-1. the effectiveness of government new investments	Midterm Report	Research obstacle	Siloed information between agencies	access limitation	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	Midterm Report	Low usage of RI	-	inefficient Information	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	Midterm Report	Exclusive use to RI	-	-	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	1st Interview	Difficulty sharing use to RIs	-	Lack of RI information	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	1st Interview	Difficulty sharing use to RIs	-	Lack of a sharing system	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	1st Interview	Difficulty sharing use to RIs	-	Poor management of research data	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	1st Interview	Difficulty sharing use to RIs	-	Cross-organizational cultures	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	1st Interview	Difficulty sharing use to RIs	-	Lack of operational capacity for RI	?
2. the effectiveness of government investments	2-2. the effectiveness of government existing Investments	1st Interview	Difficulty reinvestment of operating income	-	-	?
<i>Example that this field is case of South Korea.</i>			Duplicate purchase	Want to same RIs	Want to use the same RIs as rivals	RIs Deliberation Process
			Duplicate purchase	Do not want to borrowing	Annoying (inconvenient)	RIs Deliberation Process
			Duplicate purchase	Difficult to identify duplicate RI	Hard to find information	Construction of database
			Unnecessary purchase	Want to new RI	Just want new RIs	Strengthening the joint utilization system
			Unnecessary purchase	Feedback is not enough	No one says anything to me when make a purchase	RIs Deliberation Process
			Unnecessary purchase	To spend quickly on budget	Easy to buy RI if anyone have the money.	RIs Deliberation Process
			Reluctant to sharing use	Don't want to Share with rival	Sharing, all information will be known.	Strengthening the joint utilization system
			Reluctant to sharing use	May be Loss or failure	Someone else oneself handles the RIs	Strengthening the joint utilization system
			Reluctant to sharing use	No benefit to share.	Own the RIs	Strengthening the joint utilization system
			Managemant neglect	Researchers maintain own RI.	Lack of engineer	Training of specialist
			Managemant neglect	Maintenance is expensive.	Lack of Maintenance costs	Operation and maintenance support
			Managemant neglect	Don't know what to do	Lack of operating know-how	Training of specialist
			Unused and Neglect	Don't have anyone to take care of RIs	Move and disadvantageous	Identification and transfer of unused RI
			Unused and Neglect	Don't have much use for it	Helping to select other project	Identification and transfer of unused RI
			Unused and Neglect	Use it sometimes	Maybe use again someday.	Identification and transfer of unused RI

Source: BRIN research team's elaboration.

Accordingly, Indonesia has prepared and presented an action plan but, nevertheless, we believe that fundamental changes aimed at improving the policies are needed for the following reasons.

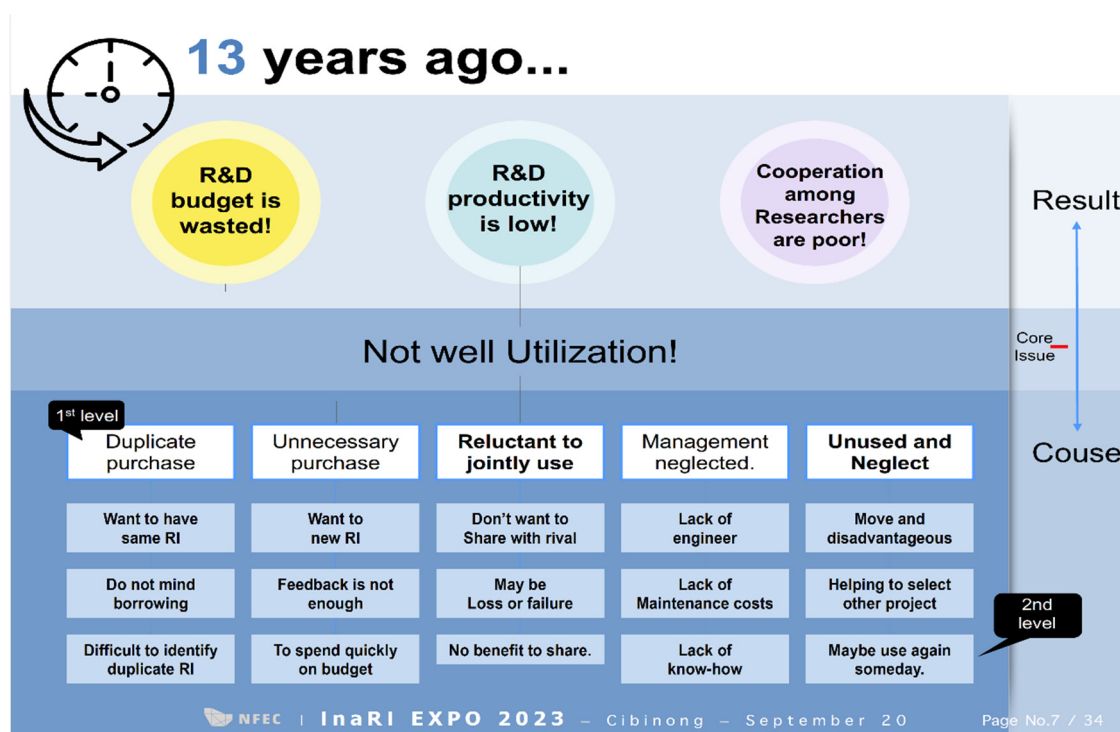
Most of the current analysis of the research infrastructure in Indonesia relies on interviews with researchers, making it difficult to analyse the quantitative causes and propose fundamental solutions. For example, a quantitative analysis of such questions as “What was the demand for research infrastructure in the past?” “How much investment was made in response to this demand?” “What areas should be focused on in the future?” and “How much investment is needed?” is not currently possible. This is inconsistent with the current trend of evidence-based policy, which is essential for systematic policy development.

As the focus is on solving visible phenomena, it is judged that the analysis of the causes is insufficient. For example, regarding inefficient maintenance, which is one of the main problems, the analysis of the current status of maintenance did not reveal any additional information about whether the problem is simply a lack of maintenance funds, a lack of manpower and the environment necessary for maintenance, or a lack of preventive maintenance and utilisation analysis for maintenance. Even if a maintenance strategy is prepared as a solution, it is judged that it will still be difficult to solve the problem if there is insufficient manpower and budgetary support, or if there is no procurement system to perform it.

As for the lack of know-how in building and operating information systems, information systems do not simply collect and provide information but must continuously maintain accurate information and discover and provide services. For example, the information on most research infrastructures is constantly changing (installation site, operator, contact information, etc.), making it necessary to continuously update the information on research infrastructures through periodic updates and surveys. In addition, researchers who want to utilise research infrastructures often lack the related operational information. Therefore, it is necessary to establish a system for equipment service clustering and providing operational support from a service perspective.

To solve these problems, Korea collected all phenomena related to research infrastructures in 2009 and analysed them in order to analyse problems that are not well utilised and propose solutions.

[Figure 4-3] Case Study of Analysis of Research Infrastructure Problems in South Korea



Source: Author's elaboration.

Therefore, it is judged that policy improvements can be made by adapting the system that Korea has been promoting to the problems with the research infrastructure proposed by Indonesia.

[Table 4-6] Logical framework for problem-solving methods

What is (Phenomena)	Why is (Reason)	Solution
Duplicated procurement	Bad procurement planning	Create an efficient deployment policy.
Unequal procurement between technologies	Bad procurement planning	
Damage to facility	Inefficient maintenance	

What is (Phenomena)	Why is (Reason)	Solution
Damage to facility	Weak monitoring and maintenance	Create a fundamental maintenance plan (standard management manuals & current status survey).
Research obstacles	Limited access	
Low use of RI	Inefficient information	
Exclusive use to RI	-	
Difficulty sharing use to RIs	Lack of RI information	Operate an information collection & management system.
Difficulty sharing use to RIs	Lack of a sharing system	
Difficulty sharing use to RIs	Poor management of research data	
Difficulty sharing use to RIs	Cross-organisational cultures	
Difficulty sharing use to RIs	Lack of operational capacity for RI	
Difficulty re-investing operating income	No separate system for managing operating expenses	Operate a cost procurement system.

Source: Author's elaboration.

■ (Improvement 1) Establish an efficient policy for investment in the research infrastructure

(Goal 1) Establish a BRIN-wide demand identification system and strategically expand the cutting-edge research infrastructure.

⇒ (Action Plan) Establish a demand survey system for researchers and institutions at BRIN, universities, and businesses/industries; and establish a system of cooperation with major countries based on the results of the survey in order to strategically expand the necessary research infrastructure.

(Goal 2) Establish and activate a system for evaluating research infrastructure investment at BRIN.

⇒ (Action Plan) Establish a deliberation system to comprehensively review investment priorities, redundancy, matching of demand, realism of performance, operability, and investment effectiveness for the research infrastructure needs of BRIN.

(Goal 3) Monitor BRIN's construction of research infrastructures and establish an investment portfolio.

⇒ (Proposal) Establish an investment strategy that comprehensively considers the input costs and planned costs of research infrastructure currently in operation at BRIN, and an investment portfolio that combines them with new demands.

■ **(Improvement 2) Establish a lifecycle management plan for fundamental research infrastructure**

(Goal 1) Systemize the operational management of research infrastructures.

⇒ (Action Plan) Establish a comprehensive system for managing the research infrastructure at BRIN, including a general management system, a management manual, and a legal system.

(Goal 2) Strengthen support for the operation of research infrastructures.

⇒ (Action Plan) Establish effective governance for the operation of research infrastructures at BRIN.

(Goal 3) Foster specialized operators of research infrastructure and discover services.

⇒ (Action Plan) Establish degree courses and training programs designed to foster professional operators, establish an information management and fostering system for career management, and discover services through professional operators.

■ **(Improvement 3) Establish an organic information system operation plan**

(Goal 1) Establish a research infrastructure information collection system.

⇒ (Action Plan) Establish the management system and standards for each stage of information generation and the lifecycle of research infrastructure, develop the related laws and regulations, and establish an information linkage system and utilization system.

(Goal 2) Establish a management system for information realization.

⇒ (Action Plan) Establish measures for information actualization and statistical calculation such as surveys, survey analysis, etc.

(Goal 3) Discover information systems.

⇒ (Action Plan) Discover a reservation system and information utilization system from the user's perspective and prepare a strategic information plan.

■ **(Improvement 4) Establish an efficient joint utilisation system**

(Goal 1) Establish a research equipment cost management utilization system

⇒ (Action Plan) Establish separate management standards and a management system for income generated by the use of research equipment, and establish a reinvestment system through this.

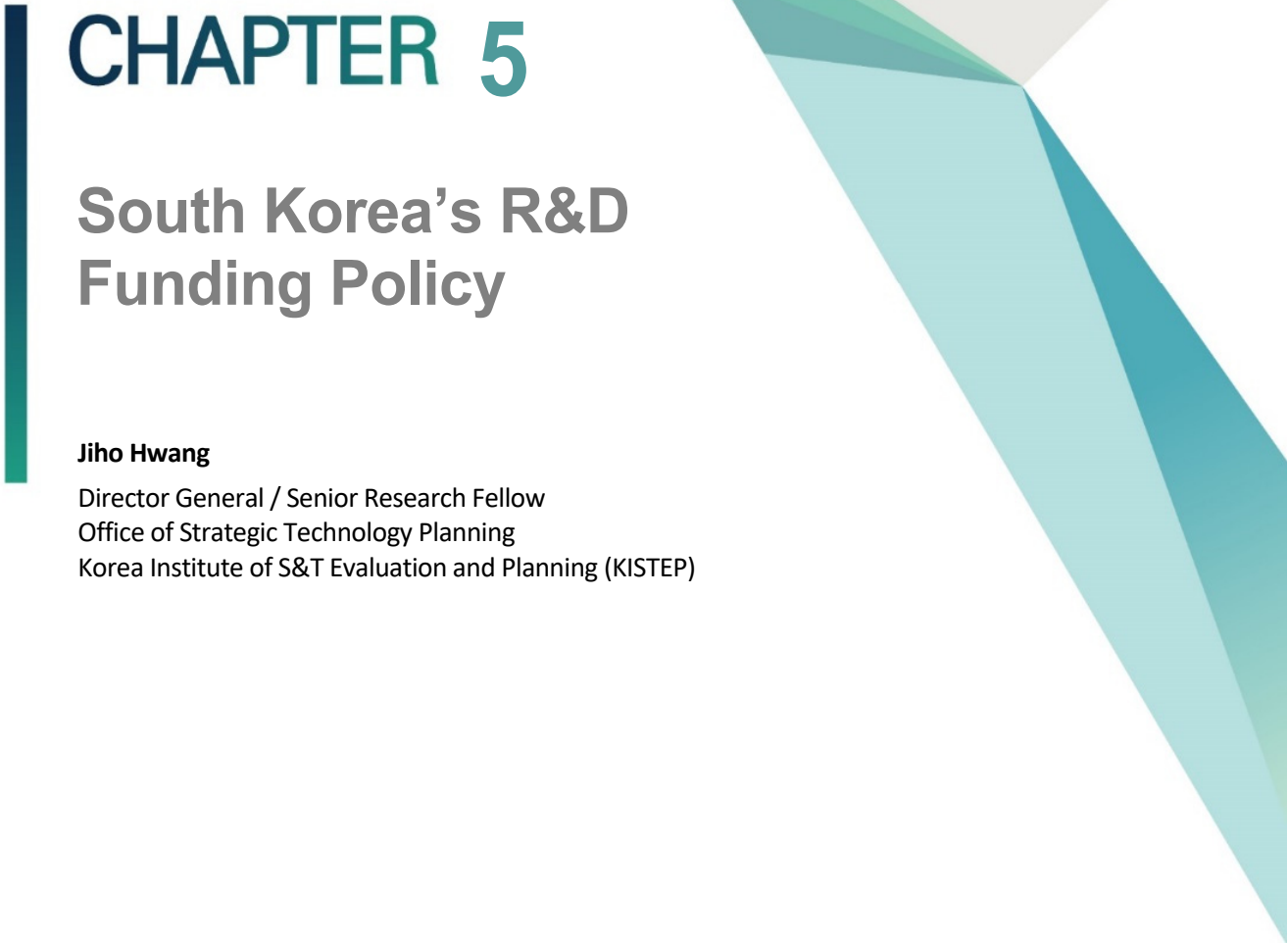
(Goal 2) Establish a performance management and compensation system for joint utilization

⇒ (Action Plan) Establish a separate system of rewards (in addition to point rewards) for voluntary joint utilization, and establish programs to promote its active use.

(Goal 3) Foster joint utilisation bases.

⇒ (Action Plan) Designate joint utilization bases according to the strategic goals of S&T, prepare and foster the necessary research infrastructure for each base, and devise strategic investment plans and staffing plans.

To prepare a detailed implementation plan for the research infrastructure improvement strategy derived in this way, it will be necessary to conduct a study on strategy formulation from next year.



CHAPTER 5

South Korea's R&D Funding Policy

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Chapter 5. South Korea's R&D Funding Policy

1. Introduction

The Indonesian government disbanded all of the country's public R&D institutions in 2021 and integrated them into the National Research and Innovation Agency (BRIN), which was launched in January 2022. All resources, including human resources, infrastructure, and budgets, from forty-eight government research organisations across all ministries and agencies were consolidated into BRIN. Henceforth, BRIN was responsible for the integrated management of all human resources, infrastructure, and budgets under its jurisdiction. As BRIN is a government agency directly under the President, it does not receive any coordination from other ministries or coordinating ministries.

As Indonesia's sole national research institute, BRIN is tasked with the integrated management of the research and development budget. As such, BRIN focuses on enhancing R&D efficiency and effectiveness at the national level by establishing systems and processes for the efficient distribution and adjustment of the R&D budget.

In light of this goal, we will begin by examining Korea's R&D funding system, and then look at the results of a survey on Indonesia's R&D funding status and compare it with the Korean system in order to identify directions for future improvements of the Indonesian funding system.

2. The R&D Funding System in Korea

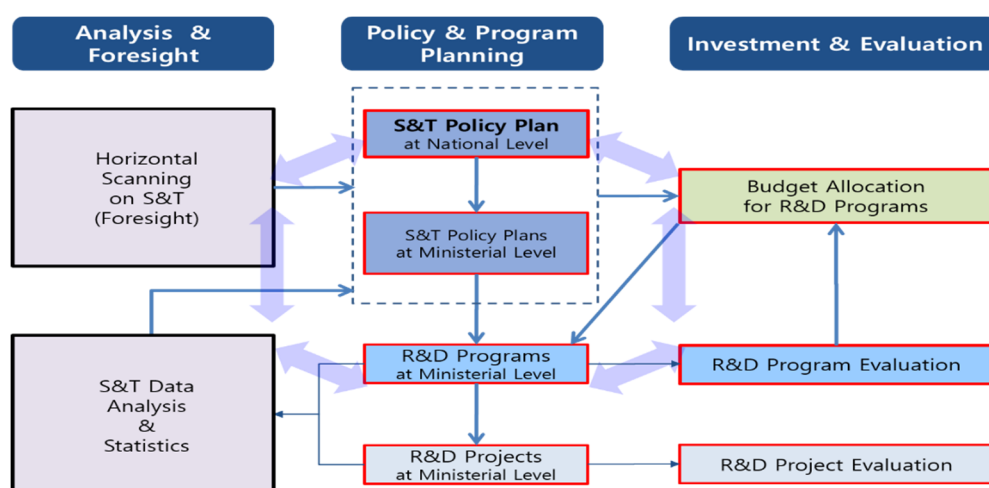
2.1 National S&T Management System

2.1.1 The Process of the National S&T Management System

The overall process of National S&T Management is as follows.

Before planning S&T policy and R&D programs, it is necessary to conduct analysis and foresight. This involves a horizontal scanning of S&T to forecast which promising future technologies will be realised and analysing S&T data to understand our current position. Using the results of analysis and foresight, the national-level S&T policy is planned first. Each ministry then plans S&T policy and R&D programs at the ministerial level with the assistance of funding agencies or other experts. The funding agency plans the R&D projects that constitute the R&D programs and the management system. If it is feasible to launch the programs, they can be funded via budget allocation. Subsequently, evaluations are conducted at both the program and project levels, and the results of the evaluation of the R&D programs serve as feedback for allocation of the budget for next year's R&D programs.

[Figure 5-1] Overall Process of National S&T Management



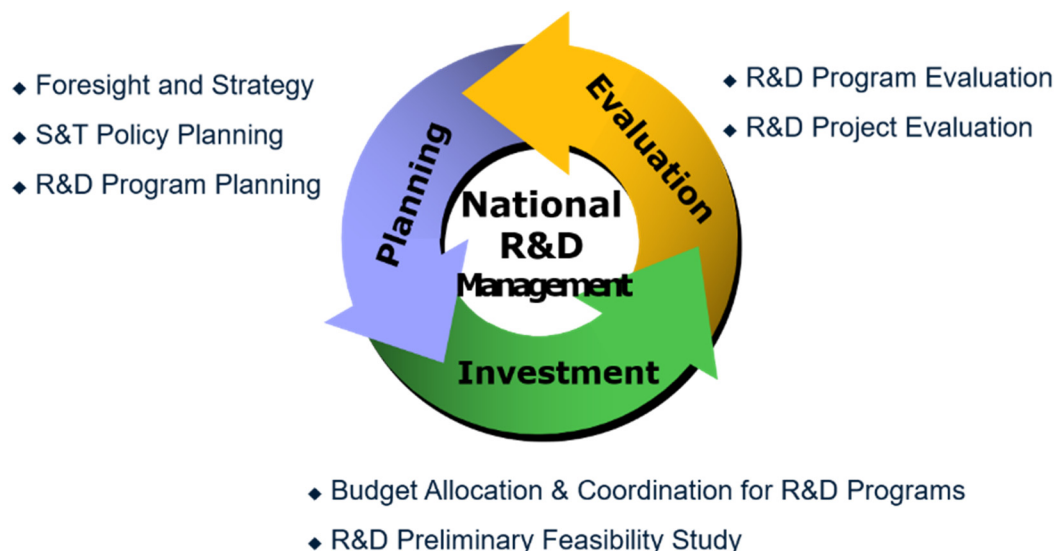
Source: KISTEP's internally prepared material.

2.1.2 Three Stages of National R&D Management

National R&D management consists of the following three stages: planning, investment, and evaluation.

In the planning stage, we conduct foresight, establish strategy, and plan the S&T policy and R&D programs. In the investment stage, we allocate budgets and coordinate the R&D programs, and also conduct a preliminary R&D feasibility study to determine whether it is feasible to launch large, newly proposed R&D programs or not. In the evaluation stage, we evaluate the R&D programs and projects.

[Figure 5-2] Three Stages of National R&D Management



Source: Author's elaboration.

2.2 Coordination of National R&D programs

2.2.1 Key Features of Current R&D Investment

In 2021, the total national R&D investment in Korea, including both government and the private sector, amounted to approximately US\$ 89.3 billion, ranking 5th in the world.

National R&D investment as a proportion of GDP amounted to 4.96%, ranking 2nd in the world.

The government's R&D budget for FY 2021 amounted to US\$ 23.8 billion, accounting for 4.9% of the total government expenditure.

2.2.2 Historical Overview of R&D Investment and S&T's Activities

In 1963, South Korea invested just US\$ one million in national R&D. Some thirty years later, in 1991, the country invested US\$ 3.8 billion in national R&D, and after another thirty years, in 2021, it invested US\$ 89.3 billion. This rate of increase is remarkable.

Now, let's look at the trend of the government R&D budget and major activities.

The first government-funded research institute, KIST, was established in 1966, while the first government R&D program was launched in 1982, with an R&D budget of US\$ 192 million. By 2021, the government's R&D budget had risen to US\$ 23.8 billion. Clearly, the Korean government has been focusing on R&D investment for our future.

2.2.3 The Necessity of Coordinating National R&D programs

As the government R&D budget has increased, we have recorded many remarkable S&T achievements, including scientific outputs, industrial products, and global contributions.

However, we have also encountered unexpected side effects.

Managing an ever-growing number of projects and programs, and involving many ministries in government R&D programs, has made efficiency and effectiveness in budget allocation increasingly critical. Furthermore, it is essential to consider specialisation in the area of science and technology budgeting because disruptive innovation occurs more frequently.

In the early 2000s, government R&D investment continued to increase, resulting in the diversification of R&D activities across various ministries and agencies. At that time, there

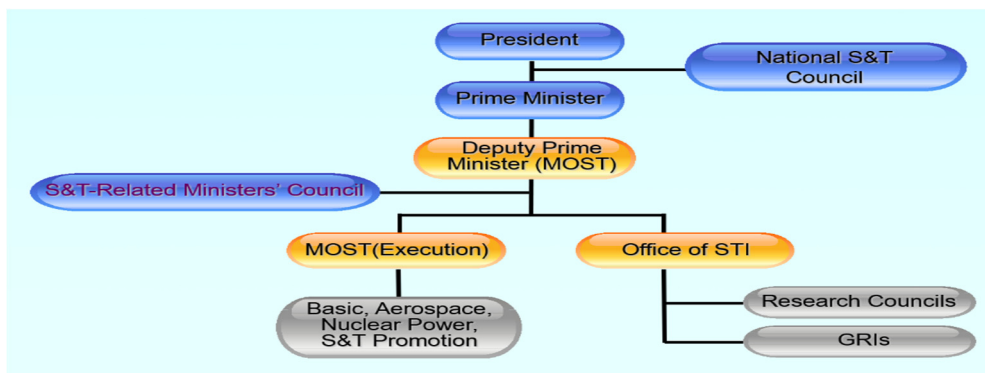
were more than 200 R&D programs involving 20 ministries and agencies. Consequently, there was a growing focus on the need for efficient and effective allocation of government R&D budgets, which led to the founding of the NSTC (National S&T Council) in 1999. The NSTC was established to develop new initiatives for the evaluation and coordination of national R&D programs.

2.2.4 S&T Governance Structure

The S&T governance was restructured in 2004. The status of the Minister of the Ministry of Science and Technology (MOST) was raised to Deputy Prime Minister. Additionally, the Office of S&T Innovation (OSTI) was newly established under MOST, which is responsible for evaluating all ministerial R&D programs and allocating their respective budgets.

Established in 1999, the NSTC was chaired by the President of Korea and composed of fourteen S&T-related cabinet ministers, the chairperson of the Presidential Advisory Council for Science and Technology (PACST), and nine non-governmental members appointed by the President. MOST served as the NSTC's secretariat. Meanwhile, the Korea Institute of Science and Technology Evaluation and Planning (KISTEP), established in 1999, played a role in national-level S&T planning, assisted with the evaluations, and helped to coordinate the national R&D programs, etc. Therefore, one of the main missions of KISTEP is to support the NSTC.

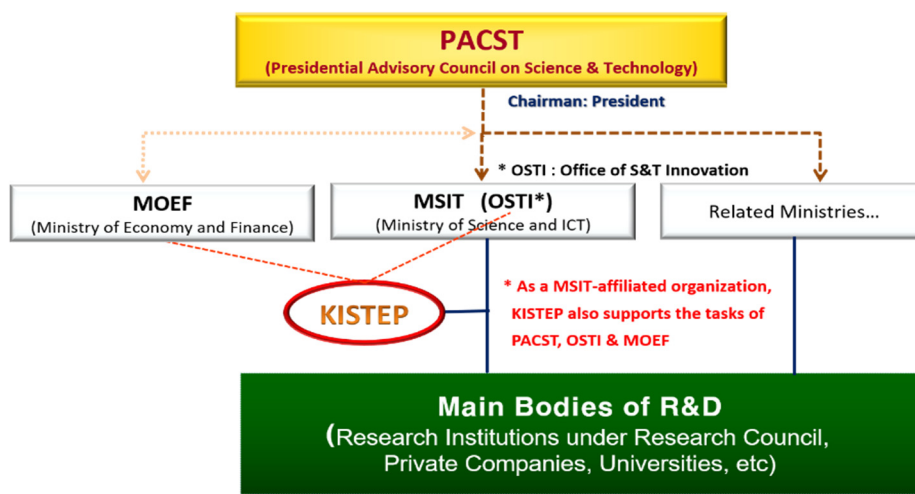
[Figure 5-3] Restructuring the S&T Governance Structure (2004)



Source: KISTEP's internally prepared material.

Figure 5-4 shows the S&T governance structure in 2020. PACST was in charge of the functions and roles of the NSTC and chaired by the President. As an organisation affiliated with the Ministry of Science and ICT (MSIT), KISTEP also supported the tasks of PACST, OSTI, and MOEF.

[Figure 5-4] Restructuring of S&T Governance Structure (2020)

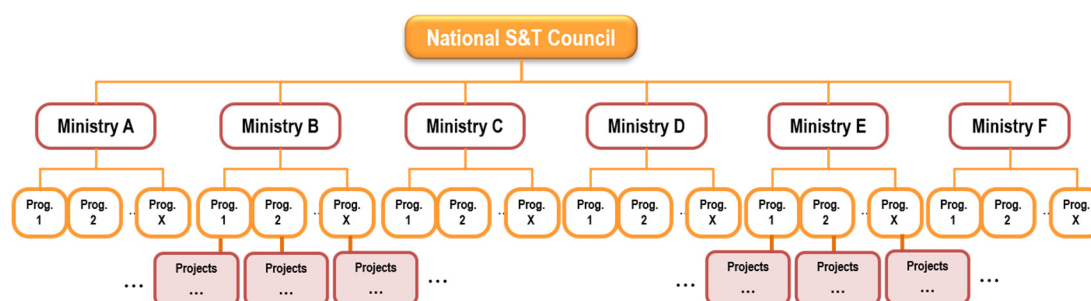


Source: KISTEP's internally prepared material.

2.2.5 Coordination of R&D programs

In 1999, the NSTC was in charge of coordinating the R&D programs, while various ministries were responsible for managing them, and R&D funding agencies were responsible for executing the R&D projects. Also, research institutes, universities and enterprises were involved in performing the R&D projects.

[Figure 5-5] Responsible Entities for Managing the R&D programs



Source: Author's elaboration.

2.3 Allocation of the Government's R&D Budget

The general budgeting process is as follows.

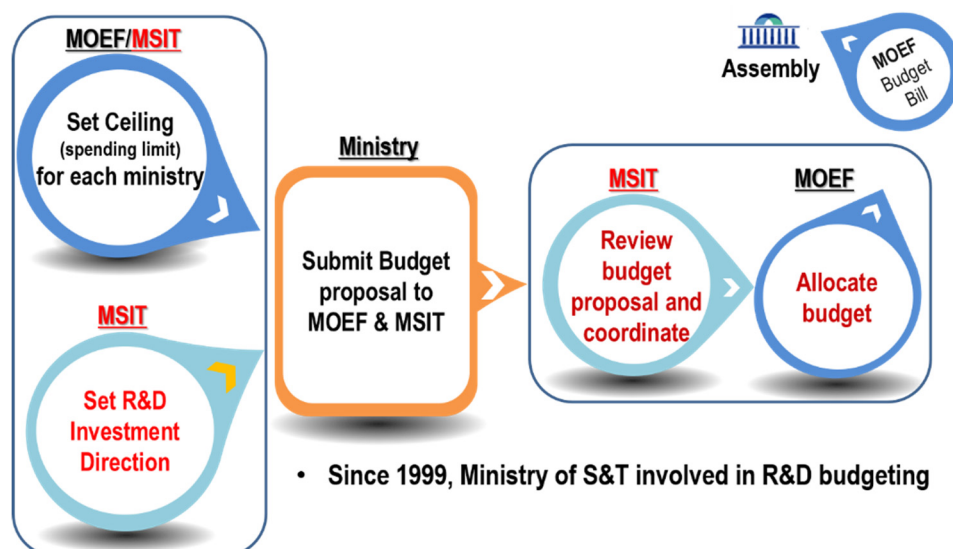
At the beginning of the year, the Ministry of Economy and Finance (MOEF), which is responsible for budget management, establishes budget ceilings for each ministry. All ministries submit their budget proposals for the next year to MOEF, which then reviews and allocates the budgets for programs across all ministries, and then submits the budget bill to the National Assembly. This constitutes the general budget process.

However, the R&D budget flow differs from that of other sectors in order to accommodate the specialisation of S&T. Since 1999, the Ministry of Science and ICT (MSIT) has been involved in R&D budgeting.

At the start of the year, MOEF, with assistance from MSIT, sets budget ceilings for each ministry, and MSIT establishes the direction of R&D investment for the upcoming year. All ministries submit their R&D budget proposals for the next year to MOEF and MSIT, which then review them. MSIT distributes and coordinates the budget for the next year's principal R&D programs and submits the results to MOEF. MOEF allocates the budget for all R&D programs, including the principal R&D programs, and submits the budget bill to the National Assembly. The R&D budget for the next year is finally deliberated and confirmed by the National Assembly.

The government's R&D programs consist of principal and general programs. The principal R&D programs account for over 80% of all R&D programs, and include basic research, applied research, development research, and the construction of large-scale facilities and equipment, etc. As for the general R&D programs, they encompass policy research, the operating expenses of research institutes and universities, and more.

[Figure 5-6] Flow of the Government's R&D Budget



Source: KISTEP's internally prepared material.

We have been operating Expert Committees within the NSTC for some years to facilitate highly scientific and rational decision-making.

The major role of the Expert Committees is to produce review reports of budget requests from the ministries involved in the execution of R&D programs. These reports mainly cover the validity of each program, the clarity of its goals, and the adequacy of the requested amount among other things.

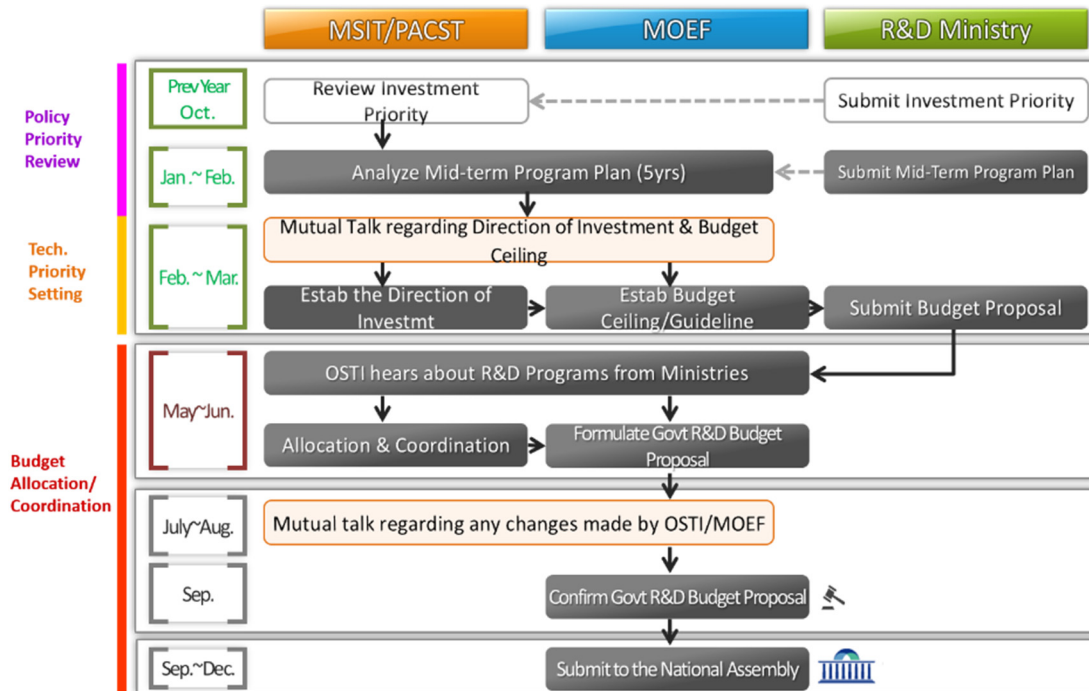
There are seven Expert Committees, each composed of sixteen experts. Since 2004, many experts from industry have participated in these committees, serving as a bridge between the government and the private sector.

During the R&D budget process, we also consider the Science and Technology Basic Plan, which outlines the fundamental direction for the national science and technology policy over the next five years. Additionally, to incorporate ongoing technological trends and changes, MSIT presents the next year's "Investment Direction and Technology Priority" in March of each year.

Figure 5-7 illustrates the budget flow process of R&D programs in Korea, providing a time

schedule and outlining the roles of the various bodies involved in the allocation of the next year's R&D budget, as explained above.

[Figure 5-7] Flow of the Government R&D Budget at a Glance



Source: KISTEP's internally prepared material.

2.4 The Law Governing Allocation of the Government's R&D Budget

In Korea, two major laws form the foundation for allocation of the budget for government research and development programs: The National Finance Act and the Framework Act on Science and Technology.

Under the National Finance Act, the Ministry of Strategy and Finance is responsible for establishing a national financial management plan and preparing the government budget plan for the following year. Initially, the Ministry of Strategy and Finance is tasked with establishing a national financial management plan. To enhance the efficiency and soundness of national financial management, a "national financial management plan" for

R&D expenditure over a five-year period must be developed and submitted to the National Assembly 120 days before the commencement of the fiscal year. Furthermore, this plan should outline the allocation of resources and the direction of investment for each of the following twelve major sectors: health-welfare-employment; education; culture-sports-tourism; environment; R&D; industry, including small and medium-sized enterprises, and energy; SOC; agriculture, forestry, fisheries, and food; national defence; diplomacy and unification; public order and safety; and general and local administration.

Furthermore, the Ministry of Strategy and Finance is responsible for preparing a government budget plan for the upcoming year. To that end, the Ministry of Strategy and Finance must issue budget preparation guidelines and expenditure limits for each ministry by March 31 of each year. Each relevant ministry is then required to submit a budget request for the following year to the Minister of Strategy and Finance by May 31, taking into account the specified expenditure limits. Subsequently, the Minister of Strategy and Finance is tasked with allocating the government budget for the next fiscal year and submitting it to the National Assembly 120 days prior to the commencement of that fiscal year.

At the same time, the Ministry of Science and ICT is responsible for developing a mid- to long-term investment strategy for national research and development, in accordance with the Framework Act on Science and Technology. This ministry is also tasked with distributing and coordinating the budget for major national research and development programs.

Initially, the Ministry of Science and ICT must establish a mid- to long-term national R&D investment strategy on a five-year basis for strategic investment of the national R&D program budget. The national mid- to long-term R&D investment strategy includes the mid- to long-term investment goals and directions of national R&D, the direction of the distribution of investment resources in each field and stage of national R&D, and the plan for the sharing of roles between national R&D programs and private R&D programs.

Furthermore, the Ministry of Science and ICT is responsible for distributing and coordinating the budgets for significant national research and development programs. To accomplish this, the Ministry of Science and ICT must announce the direction and standards for national

research and development investment by March 15 of each year. The relevant ministries are then required to submit budget requests for the following year to the Minister of Strategy and Finance by May 31 of each year. Subsequently, the Ministry of Science and ICT must inform the Minister of Strategy and Finance of the results of the distribution and coordination of the budgets for major national research and development programs by June 30 of each year.

As mentioned above, government R&D programs are categorised into principal and general programs. Principal programs account for over 80% of all R&D programs and encompass such activities as basic research, applied research, development research, and the construction of large-scale facilities and equipment. On the other hand, general R&D programs include policy research and operating expenses for research institutes and universities, among other areas.

3. R&D Funding Issues and Suggestions for Indonesia

3.1 Comparison of the Funding Systems of the Two Countries

[Table 5-1] R&D Coordination and S&T Governance System

Korea	• (1999) The NSTC was in charge of coordinating R&D programs, while various ministries were responsible for managing them, and R&D funding agencies were responsible for managing R&D projects. • (2020-present) PACST and OSTI are in charge of the functions and roles of the NSTC. KISTEP supports the tasks of PACST, OSTI and MOEF.
Indonesia	• (Before 2021) The Ministry of Research and Technology was responsible for managing R&D programs and funding, and for coordinating seven non-ministerial government institutions. • (2021-present) BRIN is responsible for managing and implementing R&D programs and for funding research in non-university institutions.

Source: Author's elaboration.

[Table 5-2] Scale and Composition of R&D Funding

Korea	• National R&D Investment (2021): US\$ 89.282 billion (4.96% of GDP): public sector (US\$ 23.856 billion; 26.7%) + private sector (US\$ 65.426 billion; 73.3%). • Government R&D Budget (2021): US\$ 23.856 billion (4.9% of total government expenditure) principal R&D (80%) + general R&D (20%).
Indonesia	• National R&D Investment (2021): 41.43 T (0,28 % of GDP) public sector (36.85 T; 88, 95%) + private sector (4.57 T; 11,03 %) • Government R&D Budget (2021): 34.69 T (0.23 % of total government expenditure). • Central government (81.58%) + local government (2.16%).

Source: Author's elaboration.

[Table 5-3] Government's R&D Budget Allocation Entities

Korea	• MOEF (Ministry of Economy and Finance): Entire Government R&D Budget (principal R&D + general R&D) MOEF is responsible for budgeting 100% of government R&D. • OSTI under MSIT (Ministry of Science and ICT): Principal R&D (basic/applied/development R&D, facility equipment, etc.) OSTI is responsible for distributing and coordinating about 80% of government R&D, and submits the results to MOEF.
Indonesia	• Bappenas: Responsible for planning and budgeting programs and activities (but limited to indicative budget allocation). • Ministry of Finance: Responsible for planning and determining the definitive budget allocation, and coordinating with other ministries and institutions in planning their programs. • Directorate General of Budget, MoF: Responsible for reviewing and approving the budget proposals submitted by each ministry and institution. • BRIN: Responsible for coordinating and funding research programs in non-university institutions.

Source: Author's elaboration.

[Table 5-4] Flow of the Government R&D Budget

Korea	• MSIT announces the direction of R&D investment while MOEF determines the budget ceiling of all ministries in March of each year. • All ministries submit their budget proposals to MOEF and MSIT in May of each year. • In June of each year, MSIT submits the results of budget allocation and coordination for the principal R&D programs to MOEF. • In August, MOEF submits the budget allocation results for all R&D programs to the National Assembly. • The National Assembly deliberates and finalises the next year's government budget in early December of each year.
Indonesia	• Bappenas plans and budgets all programs and activities. • The Ministry of Finances determines the definitive program and budget allocation based on consultation with legislative bodies. • Each ministry and institution is required to propose a plan for its budget every year and submit it to the MoF. • The Directorate General of Budget (MoF) reviews and approves the budget proposals submitted by each ministry and institution.

Source: Author's elaboration.

[Table 5-5] Reinforcing S&T Specialties

Korea	The NSTC operates seven expert committees and receives review reports with a view to strengthening S&T expertise, life science/welfare, machinery/materials, basic science/infrastructure, ICT/convergence, energy/environment, big science/public R&D, and defence.
Indonesia	BRIN consists of one executive secretary, seven deputies, and twelve research organisations. The deputies manage infrastructure, human resources and research funding. The research organisations implement R&D programs in the following areas: aeronautics and space, nuclear energy, energy and manufacturing, earth and maritime sciences, life sciences and environment, electronic and informatics, social sciences and humanities, archaeology, language and literature, health, nanotechnology and materials, agriculture and food, governance, economy, and community welfare.

Source: Author's elaboration.

[Table 5-6] Agile Budget Reflecting Trends and Changes

Korea	- R&D Budget that takes into account government plans and technology trends: - Science and Technology Basic Plan (every 5 years, MSIT) - Next-Year Investment Direction and Technology Priority (every year, MSIT) - Mid-Term Fiscal Management Plan (every year for the next five-year budget plan, MOEF) - Mid-Term R&D Investment Strategy (every 5 years, MSIT)
Indonesia	- National Research Plan (RIRN) - National Medium-Term Development Plan 2020-2024 - National Long-Term Development Plan

Source: Author's elaboration.

3.2 Issues and Suggestions

At the national level, R&D funding should be promoted while ensuring strong links with other sectors within the national science and technology management system. R&D program funding should be closely aligned with national science and technology policy, and the budget for the following year should be prepared with sufficient consideration of the results of R&D program evaluations.

At the national level, it is crucial that R&D funding adequately reflects the unique characteristics of science and technology in the budgeting process, distinguishing it from budgeting in other fields. BRIN needs to assess ways to establish a mid- to long-term national R&D investment strategy that can serve as a reference for other related organisations. Furthermore, it should announce the direction and standards for national R&D investment at the beginning of each year. Additionally, BRIN could provide the Ministry of Finance (MoF) with the results of professional reviews of the budget proposals for major R&D programs submitted by related organisations, while the MoF may need to consider these results when preparing the budget allocation for the next year.

The procedures for efficiently promoting R&D funding and the roles of each entity must be clearly established. The roles of the major implementing entities, such as Bappenas, the MoF, BRIN, and R&D performing organisations, also need to be effectively defined and established. Furthermore, it is essential to establish systems and processes aimed at enhancing the efficiency and effectiveness of R&D budget allocation and adjustment.

This is because there are organisations within BRIN that manage R&D project funding and institutions that perform R&D, making it necessary to establish a system by which the budgets can be evaluated and managed independently.

3.3 Expected Effects in the Future

The results of research and analysis of Indonesia's R&D funding system and a comparison with Korea's R&D funding system were used to derive directions for improving Indonesia's R&D funding system in the future.

It is crucial to establish a system that can adequately incorporate the characteristics of science and technology into the national R&D funding process. To efficiently promote R&D funding, it is essential to define effective roles for the major players, including Bappenas, the MoF, BRIN, and R&D performing organisations, and to establish sophisticated systems and processes for the efficient distribution and coordination of R&D budgets.

By improving both the efficiency and effectiveness of Indonesia's R&D investment, it should be possible to enhance consistency with the national science and technology policy, increase the government's R&D productivity, and enhance the contribution of R&D to national economic development.

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CHAPTER 6

Alternative Policies for Developing the Research and Innovation (R&I) Infrastructure and Funding in Indonesia: The Lessons learned from South Korea

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Chapter 6. Alternative Policies for Developing the Research and Innovation (R&I) Infrastructure and Funding In Indonesia: The Lessons Learned from South Korea

1. Supporting Factors for Improving the Performance of Research and Innovation Infrastructures in BRIN

As described previously, the overall public research and innovation (R&I) infrastructure in Indonesia is currently delivered by the national research and innovation agency of Indonesia (BRIN). This institution has three authorities in three areas, namely, as an R&I executor, as an R&I funding and facilitation institution, and as a formulator of R&I policy. Providing the R&I infrastructure is one of the aspects of R&D funding and facilitation delivered by BRIN. The new scheme for the R&I infrastructure under BRIN's management consist of creating a centralised and integrated R&I infrastructure with the aim of improving the efficiency and effectiveness of public services in delivering the R&I infrastructure in Indonesia.

Prior to BRIN's formation in 2021, each Indonesian Public Research Institution (PRI) had autonomy in managing its own research infrastructure, including its business processes, funding for procurement and maintenance, and human resource requirements. However, these different degrees of authority and the varying management processes of each PRI frequently led to complications in managing the national research infrastructure, including overlapping facility procurements, a complex bureaucracy for accessing cross-institutional facilities, and a lack of best practices in the field of infrastructure management. This decentralised system created numerous inefficiencies, particularly in government research

funding, exacerbating the already low national budget for research.

The main issue of research budget inefficiency is one of the reasons for BRIN's formation. By merging all PRIs under one big institution, it is expected that BRIN will be able to manage its research infrastructure effectively and efficiently. The current situation of R&D infrastructure governance at BRIN is now more of a centralised system, with one deputy acting as the sole authority for its management called as the Deputy of Research and Innovation Infrastructure (DIRI). This integration is expected to prevent the aforementioned problem of inefficient research infrastructure management. In addition, the integrated and centralised R&I infrastructure is related to corroborate the wider impacts on users and society of the presence of BRIN in Indonesia. Therefore, a large transformation and change has happened in the field of R&I infrastructure in Indonesia.

BRIN's Deputy of Research and Innovation Infrastructure (DIRI) deployed a new system in order to manage the research infrastructure undergoing various transformations. One particular transformation is the grouping of laboratories and their facilities by region and research cluster as defined by the central management. This clustering of research facilities, based on research fields in geographical proximity, is expected to create a critical mass in the research and innovation infrastructure. Further, this transformation is intended to improve the management and accessibility of research facilities. Researchers that previously were unable to access a research facility will henceforth have access to a more integrated one with a relatively simplified bureaucracy.

The formation of BRIN as an 'open platform' for the research and innovation ecosystem also encourages research collaboration. Most of BRIN's research facilities, apart from several large and strategic ones, are now open to the public. The ELSA system, an integrated information system for the management and utilisation of research facilities, offers researchers in industry and academia open access to BRIN's research facilities, encouraging research activities in the private sector and research collaboration across institutions. The open platform is an effective way to promote collaboration with private and overseas partners and thereby raise the quality of the R&I infrastructure in Indonesia.

It was clearly mentioned above that the purpose of the transformation of the R&I infrastructure under BRIN's public management is not restricted to creating an efficient and effective process for R&I services in the bureaucracy system. This new scheme also aims to make it easier for researchers to gain access to the most appropriate and affordable R&I infrastructures in a particular area where there are centralised and specific R&I infrastructures positioned within several of BRIN's co-working spaces. In addition, this transformation is intended to open wider global connections and partnerships for R&I activities by involving BRIN's researchers as key partners in global collaborations. In sum, the integrated and centralised R&I infrastructures initiated by BRIN are expected to have many positive and beneficial effects for R&I achievements in Indonesia in the future.

Having learned from the practical experience of South Korea and considering the input of Korean team, certain supporting factors can improve the performance of the R&I infrastructure under BRIN's management, as well as minimising or resolving the existing R&I infrastructure problems currently faced by BRIN. These supporting factors include the following:

1.1 Legal base: National Acts and Presidential Decrees

The legal regulations provided under National Act No. 11, 2019, which concerns the national system for research, development, and the application of science and technology, and Presidential Decree No. 78, 2021, which concerns national research and innovation agencies, are outstanding factors that could back up or bolster diverse policies and programs aimed at improving the performance of BRIN's R&I infrastructure. Both laws have the legal power required to mobilise BRIN's resources, including funding, personnel, and facilities among others. In addition, national laws provide the legal foundation required by BRIN's policymakers when formulating and implementing programs for supporting the performance of the R&I infrastructure.

BRIN policymakers can enhance R&I infrastructure performance by implementing cascading programs aligned with the two aforementioned laws. These laws constitute a foundational framework for public servants and BRIN personnel in the execution of

government initiatives aimed at strengthening R&I infrastructure. Also, these laws provide the legal basis for BRIN's efforts to attract external governmental and non-governmental partners in Indonesia and global connections to improve and reinforce the performance of the R&I infrastructure in Indonesia.

1.2 A new organization with new structures

BRIN is the only R&D organisation funded by the Government of Indonesia, having incorporated all the public R&D organisations previously existing in Indonesia. BRIN has a new organisational structure consisting of deputies and research organisations endowed with the following three roles: R&I funding agency, R&I policy formulator and implementer, and R&I executor. This new organisation and structure provide opportunities for new tasks and duties to be performed, including the new role played by each unit under BRIN's organisational structure. It also provides the momentum needed to bring about transformation and changes that will lead to improvements and better outcomes in the R&I field.

The reorganization and restructuring of numerous units, roles, and task forces amidst a transformative period for the organization and its R&I infrastructure presents significant challenges. New R&I infrastructure advancements must be seamlessly integrated into the organization's new structure to optimize BRIN's overall R&I infrastructure performance. Further, it will present a major opportunity for policymakers to undertake huge changes in R&I infrastructure management simultaneously.

1.3 Adaptive and dynamic institution

BRIN has established itself as an adaptive and dynamic institution that is identified with co-work spaces, work from anywhere (WFA), flexible work-time e-services for performing R&I under the ELSA system, the creation of research groups, and two channels of communication including both top-down and bottom-up modes. BRIN's establishment was informed by a benchmarking of globally renowned R&D institutions with a proven track record in swiftly addressing global challenges. This strategic approach is anticipated to significantly enhance Indonesia's R&I infrastructure performance relative to its predecessors.

Enhancing the performance of the R&I infrastructure necessitates a more strategic approach than mere continuation of existing progress. Institutional reform is paramount to optimizing R&I infrastructure performance. This often necessitates a fundamental restructuring to align with the strategic objectives of R&I policy. Conversely, incremental adjustments to the existing institutional framework may suffice to achieve R&I infrastructure policy objectives.

Nonetheless, BRIN's capacity for adaptation and dynamism is essential to navigate the evolving landscape of R&I policy and its goals.

1.4 Increasing number and capabilities of human resources

As Indonesia's preeminent R&D institution, BRIN employs approximately 15,000 highly skilled personnel encompassing researchers, engineers, policy analysts, managers, and other specialized roles. These individuals possess a diverse range of competencies and capabilities, cultivated through their expertise and education, which collectively contribute to BRIN's overall performance. Also, they are the executors and beneficiaries of the R&I infrastructures and facilities provided by BRIN to bolster their work and careers - particularly those who work extensively in non-social research fields.

The performance of the R&I infrastructures is inseparable from the human resources who use and benefit from them. The capabilities of researchers and engineers constitute an intangible capital that can strengthen the performance of BRIN's R&I infrastructure. They also use strategies that can benefit the R&I infrastructure according to their goals. As a result, their accomplishments will have an impact on BRIN's organisational performance overall. Therefore, considering the needs of BRIN's personnel is crucial to improving R&I performance in Indonesia.

1.5 Multiple national and global networks

Established by integrating numerous public R&D institutions, BRIN is now involved in multiple partnerships and collaborative ventures in diverse countries. This means that BRIN is a compilation of diverse networks dispersed across many countries. Partnerships and networks constitute an intangible capital that can benefit an organisation by reinforcing or supporting its organisation goals. Commonly, they include close relationships, similar organisations, shared interests, complementary functions, and influential units. Such organisations are needed as the components of an external supporting system that influences how an organisation works.

The performance of R&I infrastructure can be feasibly and easily improved through the involvement of extensive networks comprising national and international partners. In this occasion, BRIN is on the good moment by which all networks have been integrated into this organisation. BRIN's policymakers can benefit from and use the existing networks to support R&I infrastructure performance by means of external funding, equipment, and assistance. It is important that BRIN should exercise a strong appeal for its partners and colleagues.

2. Alternative Policies for Improving the Performance of BRIN's Research and Innovation Infrastructures

Research infrastructure plays an important role in supporting the growth of research and innovation across a wide range of fields. Better governance of the research infrastructure will lead to increased access to cutting-edge research facilities, enabling researchers to carry out high quality and collaborative research activities that drive the knowledge-based economy. BRIN's centralised system of research infrastructure management, though good-willed, is still far from its ideally expected performance, and several challenges have emerged. As such, we propose a series of alternative policies aimed at moderating these challenges by drawing on South Korea's experience in implementing R&D infrastructure policy as follows.

i) Redefine and broaden the scope of research and innovation (R&I) infrastructure, which should be not limited to the availability of equipment, and facilities, to include the availability of funding support, personnel capability, inter-unit coordination, maintenance, and time management.

BRIN's R&I infrastructures can be classified by referring to the scope of their utilisation (sole utilisation, joint utilisation), utilisation status (normal operation, idle, underused, unused), and construction type (individual research facility equipment, system research facility equipment, and development research facility equipment). In addition, the categories of early (leading) research utilisation, (middle) general research utilisation, and (late) educational research utilisation are important.

ii) Attract and induce external partners and governmental representatives or assemblies to invest in R&I infrastructures through an R&I investment scheme. External partners and government agencies can allocate budgets for the development of BRIN's R&I infrastructure. Such budgets can be evaluated and monitored accordingly, and the results shared between BRIN and its external partners.

iii) Establish an information technology system to monitor and control availability and operation of the R&I infrastructure at each unit located in a different location under BRIN's management system. Representative of South Korea, adoption of the integrated R&D infrastructure monitor system is strongly recommended to improve BRIN's R&I infrastructure management.

iv) Formulate Indonesia's Basic Plan for National Research Infrastructure with the aim of creating an efficient and effective R&I infrastructure through cross-departmental collaboration to minimise increasing inefficiency issues such as overlapping procurement and purchasing, and non-optimal joint utilisation of R&I infrastructure. In this regard, medium- and long-term planning will make it easier for policymakers to formulate and implement R&I infrastructure programs for 5-10 years in the future, including forecasting funding. In addition, sequential plans for developing R&I infrastructures such as R&I equipment, R&I facilities, governmental R&I support, and the R&I equipment industry could be implemented.

v) Enact and promulgate supporting laws in the field of medium- and long-term planning for Indonesia's R&I infrastructures development on topics. Those laws are like investment efficiency, operational advancement, maximising utilisation, and enhancement of support capabilities aimed to formulate the first R&I planning. The subsequent laws are such as improving the utilisation of research-supported research infrastructure, improving efficiency of research infrastructure utilisation, domestic research infrastructure industry development and human resource development aimed to second R&I infrastructure planning.

vi) Prioritise facilities procurement. To ensure efficient infrastructure management, one alternative policy is to prioritise infrastructure development. As such, it is necessary to establish a mid- or long-term roadmap for research infrastructure that serves as the main strategic direction. This roadmap should encompass the prioritisation of research infrastructure management and development, its objectives, and also the necessary steps and resources to ensure its operation. BRIN could prioritise the procurement of R&I

infrastructures by both top-down (from BRIN's management to researchers) and bottom-up (from researchers' needs to BRIN management) methods. All processes should pass strict reviews by the committee, taking into account personnel requirements, operational costs, and the environment.

vii) Create a facility maintenance strategy encompassing all services according to laboratory licenses and standards. This policy aims to ensure the efficient maintenance of R&I infrastructures such as laboratories, equipment and facilities, and to eliminate inefficiencies.

viii) Optimise the occupancy/usage of facilities by expanding the users of R&I infrastructures and involving external partners, and by improving the accessibility of R&I infrastructures in order to optimise the utilisation of R&I infrastructures. This process can be conducted through joint utilisation services, and it could supplement or complement the R&I infrastructure needs of each other. Establishing external partnerships with the private sector for research infrastructure development is one alternative means of further developing the Indonesian innovation ecosystem. Attracting private investment in research infrastructure is a challenging issue. BRIN will need to provide a common ground that will benefit both BRIN and its counterparts through research services or collaborations, and risk and capital sharing with industry. A dynamic and sustained relationship with the private sector will generate diverse benefits. BRIN will not only gain access to investment enabling it to develop its research infrastructure, but also attract collaborations, particularly with industry, thereby facilitating the transfer of industrial knowledge to its researchers.

ix) Form partnerships with universities and Ministry of Education, Culture, Research, and Technology (MoECRT). Partnerships with the public sector, particularly with universities and the MoECRT, possess multiple advantages for BRIN as a public research institution. With relatively equivalent characteristics, the alignment of common interests for national research and innovation may serve as a driver for a sustainable partnership. Opening access to the infrastructures of BRIN and universities through formal institutional or research collaboration will enhance national critical mass in research infrastructure, while interactions with universities will accelerate the process of transferring knowledge in the

science pool. A strategic partnership with MoERCT will strengthen Indonesia's capacity for research and innovation as part of a national partnership program or strategic infrastructure funding.

x) Upgrade facilities, services for facilities, the capacities of human resources, and the management and training of equipment operators. BRIN's efforts to provide research and innovation services could be supported by providing access to cutting-edge research infrastructures. Upgrading research facilities will not only strengthen researchers' capability for quality research activities, but will also attract multidisciplinary and cross-institution research collaborations. The upgrading of research facilities consequently requires improvements in human resources capacity. Therefore, improvements in terms of both the sophistication of facilities and the ability of human resources to operate and ensure the safety and maintenance of modern research facilities needs to be the aim of effective research infrastructure management.

xi) Provide Funding Support. A shortage of funding is a recurrent issue in the Indonesian innovation system. It is imperative for BRIN to provide not only sufficient funding, but also to provide sustainable and agile funding support in order to build and manage a modern research infrastructure effectively. This financial constraint is often the main barrier to research institutions' efforts to maintain and upgrade their research facilities. Thorough planning and adequate funding for research infrastructure and its derived resources and requirements (e.g. human resources, materials, installation, etc.) will lead to efficient management and is the key factor in optimising the research infrastructure in Indonesia.

3. Supporting Factors for Improving Research and Innovation Funding at BRIN

In the landscape of national development, research and innovation play pivotal roles, driving economic growth, fostering societal advancement, and ensuring global competitiveness. The National Research and Innovation Agency (BRIN) in Indonesia serves as a key institution tasked with promoting and supporting research and innovation activities. However, enhancing research and innovation funding within BRIN requires a comprehensive and strategic approach. Based on the lessons learned from South Korea, there are several supporting factors for optimising the research and innovation funding mechanisms at BRIN as follows.

Efficient governance structures within BRIN are fundamental for the effective allocation of funds and project oversight. A well-defined governance framework outlines roles and responsibilities, establishes clear decision-making processes, and ensures transparency and accountability. By adhering to robust governance practices, BRIN can streamline its administrative processes, thereby reducing bureaucratic hurdles that often impede the timely execution of research and innovation projects. Transparency in decision-making also builds trust among stakeholders, inducing both public and private partners to contribute to funding initiatives.

Collaboration between BRIN, academia, industry, and other stakeholders is indispensable for the success of its research and innovation endeavours. This collaborative approach ensures that research initiatives are not isolated but instead reflect the interconnected needs of society and industry. Public-private partnerships, for example, can bring diverse perspectives and resources to the table, enhancing the relevance and impact of research projects. Through continuous dialogue and cooperation, stakeholders can align their efforts, creating a synergistic ecosystem that fosters innovation while addressing real-world challenges.

Investing in human capital is a strategic imperative for fostering a thriving research and

innovation environment. BRIN should prioritise capacity-building initiatives, providing researchers and innovators with opportunities for training, skills development, and access to cutting-edge technologies. A well-trained workforce enhances the quality of research outputs and innovation outcomes, contributing to the overall success and competitiveness of the research ecosystem.

The efficiency of funding processes has a significant impact on the pace at which research and innovation projects can be initiated and completed. By streamlining the application and approval processes, BRIN can reduce delays and uncertainties, allowing researchers to focus on their work rather than wasting time having to navigate bureaucratic obstacles. An agile and responsive funding mechanism is critical for maintaining momentum in the rapidly evolving landscape of research and innovation.

Cultivating a culture that encourages risk-taking and innovation is pivotal for driving groundbreaking advancements. As such, BRIN should foster an environment in which researchers and innovators feel empowered to take calculated risks, experiment with unconventional ideas, and learn from their failures. Embracing risk is often the catalyst for transformative breakthroughs, and the adoption of a culture that encourages innovation would position BRIN as a leader in driving technological and scientific progress.

Global collaboration is essential for research and innovation agencies to stay at the forefront of technological advances. By actively engaging with the international research community, BRIN can access diverse expertise, share knowledge, and leverage resources beyond national borders. Collaborating with international partners broadens the scope of research, exposes researchers to different perspectives, and facilitates the exchange of best practices.

Implementing robust monitoring and evaluation mechanisms is crucial for assessing the impact of funded projects and informing future decisions. BRIN should establish comprehensive evaluation frameworks that measure the success and effectiveness of research and innovation initiatives. Regular assessments help identify successful strategies, understand challenges, and ensure that funding is directed towards those projects with the greatest potential for societal and economic impact.

Relying on a diverse range of funding sources is essential for building resilience within the research and innovation ecosystem. BRIN should explore avenues such as public funds, private investments, philanthropy, and international grants to reduce dependence on a single source of funding. Diversification not only ensures financial stability but also opens up opportunities for collaboration with different sectors, enriching the research landscape.

Building public awareness and engagement is critical for garnering support for research and innovation funding. BRIN should implement targeted communication strategies that highlight the societal benefits of research initiatives. Engaging the public in this process would foster a sense of ownership and underscore the importance of research and innovation in driving national development.

To improve research and innovation funding at BRIN, there are several factors which could have an influence in this regard. These include adjusting the composition of private sector involvement in research through the introduction of funding and facilitation schemes that are open to private parties. This could be done by issuing open calls for proposals that invite collaboration with BRIN's researchers. Currently, BRIN's competitive open-platform funding schemes accommodate such endeavours. Enhancing R&D funding with collaboration and cooperation through the enhancement of funding partnerships with non-governmental entities is achieved through an open-platform scheme encompassing human resources, infrastructures, networking, and shared research spaces. Additionally, to increase R&D funding, BRIN is opening and engaging in joint funding schemes with various global institutions in Japan, China, Australia, South Korea, ASEAN countries, etc. The aim here is to have the government to act as a facilitator that enables local business entities to carry out R&D-based product development with appropriate investment and to support global engagement and collaboration, thereby accelerating the transfer of knowledge and skills through research collaborations.

4. Alternative Policies for Improving Research and Innovation Funding at BRIN

In determining research funding by learning from the example of South Korea, Indonesia will need to establish medium- and long-term plans, and invest strategically in national research and development. It is essential to align these strategies with Indonesia's science and technology framework, namely the law on the national science and technology system. Indonesia needs to clearly define a mid- to long-term national R&D investment strategy on a five-year basis for strategic allocation of the national R&D program budget. Furthermore, BRIN is responsible for allocating and coordinating the budgets for significant national research and development programs. BRIN needs to allocate funds to research areas deemed to be priorities. In this case, funding for priority research will not be subject to competitive selection but will receive specific budgeting. Additionally, BRIN needs to form a funding committee tasked with coordinating and reviewing budget proposals for major R&D programs with the Ministry of Finance. This initiative is aimed at establishing a system for incorporating the characteristics of science and technology into the national R&D funding process

The need for research and innovation to promote sustainable development is becoming increasingly apparent. This is proven by various examples of countries that have succeeded in becoming developed countries due to the success of their science and technology policies, which have become the backbone of their development. In the case of the National in Indonesia, alternative policies can be explored to enhance research and innovation funding.

■ Venture Capital and Public-Private Partnerships (PPPs)

One alternative policy involves fostering a more dynamic relationship between BRIN and the private sector through the utilisation of venture capital and PPPs. By establishing partnerships with venture capital firms, BRIN can attract private investment in research and innovation projects, providing not only financial support but also industry expertise. Public-Private Partnerships, on the other hand, enable joint ventures with private entities, sharing

the risks and rewards in research initiatives. This approach not only diversifies the sources of funding but also encourages the alignment of research goals with market needs.

■ **Tax Incentives for Corporate Research and Development (R&D) Spending**

Introducing tax incentives for corporations engaging in research and development activities can stimulate increased private-sector investment in innovation. By offering tax breaks or credits to companies that contribute to collaborative research projects with BRIN, the government can encourage a more vibrant and mutually beneficial partnership between the public and private sectors. Such incentives serve as powerful equipment for stimulating innovation while reducing the financial burden on participating companies.

■ **Upgrading of the Intellectual Property (IP) Regulations**

A re-evaluation of the intellectual property regulations could be instrumental in incentivizing innovation. Streamlining and modernizing the IP laws can make it easier for researchers and innovators to protect their intellectual assets, fostering an environment in which stakeholders feel more secure about their investments. This, in turn, can attract increased private funding and partnerships, as companies may be more inclined to invest when they are confident about the protection of their innovations.

■ **Agile Funding Mechanisms**

The introduction of more agile and flexible funding mechanisms within BRIN can expedite the research and innovation process. By adopting such mechanisms as challenge-based funding or agile project financing, BRIN will be able to respond more swiftly to emerging priorities and opportunities. This approach can also enhance adaptability and ensure that funding is directed to projects with immediate relevance and potential impacts.

■ **Open Innovation Platforms**

The creation of open innovation platforms can serve as an alternative policy for stimulating collaboration among diverse stakeholders. By establishing platforms that facilitate collaboration between academia, industry, and the non-government organizations (NGOs),

BRIN can harness a broader range of expertise and resources. In addition, these platforms can act as intermediaries, connecting researchers with industry and (NGOs) needs and promoting a more collaborative and responsive innovation ecosystem.

■ Performance-Based Funding

Shifting towards a performance-based funding model can incentivise researchers and institutions to deliver tangible results. By linking funding to key performance indicators (KPIs) and project milestones, BRIN can ensure that resources are directed towards projects with a higher likelihood of success and societal impact. This approach aligns the interests of researchers with the broader goals of national development.

■ Innovation Procurement Policies

The government can leverage its procurement power to drive innovation by incorporating innovation procurement policies. By favouring products and services developed through innovative processes or technologies, BRIN can incentivise companies to invest in research and development activities. This policy encourages collaboration between the public and private sectors while ensuring that public procurement contributes to technological advancements.

■ International Collaboration Frameworks:

Strengthening international collaboration frameworks can expand funding opportunities for BRIN, while actively participating in global research consortia, joint projects, and initiatives can bring additional resources and expertise. Moreover, participation in international funding programs and grants can provide BRIN researchers with access to a broader pool of financial support.

■ Inclusive Innovation Policies

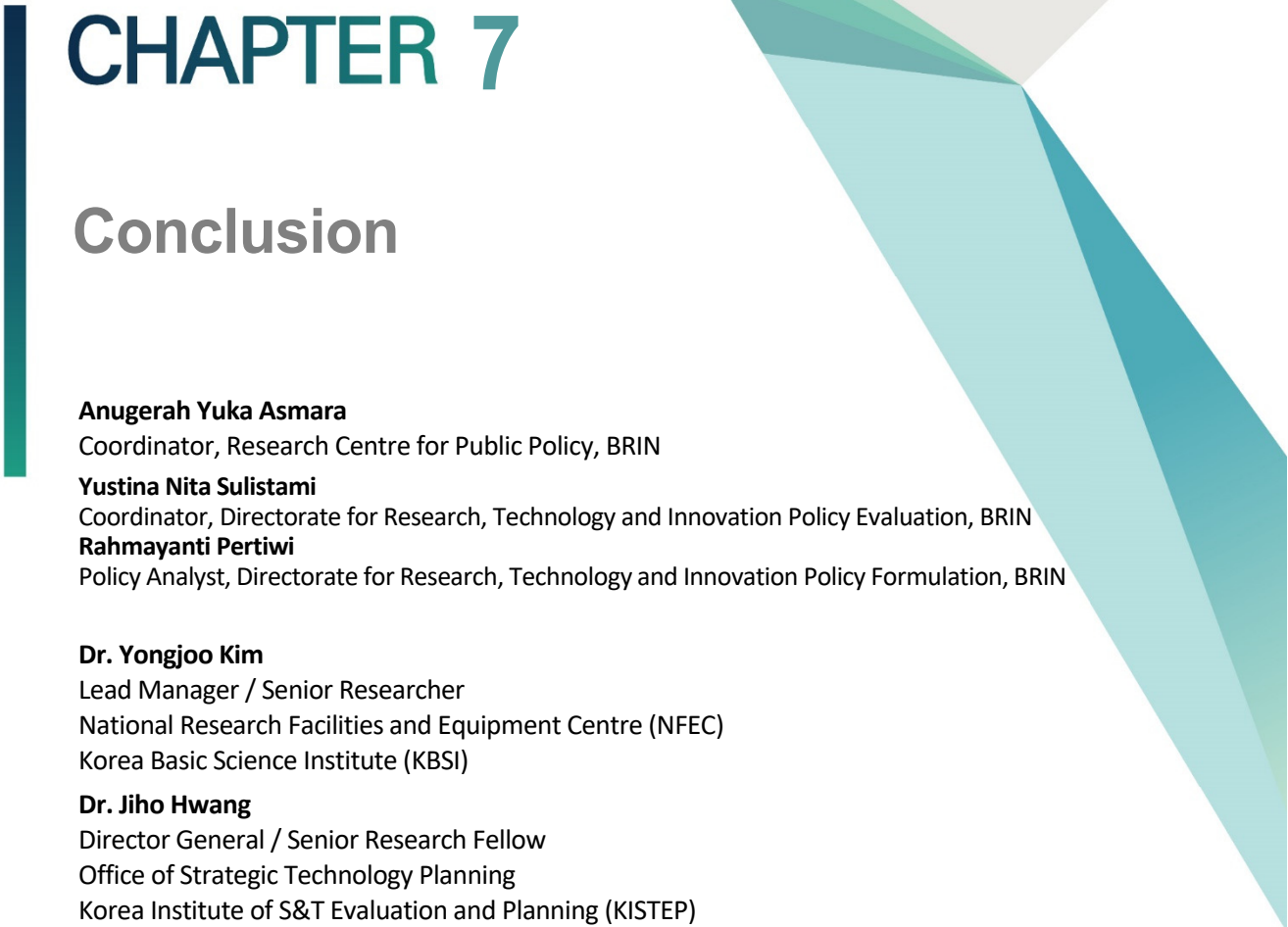
The development of inclusive innovation policies will ensure that funding is accessible to a diverse range of researchers, including those from underrepresented groups. By actively promoting diversity and inclusivity in research projects, BRIN can tap into a broader talent

pool and address a wider spectrum of societal challenges. Furthermore, inclusive policies foster creativity and ensure that innovative solutions are representative of diverse perspectives.

■ **Strategic Communication and Advocacy**

A robust communication and advocacy strategy is crucial for creating awareness and generating support for research and innovation funding. As such, BRIN should invest in strategic communication campaigns that highlight the societal impact of research initiatives. Engaging the public, policymakers, and industry stakeholders will foster a shared understanding of the importance of research and innovation, potentially leading to increased funding support.

These alternative policies for improving research and innovation funding at BRIN encompass a broad spectrum of strategies ranging from financial instruments and collaboration frameworks to regulatory adjustments and incentive mechanisms. By exploring these alternatives, BRIN can diversify its funding sources, stimulate private-sector engagement, and foster a more responsive and inclusive innovation ecosystem. The adoption of these policies would align with the dynamic and adaptive nature of the research and innovation landscape, ensuring that BRIN remains at the forefront of driving progress and contributing to the Indonesia's socio-economy development.

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CHAPTER 7

Conclusion

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Chapter 7. Conclusion

1. Summary

Research and innovation (R&I) infrastructure and funding have become critical factors in developing the science-technology-innovation (STI) environment in many countries around the world. South Korea, as one of the most advanced, cutting-edge countries, has implemented the best practices of R&I infrastructure and funding in the world. Indeed, South Korea has often been referred to as the most innovative country in one decade. Similar to Indonesia, South Korea has a long institutional history concerning the transformation of STI organizations. Currently, Indonesia's STI organisation is solely delivered by BRIN. As a new entity, BRIN should improve and accelerate the STI institution to achieve the national development goals. It is the reason why Indonesia should learn more from the South Korea's practice in implementing R&I infrastructures and funding to bolster and accelerate national STI development.

As regards the R&I infrastructure in Indonesia, especially at BRIN, one can refer to the aforementioned definition of R&I infrastructure as "facilities, resources and related services used to conduct top-level research in a scientific field, including major scientific equipment or facilities, or databases for disseminating research data." As a newly integrated R&D institution in Indonesia, BRIN is on the adaptation and consolidation phase. BRIN has faced various obstacles and challenges since October 2021, and has resolved several of them, but some still remain as major issues. An integrated and centralised R&I infrastructure delivered by BRIN is one of the issues that urgently needs to be resolved as soon as possible.

According to the results of empirical studies and a survey of 400 of BRIN's researchers, the three top issues are "appropriate structure", "clarity of roles", and "key individuals", while the last issues are "knowledge sharing", "external focus", "standard procedures", and "IT

support”. These issues are classified into three terms, namely, physical, management, and human resources. The physical obstacles of R&I infrastructures include aging equipment, insufficient supply of equipment, and concentrated or centralised R&I infrastructures. As regards the last obstacle, it is not linear with the early goals, i.e. that centralised or concentrated R&I infrastructures would be an efficient and effective way of managing R&I infrastructures.

On the adaption phase, the managerial obstacles to the R&I infrastructure under BRIN’s management are overlapping procurement and planning, complex maintenance, limited access with sectoral coordination among researchers, the under-utilisation of equipment due to unclear reasons, modest surveillance and monitoring of R&I infrastructures, the distant locations of laboratories, and the absence of supply and demand data for monitoring the R&I infrastructure under BRIN’s supervision areas. Other issues related to human resources management have emerged, such as the weak coordination and communication among human resources across units and laboratories, non-optimal socialisation on how to use R&I infrastructures, insufficient knowledge for managing laboratories, and the rapid mobilization of human resources at each unit. Therefore, all the emergent R&I infrastructure problems faced by BRIN should be solved as soon as possible.

As regards R&I funding, especially at BRIN, the efficiency and effectiveness are the main goals, in line with the expectations of national stakeholders. BRIN is the sole R&D organisation tasked with consolidating all public R&I funding sourced from government agencies and the private sector. In addition, overseas funding sources constitute one of BRIN’s main objectives for creating a huge R&I in achievement Indonesia. BRIN has a large authority to spend the national R&I funding in Indonesia. Besides, the Ministry of Education, Culture, Research and Technology also has R&D funding specifically aimed to universities in Indonesia. As a result, BRIN is able to mobilise most of the Indonesia’s R&I resources by using the availability of funding under BRIN’s management.

The R&I funding issues are more dynamic and adaptive, since October 2021-December 2023, the expected goals of BRIN are hard to achieve at the short run. The accomplishment of

efficient and effective funding goals requires more efforts and struggles. BRIN also faces with a series of classic funding challenges namely, bureaucratic funding mechanism and relatively small budgets. Although BRIN has implemented an integrated R&I funding scheme to simplify the process of R&I funding management, it will still require further improvements in the future.

The emerging challenges with the R&I infrastructure and funding reveal that BRIN are part of a long process of organisational change. Currently, BRIN is being on the adaptation and consolidation phase which requires more time and efforts to decipher existing challenges and obstacles. Accordingly, the practices of South Korea - such as the National Research Facilities and Equipment Centre's (NFEC) focus on improving the R&D infrastructure, and the Korea Institute of S&T Evaluation and Planning's (KISTEP) efforts to improve the R&D funding –represent a “cutting-edge” lesson for BRIN in terms of improving R&I infrastructure performance and funding schemes in Indonesia. In Addition, governmental policy supports from outside BRIN are essential for accomplishing STI development in Indonesia.

2. Policy Recommendations

South Korea's explanation has given much insight for thought for Indonesia, especially for BRIN as the sole R&D institution funded by the government. We have prepared two typical policy recommendations aimed at improving the performance of BRIN in this regard: one concerns the research and innovation infrastructure, and the other research and innovation funding. The majority of our policy recommendations for Indonesia are taken from empirical results and South Korea's experience and modified accordingly.

A. Improvement of Research & Innovation (R&I) Infrastructure Performance

Establishing an R&I infrastructure ecosystem is the most appropriate policy that BRIN can execute.

1. As a first measure, BRIN should create three plans by referring to South Korea's practical experience: i) establish a system for investing and managing the R&I infrastructure and unifying the decentralised management system across government ministries; ii) Lay the foundations for utilising the research infrastructure by linking and integrating the utilisation systems of ministries and technologies into one; iii) establish a field-centred management and utilisation system.
2. Establish a system and management direction for providing comprehensive facilities and equipment to support the R&I infrastructure; create and decentralise duties and responsibilities autonomously; and improve the quality and quantity of personnel who operate R&I infrastructures, such as training, career management, and specialised jobs.
3. Arrange sequential planning for the medium to long term (5-20 years), starting with the identification of recent issues and the status of R&I infrastructures; create distinctive roles for stakeholders by involving central and local government agencies, R&D institutes, researchers, engineers, policy analysts, and other related users; accommodate and prioritise the diverse demands of researchers and users of R&I infrastructures including R&D

laboratories, R&D facilities, etc.; and prioritise and establish sustainable R&I infrastructure management planning supported by the legal system and a virtuous management cycle.

4. Promote R&I infrastructure investment by attracting private sectors, universities, and overseas partners efficiently and effectively. This will require the creation of an efficient and effective joint utilisation system, as well as the establishment of a research equipment cost management utilisation system and a performance management and compensation system for joint utilisation, and the fostering of joint utilisation bases.

5. Monitor and evaluate existing and new schemes of R&I infrastructure investment at BRIN and other R&I infrastructure schemes conducted by BRIN. All such processes need to be reviewed and supervised to minimise problems and newly emerging issues.

6. Reinforce support for the R&I infrastructure at the technical and operational levels in order to strengthen the R&I infrastructure efficiently and effectively.

7. Establish an agile, adaptive and organic information system capable of executing plans and monitoring R&I infrastructure management. This will cover the establishment of an R&I infrastructure information collection system, and a management system for information realisation, and the discovery of information systems

B. Improvement of Research & Innovation (R&I) Funding Performance

1. BRIN should establish and strengthen an interlinked system with the Ministry of Finance, other ministries, assemblies, and public boards in proposing R&I funding allocation. This system is fruitful to accelerating in managing public R&I funding in Indonesia.

2. It is essential to plan a sequential budget process: This can be divided into short-term funding planning (1-2 years), medium-term funding planning (3-5 years), and long-term funding planning (more than 5 years). The master plan and budget priorities are the basic elements to be formulated by BRIN in order to provide clear and responsible information to stakeholders and society.

3. Monitoring and evaluating the system of R&I budget planning and implementation are essential to supervise all funding processes in accordance with public accountability to and responsibility for its stakeholders. Again, such a system can be used to overview the progress or lack of progress requiring improvement.
4. BRIN should establish an autonomous and independent agency or unit to supervise and monitor budget planning throughout all process stages. This unit would be composed of professional personnel from academia/universities, the private sector, non-governmental organisations, and related partners who have committed themselves to acting independently and autonomously.
5. BRIN and its external partners should collaborate on the creation of a special funding system that can guarantee the use of funding sourced from the private sector, overseas partners, and governmental sources among others.
6. The use of information technology (IT) is a prerequisite to corroborating the R&I funding system established by BRIN and its partners. Indeed, IT is one of means to ensure fair accountability and boosting the trust of stakeholders and public society, including to BRIN's external partners.

REFERENCES

Chapter 1. Project Overview

- European Commission (2015), "Supply and Demand Side Innovation Policies Annexes of First Policy Brief, Directorate-General for Research and Innovation, Luxembourg: Publications Office of the European Union", doi: 10.2777/43249.
- Mani, S. (2002), "Government, Innovation, and Technology Policy, Cheltenham-UK: Edward Elgar"
- Patanakul, P. and Pinto, J.K. (2017), "Examining the roles of government policy on innovation", *Journal of High Technology Management Research* 25 (2014), pp. 97-107
- Schot, Johan & Steinmueller and W. Edward (2018), "Three frames for innovation policy: R&D, systems of innovation and transformative change", *Research Policy*, pp. 1554-1567.
- Sun, S. L., et al. (2019), *Enriching Innovation Ecosystems: The Role of Government in a University Science Park*, *Global Transitions* 1: pp. 104-119
- Wang, J. (2018), "Innovation and government intervention: A comparison of Singapore and Hong Kong", *Research Policy* 47(2), pp. 399-412.
- World Bank (2010), "Innovation Policy: A Guide for Developing Countries, The Washington, D.C.: World Bank".
- Yoon, J. W.. (2018), "Evolution of Science and Technology Innovation Policy in Korea -From the Innovation System Perspective", Presentation of the public lecture in the Faculty of Administrative Science – University of Indonesia in 2018 (Online).

Chapter 2. Current Status of the R&I Infrastructure Policy in Indonesia

- Danica Zendulková, Gabriela Gavurníková and Ján Turňa (2022) "Map of the Research Infrastructure as a Tool of the Information System for Research, Development, and Innovation," 15th International Conference on Current Research Information Systems, pp. 47–56.
- Eder, M. and J. R. Pierce Jr., (2011) "Innovations in Faculty Development: Study of a Research Assistance Unit Designed to Assist Clinician-Educators with Research," *Southern Medical Journal*, 104(9), pp. 647–650.
- Gianluca Fabiano, Andrea Marcellusi, and Giampiero Favato (2020) "Channels and processes of knowledge transfer: How does knowledge move between universities and industry?" *Science and Public Policy*, 47(2), pp. 256–270.
- Juhlin, M. et al. (2013), *Interactions between research capital and other research resources in UK Higher Education Institutions*, Department for Business Innovation and Skills.

- Kurniawan, T. (2017) "Addressing Barriers to University Research: A Case Study of the University of Indonesia, Centre for Study of Governance and Administrative Reforms," <https://www.Ksi-Indonesia.Org/Old/En/News/Detail/Addressing-Barriers-to-University-Research-a-Case-Study-of-University-of-Indonesia>.
- Law of BRIN No. 6, 2023, About Strategic Planning of the National Research and Innovation Agency (BRIN).
- Širá, E. et al. (2020) Knowledge Economy Indicators and Their Impact on the Sustainable Competitiveness of EU Countries, *Journal of Sustainability*, 12(10), doi: 10.3390/su12104172.

Chapter 3. Current Status of R&I Funding Policy in Indonesia

- Andhina Ratri Aryani, M. Rinaldi Camil, and Mona Luthfina Usmani(2021) "Reformasi Tata Kelola Pendanaan: Peningkatan Efektivitas dan Efisiensi Skema Pendanaan Kementerian Ristek/BRIN", KSI.
- Dzulfikar, L. T. (2019) "A Better Research Funding Model for Indonesia: Learning from Singapore," www.theconversation.com
- <https://www.statista.com/statistics/732247/worldwide-research-and-development-gross-expenditure-top-countries/>.
- Lepori, B. (2007) "The Power of Indicators: Introduction to the special issue on public project funding for research," *Science and Public Policy*, Vol. 34, Issue No. 6, July 2007, Pages 370–371, <https://doi.org/10.3152/030234207X234596>
- Sari, A.A. (2023), "Challenges of Research and Innovation Funding," STEPI Korea- BRIN Mini Workshop R&D Funding Policy.
- Steen, J. (2012) "Modes of Public Funding of Research and Development: Towards Internationally Comparable Indicators," *OECD Science, Technology and Industry Working Papers*, No. 2012/04, OECD Publishing, Paris, <https://doi.org/10.1787/5k98ssns1gzs-en>.World Intellectual Property.
- Organisation (WIPO) (2023), "Global Innovation Index 2023: Innovation in the face of uncertainty, Geneva: WIPO", doi:10.34667/tind.48220.

Chapter 4. R&D Infrastructure Policy in Korea

- Paula E. Stephen and Sharon G. Levin(1992) "Striking the Mother Lode" in *Scienc*. Oxford University Press, NY.(ISBN: 0-19-506405-4).

Dickinson, John P.(1989) "Science and scientific researchers in modern society," Unesco Press.
(ISBN : 978-92-3-102427-6).

OECD(2019) "Reference framework for assessing the scientific and socio-economic impacts of research infrastructures," OECD Science, Technology and Industry Policy Papers, No. 65, OECD Publishing, Paris, <https://doi.org/10.1787/3ffee43b-en>.

Ministry of Science and ICT(2021) Administrative rules: Standard Guidelines for the Management and Utilisation of National R&D Facilities and Equipment, No 2021-92.

Chapter 5. R&D Funding Policy in Korea

Ministry of Science and ICT (2020) Framework Act on Science and Technology, Act No. 18863, June 10, 2022.

Ministry of Strategy and Finance (2023) National Finance Act, Act No. 19430, June 9, 2023.